StandardEyes — Statistical Reference and Walkthrough

Laser Vision Correction Analysis Software

Companion document to the main manuscript — what the software does statistically for every option you can choose, how behaviour changes with more than two groups, and an illustrated walkthrough of a complete analysis.

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Every screenshot and figure in Parts 3 to 5 is an unmodified capture of the software, or of its output, running on the example dataset distributed with the app (`Example Data\Example_Refractive_Dataset.xlsx`). That dataset is made up: it contains no patient information, and the numbers illustrate the software's behaviour rather than any clinical result. It was built deliberately so that every feature has something to act on — three cohorts, three procedures, both eyes of most patients, retreatment rows, mixed plus/minus cylinder notation, all four variable types, decimal acuities and some missing values.

PART 1 · WHAT EACH OPTION DOES, STATISTICALLY

1.1 The data model

The software expects **one spreadsheet row per eye, per surgical procedure**. Everything else follows from that convention.

Role you assign	What it is used for
Patient / cluster ID	Links the two eyes of one patient. This is the clustering unit for GEE and the denominator for patient-level statistics.
Eye / case ID	Row identity; used for traceability and for the Alpins per-eye calculations.
Group source column	The column whose values define the groups being compared.
Reference group	The comparator. Every effect estimate, matching pairing and adjusted model is expressed *relative to this group*.
Patient-level parameters	Variables that describe the patient, not the eye (age, sex, systemic disease).

Why patient-level parameters matter. If a patient contributes two eyes, their age appears twice. Summarizing it per eye inflates the sample and understates the standard error. When a variable is flagged as patient-level, the software collapses it to one observation per patient before computing its descriptive statistics — so demographic rows report *patients* while surgical rows report *eyes*. Such rows are tagged **[per patient]**, and as soon as any variable is patient-level the column headers print both denominators, e.g. `Control (n = 108 eyes; 73 patients)`.

Missing patient identifiers. A row whose patient ID is blank is given its own token, `(no ID) row <n>`, so that eye counts as its own patient everywhere — tables, GEE, figure *P* values and the bootstrap — and the number of such rows is written to the run log and the Manifest sheet. The tokens are given after the inclusion rules have run (n is the row's position in the original sheet), so a rule that excludes rows without a patient ID still works. When no patient-ID column is mapped at all, GEE and the adjusted models treat every row as its own cluster (eyes assumed independent) and say so in the sheet note, rather than refusing to run.

Reading the spreadsheet. The sheet is read by R's `readxl` package, and the interface types the columns it shows in exactly the same way (a column is numeric, text, date or logical according to the cells it holds), so the live counts and previews see the same values as the analysis. A cell formatted as a date inside a numeric column is a data-entry error whose serial number would otherwise pass numeric filters and distort means: it is set to missing, and the run log names the column, the number of such cells and an example, so the cell can be corrected in Excel.

Column names in the output. Internally each column is given a plain ASCII key, which the model formulas need, but everything you read uses the header **as typed in your file** — Hebrew, accents, spaces and symbols included: the *Parameter* labels of every table, the GEE and regression sheets, the within-group sections, the PSM covariate lists, the love-plot labels, figure text that names a column, the run-log lines, and the column headers of the filtered CSV and of the *Matched_Data* sheet. The Publication Table uses the same headers with each underscore printed as a space (`Age_Years` reads `Age Years`). A header that occurs twice is kept apart as the data preview shows it (`Age`, `Age.1`); in the CSV and *Matched_Data* the same is done for two headers that differ only in letter case, because Excel tables refuse such a pair.

Retreatments. A second procedure on the same eye is simply another row. A treatment-number column can therefore be used as a filter (`== 1`) to restrict the analysis to primary procedures, which is the usual choice.

1.2 Refraction standardization (plus → minus cylinder)

Exports frequently mix plus- and minus-cylinder notation. When enabled, each nominated sphere/cylinder/axis triple is transposed to minus-cylinder form by the standard rule:

new sphere = sphere + cylinder · new cylinder = −cylinder · new axis = axis ± 90°

Two points of practical importance:

1. It is applied to **every triple you nominate** — preoperative refraction, treatment

parameters and postoperative refraction — not to a single hard-coded pair. Sets are never guessed from column names. Once the preoperative and postoperative sphere, cylinder and axis roles are all mapped (*Column roles*, Special outputs tab), those two sets are pre-filled in the list; any other triple, such as the treatment parameters, you add yourself. A set you remove is not filled in again, and the sets a loaded preset carries are used as they are.

2. Only rows recorded in plus notation are altered; rows already in minus form are left

alone, including their axis (an axis recorded as 180 stays 180). A transposed row's new axis is written in the clinical range 1–180, so the horizontal meridian is 180, never 0. An axis outside 0–180 in an untouched row is a data-entry problem: it is reported in the run log and left as recorded (the figures reduce every axis modulo 180 themselves). All downstream calculations, including the Alpins vectors, then operate on one consistent notation.

1.3 Visual acuity: the notation in your file, and logMAR (optional)

Declare the notation. One setting, *VA notation in your file*, states how every acuity column is recorded — the four VA roles (preoperative and postoperative UDVA and CDVA) and every column listed for logMAR analysis. Four notations are read:

Notation	Example	How it is read
Decimal (default)	1.0, 0.5, 0,8	as written; a comma decimal is accepted; a value ≤ 0 is impossible

Notation	Example	How it is read
logMAR	0.00, 0.30, -0.10	decimal = 10^(−logMAR); values outside −0.5 to 3 are impossible
Snellen fraction	20/40, 6/12, 20/25-2, 6/9+1	20/x and 6/x alike; logMAR = $\log_{10}$(b/a) + 0.02 per letter missed ("−k"), − 0.02 per extra letter ("+k")
ETDRS letter score	85	logMAR = 1.7 − 0.02 × letters (0–100)

- **Off-chart entries** (CF, HM, LP, NLP, "counting fingers" and similar) become missing and

are counted in the run log. They are never given an invented acuity; the mEYEstro tab shows the recorded label (CF, HM, LP or NLP) in its Snellen column for those eyes.

- For every acuity column the run log states how many values were read, how many could not

be parsed (with examples) and how many were off-chart. **The run stops with a clear message** when the declared notation is contradicted: more than half of the non-blank cells unreadable, any decimal value ≤ 0, or a logMAR value outside its range. A decimal value above 4.0 is flagged as implausible. The check is made on the rows that are analysed; a contradiction found only in rows the inclusion rules leave out is reported in the log, and those cells stay missing.

- **Decimal acuity declared as ETDRS letters.** A decimal acuity (0.05–2.0) is also a valid

letter score, and read as letters it would mean about 20/1000 for every eye. A column declared as ETDRS letters therefore stops the run when, among its analysed non-blank numeric cells (a comma decimal such as `0,8` counts as 0.8), more than half are not whole numbers or the largest value is below 5: `VA column 'Postop_UDVA' is declared as ETDRS letters, but its values (e.g. 0.8, 1.0) look like decimal acuity. Choose the notation the file uses (Variables tab).`

- The interface also inspects the mapped columns and warns when they look like a different

notation from the one selected, including the same decimal-as-ETDRS condition, shown under *VA notation in your file*.

- In Excel a typed Snellen fraction such as 6/12 becomes a date; keep Snellen columns

formatted as text.

Filters see your own units. Exclusions and filter rules are applied to the values *as recorded in the file*, before any conversion, so a filter typed in your file's notation means what it says.

What the statistics see. After filtering, each acuity column is analysed as follows:

- **logMAR on** (the column is listed for conversion): logMAR = $-\log_{10}$(decimal acuity), so

1.0 decimal → 0.00 logMAR and lower values mean better vision. All summaries, tests, GEE and regression then run on the logMAR values.

- **File already in logMAR:** the recorded logMAR numbers are analysed as they are; they are

never converted a second time, and the logMAR switch changes nothing (the log says so).

- **Otherwise** the column is analysed as decimal acuity; Snellen and ETDRS entries are

replaced by their decimal equivalent.

The Overview sheet names the columns analysed in logMAR and in decimal, the Manifest records the *VA notation* and the *LogMAR conversion*, and the optional filtered CSV and the *Matched_Data* sheet mark the

converted columns with a `_logMAR` or `_decimal` suffix (left off only when the file already has a column of that exact name).

The figures and the export tabs always use decimal acuity derived from the declared notation, whatever the logMAR setting. The acuity panels are computed from decimal acuity, binned at the standard Snellen-equivalent cut-offs, so enabling logMAR for the tables does not change them. A decimal-to-Snellen lookup table is embedded, so the Snellen equivalents of the mEYEstro tab are produced without any external workbook.

1.4 Variable types → what is reported and which test is used

Every variable you place in Baseline, Preop or Outcome is given one of four types. The type determines both the presentation and the test.

Type	Reported as	Two groups	More than two groups
Continuous	mean ± SD	Welch *t*-test	Welch ANOVA
Ordinal	median [IQR]	Wilcoxon rank-sum	Kruskal–Wallis
Dichotomous	n (%)	χ^2, or Fisher exact	χ^2, or Fisher exact
Categorical	n (%) per level	χ^2, or Fisher exact	χ^2, or Fisher exact

Details that matter for reporting:

- **Welch forms are the default**, not the classical Student *t*-test or Fisher ANOVA.

Welch does not assume equal variances, which refractive groups routinely violate — they usually differ in both size and spread. This is the conservative, modern default. Welch ANOVA is undefined when a group has zero variance; three or more groups are then compared by Kruskal–Wallis, and the Test column says so.

- **Ordinal variables** use rank-based tests. Two groups use the Wilcoxon rank-sum

(Mann–Whitney) test as implemented in R's `wilcox.test` (exact distribution for small samples without ties, otherwise a continuity-corrected normal approximation); three or more groups use Kruskal–Wallis. Kruskal–Wallis on two groups equals only the uncorrected asymptotic rank-sum test, so it is not used there.

- **Categorical tests switch automatically.** Empty rows and columns of the contingency

table are dropped first. If **every expected count is at least 5**, χ^2 is applied **without continuity correction**; otherwise the software uses **Fisher's exact test**. For tables too large for the exact network algorithm it runs a Monte-Carlo Fisher test (20,000 replicates, with a locally fixed seed so identical runs give identical *p* values), labelled *Fisher's exact test (Monte Carlo, 20,000 replicates)*.

- **A test that cannot be performed is never reported as "no difference".** A *p* value is

computed only when every compared group has at least two usable values (continuous and ordinal variables) or at least one eye with data, with both outcomes present overall (proportions). Otherwise — a variable that is constant in the cohort ("variable is constant in this cohort"), or a group with a single value ("Diabetes has n = 1") — no test is run, and the row gives no *p* value and a *Not estimable* interpretation stating the reason; in the Publication Table such a cell reads **NE**, explained in a footnote.

- **The test actually used is printed beside every *p* value** in the output. A reviewer

can see which test produced each number without asking you.

- **Matched cohorts with non-unit matching weights** are summarised with the weights (see

1.8): weighted mean ± SD, weighted proportions, weighted median [IQR], and a *Weighted Wald test (GEE, robust SE)* in place of the tests above. For an ordinal variable the same model is fitted to the ranks and is labelled *Weighted Wald test on ranks (GEE, robust SE)*. That test is a GEE of the variable on the group, weighted by the matching weights, with an **independence** working correlation and robust (sandwich) standard errors clustered on the patient — the estimator MatchIt recommends after matching. Its estimate is therefore exactly the difference of the weighted means (for a proportion, the odds ratio of the weighted proportions) printed in the same row, so a per-eye variable that was matched exactly shows identical weighted values and $p = 1$. Some cells keep an honest limit: a categorical variable with more than two levels has no weighted test, nor has a proportion for which one group has no events (or only events), and each such cell states that reason; a constant variable says "variable is constant in this cohort", weighted or not. The weighted test also follows the minimum-data rule of every clustered model (1.9): below it the cell reads `Not estimable - too few eyes for this model (n = 5).`, and when the model's covariance is numerically singular (the rule of 1.12) it says that the weighted test could not be computed, rather than printing a meaningless *p* value.

- **Summary-table *p* values are not adjusted for multiple comparisons** (see Part 2).

1.5 Dichotomous "positive" level

For every dichotomous variable you choose which level is treated as the event (Sex = F, Complication = 1, and so on). This affects the reported n (%) and the direction of the odds ratio in any model. Sensible defaults are guessed from the data and can be overridden. The chosen level is matched to the data regardless of case, surrounding spaces and the usual spellings of a logical value (TRUE / True / 1 / 1.0). A level that cannot be found is reported in the run log, and the same documented default is then used in the tables and in the models alike.

1.6 Filters — inclusion rules

Rules are combined with AND. Sixteen operators are available:

Operator	Meaning	Typical use
in, notin	value is / is not in a chosen set	treatment type, monovision flag
==, !=	equal / not equal	treatment number = 1
<, <=, >, >=	numeric comparison	age ≥ 18, follow-up ≥ 30 days
between	inclusive numeric range	age 20–40
abs_le, abs_ge	absolute value ≤ / ≥	**cylinder magnitude ≤ 5.00 D regardless of sign**
abs_between	absolute value in range	cylinder 1.00–3.00 D either sign
contains, not_contains	text match	free-text surgery description
isna, notna	missing / not missing	require a recorded postoperative refraction

The absolute-value operators exist because cylinder and axis data are sign-sensitive: `abs_le 5` keeps −5.00 D and +5.00 D alike, which is almost always what is meant.

Rules act on the values **as recorded in the file** (after the plus → minus transposition of 1.2, before any logMAR conversion), so an acuity rule is typed in your file's own notation.

Live counts. As each rule is added the number of eyes remaining is shown **per group**, so the effect of every criterion on every arm is visible before anything is run. This is a live equivalent of the participant-flow diagram, and it is the fastest way to discover that an inclusion rule has silently removed one arm.

1.7 Subsets

Nominating a subset column (typically treatment type) makes the software run a **complete, independent analysis for each subset** — a full PRK analysis and a full LASIK analysis from one configuration — with an optional pooled run. Each subset produces its own workbook and its own figure directory. Subsets are *not* a grouping variable: they partition the data, and the group comparison is repeated within each partition.

1.8 Matching — every parameter

Matching is performed by `MatchIt` against the reference group. All parameters are exposed:

Parameter	Options	What it does
Enable matching	on / off	If off, the whole matched branch of the output is skipped.
Matching covariates	any baseline variables	The variables matched on (with a propensity-score distance, the variables of the propensity model). A row must have **all** of them present to be eligible.
Method	nearest (default), exact, cem, subclass	Nearest = greedy nearest-neighbour matching on the distance. Exact = keeps the eyes whose covariate values exactly equal those of an eye in the other group, so it needs discrete covariates (sex, eye, a category); with a continuous covariate almost nothing is matched. CEM = coarsened exact matching: each covariate is binned and eyes sharing a bin with the other group are kept. Subclass = strata of the propensity score, with the eyes weighted within each stratum. These four are the methods the bundled R library runs; exact, CEM and subclassification produce weights that are generally not 1.
Distance	logit, glm, mahalanobis, robust_mahalanobis	How closeness is measured, for nearest and subclass only (exact and CEM use none). Logit/glm = propensity score. Mahalanobis = covariate-space distance, ignoring the propensity score; subclassification needs a propensity score, so it cannot use the Mahalanobis distances.
Ratio	integer (default 1)	nearest only. Controls matched per treated eye. Higher ratios gain precision but admit poorer matches.
Replacement	on / off (default off)	nearest only. Whether a control eye may serve more than once. With replacement, matches are closer but the effective sample is smaller and weights become unequal.
Caliper	numeric (default 0.2; blank = none)	nearest with a propensity-score distance only. Maximum acceptable distance. Prevents bad matches at the cost of discarding treated eyes.
Standardized caliper	on / off	When on, the caliper is expressed in **standard deviations of the propensity score** (0.2 is a common choice) rather than raw units. MatchIt's logit distance is the estimated probability from a logistic model, so the standard deviation is that of the probability, not of its logit.

Parameter	Options	What it does
Match order	largest, smallest, random, data, closest, farthest	nearest only. The order in which treated units are matched; largest and closest are the usual choices. largest and smallest rank eyes by their propensity score, so they are not offered with a Mahalanobis distance.

The interface greys out every setting the chosen method does not use, and the Manifest and the PSM sheet list only the settings that method uses.

What was actually run is what is reported. The Manifest's *Matching* row names the method that ran, followed by "(MatchIt)": *nearest-neighbour on the propensity score*, *nearest-neighbour on Mahalanobis distance*, *exact*, *coarsened exact (cem)*, *coarsened exact requested; run as exact (all covariates categorical)* — coarsened exact matching on categorical covariates only is exact matching — or *subclassification on the propensity score*. The covariate lists of the PSM sheet, the regression title and the Manifest name only the covariates that were found in the file and used; a name that does not match a column is listed separately as "not found in the file (ignored): 'X'", and the run log carries one WARNING line naming them. When matching is on but **none** of its covariates is found, matching is skipped with a WARNING line in the log, no matched cohort is built (no matched sheets, love plots, matched figures or matched-only workbook), and the Manifest reads *Matching | skipped - no matching covariate found in the file*.

What the software reports back. Balance is assessed with standardized mean differences (via `cobalt`) before and after matching, against the conventional 0.1 threshold, together with a love plot and, when a propensity score is used, the propensity-score distribution of the matched rows. Binary covariates are standardized too, so every covariate is judged on the same SMD scale: the difference in means (for a binary covariate, in proportions) is divided by the standard deviation of the **treated (non-reference) group**, cobalt's convention for the ATT that the matching estimates.

Matching weights. MatchIt gives every matched row a weight. The weights are all 1 only for 1:1 nearest-neighbour matching without replacement; they differ from 1 whenever a control is re-used (replacement), whenever k:1 matching (k > 1) leaves some treated eyes with fewer than k controls inside the caliper, and generally with exact, coarsened exact and subclassification matching. When any weight differs from 1, the matched cohort is summarised **with** the weights: the matched summary tables show weighted cells and weighted Wald tests, the group headers add the effective sample size ESS = $(\Sigma w)^2 / \Sigma w^2$ (for example `Control (n = 157, ESS = 71)`), a note row above each matched table says so, the Manifest records "Matching weights: non-unit (min–max)", and the *P* values on the matched figures use the weights (the bars and means show the matched eyes unweighted, as the figure states). Every model fitted with such weights — the weighted test of the matched tables, the matched GEE and regression when *Use matching weights* is on (1.9), and the weighted figure *P* values — is a weighted GEE with an **independence** working correlation and robust standard errors clustered on the patient, so the three agree with one another and with the weighted means and proportions in the tables; the Manifest adds "(working correlation: independence (weighted))" to its matching-weights entry when the matched models use them. The within-group sections and the mEYEstro / AstigMATIC tabs list the matched rows unweighted. The matched analysis estimates the average treatment effect in the non-reference (treated) group (ATT).

Per-patient rows of a weighted matched table. Matching is done per eye. A row summarised per patient gives each patient the mean weight of that patient's matched eyes, so a covariate matched exactly per eye can still show slightly different weighted percentages per patient, while the eye-level GEE column of the same row shows no difference.

The live matched-count preview (nearest-neighbour matching; the other methods are not previewed, though the run uses them) runs the *real* matching algorithm with your exact settings on a slim copy of the data and reports, per pairing: group rows → rows with complete covariates → rows actually matched, plus the reference pool used and the patient IDs of unmatched eyes. This exposes the commonest and otherwise invisible failure: an eye discarded because no control lies within the caliper, even when thousands of controls exist. If matching is losing eyes unexpectedly, the three usual causes are a sparse covariate (reducing eligible rows), too tight a caliper, or too high a ratio.

> **Caution.** Matching addresses *measured* confounding only. It cannot repair unmeasured confounding, and it never converts an observational comparison into a randomized one.

1.9 GEE — every parameter

Generalized estimating equations account for the correlation between two eyes of one patient. Ordinary tests treat eyes as independent, which understates standard errors and inflates the type I error rate whenever fellow eyes are present.

Parameter	Options	What it does
Stage	none, unmatched, matched, both	Which cohort(s) the models are fitted on.
Main GEE	on / off	The unadjusted clustered comparison: outcome ~ group.
Adjusted GEE	on / off	Adds covariates: outcome ~ group + covariates.
Adjustment covariates	any variables	The covariates for the adjusted model.
Use matching weights	on / off	Carries the matching weights into the model. Should be **on** whenever the weights are not all 1 (see 1.8) — with replacement, with k:1 matching under a caliper, and with exact, coarsened exact or subclassification matching — so the model describes the same matched design as the weighted matched tables. Weights that are not all 1 make the model a weighted fit with an independence working correlation (below).
Add GEE columns to tables	on / off	Merges the GEE effect and *p* value into the summary tables as extra columns — the unmatched tables when the stage includes the unmatched cohort, the matched tables when it includes the matched cohort.
Source for tables	main, adjusted	Which of the two models supplies those columns.

Technical specification. Models are fitted with `geepack::geeglm` using **robust (sandwich) standard errors**, with the patient as the cluster. Unweighted models (and models whose matching weights are all 1) use an **exchangeable** working correlation structure. A model fitted with matching weights that are not all 1 uses an **independence** working correlation instead: the weighted fit with cluster-robust standard errors that MatchIt recommends after matching, whose estimate reproduces the weighted means and proportions of the matched tables. Its Model column says so (`GEE Gaussian, working correlation: independence (weighted)`; in the full regression table `Linear GEE (beta), working correlation: independence (weighted)`), and the note row of the sheet reads "Models weighted by the matching weights - working correlation: independence (weighted), robust (sandwich) standard errors." Continuous outcomes use the

Gaussian family and yield an adjusted mean difference (β); dichotomous outcomes use the binomial family and yield an odds ratio. Confidence intervals are Wald intervals (estimate ± 1.96 × robust SE, exponentiated for odds ratios), without small-sample correction.

Minimum data — one rule for every clustered model. The GEE sheets, the adjusted regression, the weighted test of the matched tables (1.4) and the clustered or weighted GEE of the figures (1.12) all apply the same rule: a model is fitted only when it has at least **6 usable eyes**, at least **3 patients** (clusters) overall, and more usable eyes than its coefficients (intercept included) + 2. Otherwise nothing is fitted and every comparison of that outcome says why — `Skipped: not estimable - too few eyes for this model (n = 5)` in the GEE and regression sheets, `Not estimable - too few eyes for this model (n = 5).` in a weighted table cell, `P not computed: too few eyes for this model (n = 5)` on a figure. On top of that rule every compared group needs at least two usable eyes and two patients (below).

A group with no data for an outcome. Each outcome is fitted on its complete cases, and groups left without a usable row are dropped from that outcome's model; the reference group is never replaced silently. If the reference group has no data for the outcome, every comparison reads `Skipped: not estimable - Control has no data for this outcome`; a non-reference group without data gets its own row, `Not estimable - Autoimmune has no data for this outcome`, and the other comparisons are still estimated. A dichotomous outcome with a single recorded value in the analysed rows reads `Skipped: not estimable - outcome is constant in this cohort (only '0' recorded)`. Ordinal and categorical outcomes are not modelled by GEE by design (`Skipped: ordinal outcome not modeled in GEE`); they are not counted as failures in the sheet's note row.

Few patients. Because robust standard errors can be too small when a group has few patients, a comparison in which either group has a single patient is not estimated (`Not estimable - <group> has 1 patient`, in the GEE and regression sheets and the figures alike), and one in which a group has fewer than 10 patients is still reported but carries a caution in the sheets' Note column (`Caution: Autoimmune has 4 patients - robust standard errors can be too small with few patients`), under the figure's *P* values, and once per run as a WARNING line in the run log, which the interface lists under "Check before using the results". That WARNING line names only the outputs that actually carry a caution. With no patient-ID column mapped, every row is its own cluster and the sheet note says so; the clusters are then eyes, and these texts count eyes (`Autoimmune has 8 eyes`). On the matched cohort the cluster is the patient (fellow eyes); the matched sets themselves are not modelled as clusters.

Why a table and a figure can show different *p* values. The *p* value in a summary table comes from the eye-level test of 1.4. With *Cluster p-values by patient* on (the default) the figures print the clustered GEE *p* value instead, which is the one the GEE columns of the tables carry. Switch the GEE columns on when you want the table to show the same clustered *p* value as the figure. This match is exact for two groups when a patient ID is mapped and clustering is on; on a matched cohort whose weights are not all 1 it also needs *Use matching weights* on, because the figure always uses those weights. With three or more groups, the figure's headline *P* is an omnibus GEE Wald test across all groups, while the GEE columns give one *p* value per comparison with the reference.

When you do not need it. If your dataset genuinely contains one eye per patient, the clustered model adds nothing and the ordinary tests are appropriate. GEE is offered as an option for exactly this reason.

1.10 Regression (covariate-adjusted models) — every parameter

Parameter	Options	What it does
Enable regression	on / off	Produces the adjusted-effect columns and the model appendix.

Estimation is by GEE, so clustering is respected in the adjusted models too, with the working-correlation, minimum-data, no-data and few-patients rules of 1.9 (a matched model weighted by matching weights that are not all 1 uses the independence working correlation). The adjusted effect (β or OR) with its 95% Wald CI and *p* value is **merged into the main outcome table** as two extra columns beside the crude comparison, with the full models written to an appendix sheet.

Choose covariates by imbalance, not by their effect on the result. Adding a covariate because it is still imbalanced is correct; adding it because it changes your *p* value is fishing. This is why the suggestion mechanism ranks by standardized mean difference.

Two safeguards you should know about:

- **Constant covariates are dropped automatically.** A covariate that takes a single value

within the filtered cohort — a frequent consequence of an inclusion rule such as "primary treatments only" — would otherwise cause the model to fail. It is removed with an explicit note.

- **Small-sample and zero-event situations are flagged.** With fewer than about ten events

per variable the estimates are unstable, and a dichotomous outcome with **zero events (or only events) in one group** produces complete separation, where the odds ratio does not exist. The software detects this before fitting: that group's comparison is reported as *not estimable*, with the reason, and the group is left out of that outcome's model, so the other comparisons of the same outcome are still estimated (with three or more groups). A continuous outcome with a single eye in a group is handled the same way. An odds ratio for the group, or for a categorical covariate, outside 1/10,000 to 10,000 (quasi-separation that the check cannot see) is not reported either; a continuous covariate's odds ratio is per unit of its own scale and is never suppressed for its size. Report the counts and the exact-test *p* value for such rows.

1.11 The recommended strategy: match, then adjust

Because both are available, **regression adjustment after matching** (described as doubly robust by Ho et al.) can be run in one pass: match on the measured confounders, then adjust in the outcome model for whatever remains imbalanced after matching. The adjustment removes residual imbalance that the matching left behind, and the matching makes the result less dependent on the form of the outcome model. It is not a doubly robust estimator in the AIPW / TMLE sense.

The workflow is: run matching → read the post-matching standardized mean differences → put any covariate still above 0.1 into the regression covariate box → report that single adjusted estimate. The matched analysis estimates the average treatment effect in the non-reference (treated) group (ATT); its standard errors are cluster-robust at the patient level, accounting for fellow eyes, but do not treat the matched sets as clusters.

1.12 Figures

The **StandardEyes figures** are drawn natively by the R engine from your data - no macro workbook, no external plotting program. Every cohort appears on one chart in its own colour, however many cohorts there are, so nothing is split into pairs or drawn per group alone. The metrics and bins are those of the published standardized graphs, so a reader familiar with the consensus layout can read them without relearning anything. With two cohorts the printed statistic is the familiar pairwise test, and the header names it — for example `GEE (clustered): P = 0.0211 (d = -0.08)` or `Welch t: P = 0.4546 (d = 0.24)`, where d is

Cohen's d on the pooled SD, computed per eye and unweighted — so it is not clustered by patient or matching-weighted even when the *P* value beside it is. With more cohorts it becomes an omnibus test plus each cohort against the reference. If a clustered model cannot be fitted for a panel — it does not converge, or the covariance of its group terms is numerically singular (reciprocal condition number below 10^{-8}, or a robust variance below 10^{-8} times the model-based one), where the robust standard error would be meaninglessly small — the classical test runs instead and is labelled *(GEE not fitted)*; the run log counts such panels, double-angle centroid tests included. A weighted GEE has no unweighted fallback: the panel says `P not computed: the weighted test could not be computed`, the matched table's wording. A clustered or weighted centroid test on the double-angle plots also needs at least three patients in every compared cohort (with two, the cohort's robust covariance cannot be estimated in two dimensions): the plot says `P not computed: Diabetes has 2 patients (a centroid test needs at least 3 patients per group)`. When a cohort cannot support a *P* value (fewer than two usable values in a cohort, or a constant measure — the rule of 1.4), no test or model is run and the panel says why, naming the cohort, e.g. `P not computed: Diabetes has n = 1`. A clustered or weighted GEE also needs the minimum data of every clustered model (6 usable eyes and 3 patients overall, 1.9) — below it the panel says `P not computed: too few eyes for this model (n = 5)`, the tables' own reason, and no classical test is put in its place — and two patients in every compared cohort, and a cohort with fewer than 10 gets a caution under the *P* values; these rules are set out under *Correlated eyes* below. A weighted GEE uses the independence working correlation of 1.8, so for the same measure a matched figure's weighted *P* equals the weighted test of the matched table. A stat line, a refusal or a caution that is too long for its panel is wrapped onto a second line, never cut off, in the composite and in the panel's own file alike.

Multiplicity on the figures. Every "vs reference" line is Holm-adjusted across that panel's comparisons and tagged **[Holm]** — but only when it holds two or more comparisons. With exactly two cohorts there is one test and no "vs reference" line at all. The omnibus *P* values and all table, GEE and regression *p* values are not adjusted.

Reference acuity of the efficacy panels. The second series of the cumulative visual acuity panel (panel A), the efficacy bars and the Efficacy Index all use the same comparator: **preoperative CDVA** for the corneal panel sets (Full and Refractive, following the corneal refractive-surgery standard) and **postoperative CDVA** for the Cataract set (the IOL-surgery convention). The legend, the axis title and the header name the comparator, e.g. "Postop UDVA vs Preop CDVA (Snellen lines)".

Intended target. An optional *Target SEQ* role holds the intended postoperative spherical equivalent of each eye (monovision, a myopic cataract target). When it is mapped, SEQ accuracy is postoperative SEQ minus target, the attempted correction is preoperative SEQ minus target (achieved = preoperative minus postoperative SEQ), and the defocus equivalent is |postoperative SEQ − target| + |postoperative cylinder| / 2; when the role is not mapped, or an eye's target cell is blank, that eye is taken as aimed at plano. The mEYEstro tab fills its *Intended* columns from the same target (and from the astigmatism target for cylinder and axis), reading a blank target cell as plano in the same way and counting such eyes in the run log.

The Alpins vectors - target induced, surgically induced, difference vector and correction index - are displayed both on the JCRS-standard **double-angle plots** and, for familiarity, as semicircular single-angle polar diagrams with the groups superimposed; vector means are computed in double-angle space and back-transformed for display, the mathematically correct treatment of axis data.

Panel sets

The **Panel set** selector follows the published standard set sizes:

Set	What it contains

A separate **Include astigmatism panels** switch drops the astigmatism panels from any set, so a purely refractive figure set needs no hand editing. Before you run, the app prints exactly which panels the current configuration will produce.

Stability over time

Map each follow-up visit's columns and the app plots mean +/- SD spherical equivalent across visits. At least one mapped follow-up visit is required (two or more for an informative curve); the preoperative point is added by default. Per visit you either pick a ready SEQ column or give sphere and cylinder and let the app compute it. Nothing is guessed from column names.

The number of eyes at each time point is printed beneath the axis, because attrition - not true regression - is the usual reason a stability curve bends. Nothing is carried forward, each visit is plotted at the **median** day actually observed, and points with fewer than three eyes are dropped. When no visit has at least three eyes in a group, no stability panel is drawn and the run log says so ("no visit with at least 3 eyes in a group"). The "% changed by more than x D" statistic is **paired**: only eyes measured at both nominated visits contribute.

Stability joins the composite as an **extra** panel of the Full set, so nothing is displaced by it; the Refractive and Cataract sets have no stability slot.

The JCRS 2025 astigmatism standard

The astigmatism figures follow the JCRS astigmatism working group (Koch et al., *J Cataract Refract Surg* 2025;51:447-455). Astigmatic vectors are shown on **double-angle plots**, which that paper establishes as the journal's reporting standard. Each cohort reports its centroid, **total SD** (sqrt of the summed component variances, invariant to rotation of the frame), the mean, median and RMS vector magnitude, and the proportion of eyes within 0.25 / 0.50 / 0.75 / 1.00 D.

The 95% boundary is drawn by default as a **nonparametric convex polygon** found by hull peeling, because that paper notes astigmatism data are usually not Gaussian. Whole convex hulls are peeled while a full layer still leaves at least 95% of the eyes; the last layer is then completed vertex by vertex (most extreme first, by Mahalanobis distance) so that exactly $\lfloor 0.95\ n \rfloor$ eyes are enclosed, and the text above the plot states the count per cohort (e.g. "107/113"). A bivariate-normal ellipse stays available for data that are Gaussian.

The dashed **95% confidence region of the centroid** uses the exact Hotelling T^2 radius, $r^2 = 2(n-1)/(n-2) \cdot F(0.95;\ 2,\ n-2)$, rather than its large-sample chi-square limit, which under-covers in small groups. With patient clustering on, its covariance is the patient-clustered sandwich estimate and the F degrees of freedom use the number of patients, so the fellow-eye correlation widens it. The text above the plot says which form was drawn.

Cohort centroids are compared **bivariately**, not by two separate tests on the x and y components. Three tests are offered:

Test	Level held	Notes
Robust trimmed + bootstrap	yes	the default; robust and cluster-aware, but not the exact JCRS Part II variant
Clustered GEE Wald	yes	more power
Hotelling T2 / MANOVA	no, if eyes are correlated	eyes treated as independent

The robust test is a Hotelling-type statistic: the sum over cohorts of $n_i \cdot d_i' \, S^{-1} \, d_i$, where d_i is the difference between a cohort's 20%-trimmed mean vector and the unweighted mean of the cohorts' trimmed mean

vectors, and S is the pooled 20%-winsorized covariance. Its null distribution comes from B = 999 bootstrap resamples — of patients when patient clustering is on, otherwise of eyes — after each cohort is recentred on its trimmed mean, with a fixed seed. It differs from the JCRS Part II procedure in using a between-group quadratic form weighted by the group sizes, rather than T^2 with trimmed effective sample sizes. The robust test has no weighted form, so on a matched cohort with non-unit weights the figure states that it does not use the weights. The text above the plot prints the method, e.g. "[robust 20%-trimmed T^2, B = 999, cluster bootstrap]", and the Manifest records the trimming fraction, B and the seed.

With three or more cohorts, the comparisons against the reference cohort are Holm-adjusted and tagged [Holm]. **Every figure prints which method produced its numbers**, so a reader can tell at a glance.

Correlated eyes

Two eyes of one patient are not independent. Tests printed on the figures are clustered on the patient using GEE with robust standard errors, and the astigmatism bootstrap resamples patients rather than eyes. In a simulation with no true group difference (two groups, both eyes of every patient, 30 to 120 patients per group, 5,000 replicates per setting) the eye-level Welch test rejected at 13.7% to 14.9% against a nominal 5% when the correlation between fellow eyes was 0.8, and at 10.5% to 11.1% when it was 0.5. The clustered test rejected at 4.8% to 6.3% throughout. The script and its results are in the `validation` folder of the source code repository (`simulate_fellow_eyes.R`). Every figure prints which inference produced its P values, so the clustering is visible on the face of the output. If a clustered model cannot be fitted (for example it does not converge, or its covariance is numerically singular), the classical test runs instead and is labelled *(GEE not fitted)*.

The robust standard error estimates each cohort's variance from its patients, so the number of patients per cohort - not the number of eyes - limits it:

- **A cohort whose eyes all belong to one patient** (or with fewer than two usable values,

the rule of 1.4) gets no clustered *P* value at all: nothing is fitted and the panel says why, e.g. `P not computed: Diabetes has 1 patient`. The comparisons that can be computed are still shown. The double-angle centroid tests (clustered and weighted GEE, and the cluster bootstrap) follow the same rule.

- **Too little data for any clustered model** — fewer than 6 usable eyes or 3 patients

overall (the rule of 1.9): no GEE is fitted and no classical test is put in its place; the panel says `P not computed: too few eyes for this model (n = 5)`, and a double-angle plot whose centroid test is the clustered or weighted GEE says `Centroid comparison: P not computed: too few eyes for this model (n = 5)`. The centroid model has two parameters per cohort (one for each component), so it needs more eyes than a single-outcome model: two cohorts need at least 7.

- **A cohort with 2 to 9 patients** still gets its GEE *P* value (the clustered or weighted

centroid test of the double-angle plots starts at 3 patients), but robust standard errors can be too small with so few clusters, so the panel prints a caution under its *P* values, e.g. `Caution: Autoimmune has 4 patients - robust SE can be too small`. There is no exception: the two-cohort Correction Index and Angle of Error panels, whose *P* is printed in the rotated layout, carry the caution under their N-boxes, wrapped to the panel. The robust centroid test with the patient bootstrap has no standard error, but a *P* value from so few patients is just as unreliable, so it carries its own caution: `Caution: Autoimmune has 4 patients - a P value from so few patients can be too small`. The run log names those cohorts once per run in a WARNING line, which the interface lists under "Check before using the results", and which

names only the outputs that carry a caution. The GEE and regression sheets carry the same caution in their Note column.

- **No patient-ID column mapped** (or *Cluster p-values by patient* switched off for the

figures). The figures then treat eyes as independent and print the classical tests, with no caution; only the weighted GEE of a matched cohort is still a robust-SE model, with each eye its own cluster. The refusal and the caution then count eyes, not patients (`Caution: Autoimmune has 8 eyes - robust SE can be too small`); without a patient ID the GEE and regression sheets, which then also make each eye its own cluster, word them the same way, and the run-log WARNING names only the outputs that carry such a caution. With a patient ID mapped they say "patient(s)".

On a matched cohort whose matching weights are not all 1, the figure *P* values are weighted GEE Wald tests with an independence working correlation and robust standard errors clustered on the patient — the model of the matched tables' weighted test (1.4) — and the composite carries the line "Matched cohort: P values use the matching weights; bars and means show the matched eyes unweighted."

Every panel is written both as its own file and inside the composite. The interface always writes LZW-compressed TIFF, at 300 dpi unless you change the resolution, and PNG, JPEG and PDF (Cairo, fonts embedded) can be added. A settings file run from the command line writes the formats it lists. Group colours and display labels are configurable. The PDF device of the bundled R build cannot shape right-to-left text (Hebrew, Arabic): when PDF is chosen and a label contains such text, the run log says so once and recommends PNG or TIFF for that run. Whatever formats are chosen, a low-resolution PNG copy of every composite is also written to a `_preview` subfolder, and these previews are shown in a figure gallery on the Run & Results tab, so the figures can be checked without leaving the app.

1.13 Optional analysis sheets

Option	What it produces
Publication-style table	One manuscript-ready sheet with Baseline / Preop / Outcome stacked under section headers, group columns and *p* values.
mEYEstro tabs	Worksheets formatted for direct entry into mEYEstro, for users who wish to verify the figures in that tool. Refraction is written in minus-cylinder form; acuity is decimal with Snellen equivalents (an eye recorded as CF, HM, LP or NLP shows that label); the *Intended* columns come from the mapped targets (1.12).
Alpins / AstigMATIC tabs	One sheet per group whose columns A–E load directly into AstigMATIC: eye (OD/OS), preoperative cylinder, preoperative axis, postoperative cylinder, postoperative axis. Eye codes R / RE / Right / 1 and L / LE / Left / 2 become OD / OS; an unrecognised code is left blank and counted in the run log.

1.14 Run options and reproducibility

- **Preview only** — results are held in the app for inspection and nothing is written to

your output folder until you choose to save. Meanwhile a temporary copy of the preview (workbooks, figures and, when requested, the filtered CSV) is kept inside the application folder (`webapp\.runs`); it is deleted when the next run starts or when the app is closed, and a copy left behind by a crash is deleted the next time the app starts.

- **Export filtered CSV** — writes the rows left after the exclusions and filter rules

(before the split into groups and subsets), which is worth keeping with the manuscript.

- **Output folder / file prefix** — where results are saved and how they are named.
- **Study presets** — every setting of a study is saved under one name inside the app, and

Download config JSON writes the same settings to one portable file. Loading that file (**Load config JSON**) restores the analysis exactly, and sending it to a colleague reproduces it on their machine. This is the practical answer to "how did we analyse this a year ago?"

Reproducing the example in this document. The two studies shown here are among the seven worked examples supplied as ready-to-run settings files beside the example dataset (and as the matching *Example - …* presets inside the app):

File	Study
Example Data\Settings_02_Two_cohorts_full.json	Two cohorts, every statistical step (steps 1–9, Part 4 and Part 5)
Example Data\Settings_03_Three_cohorts.json	Three cohorts (steps 10–11 and 5.3; the *Example - 07_Full_figure_suite* preset of the captures runs the same analysis)

They can be opened in the app with **Load config JSON**, or handed straight to the statistical engine without the interface at all, from the application folder:

```
Rscript --vanilla engine\run_pipeline.R "Example Data\Settings_02_Two_cohorts_full.json"
```

Both use paths *relative to the application folder*, so they keep working wherever the folder is copied, and both reproduce the analyses shown in this document — including the balance figures in Part 4 and the outcome table in Part 5. Every number is produced by the R engine; that the same configuration file drives the interface and the command line is the practical guarantee that the interface is not doing any of the arithmetic.

PART 2 · TWO GROUPS versus MORE THAN TWO

Everything scales to any number of groups, but several behaviours change. This is the complete list.

Aspect	Exactly two groups	Three or more groups
Continuous variables	Welch *t*-test	Welch ANOVA (a single omnibus *p* across all groups)
Ordinal variables	Wilcoxon rank-sum (wilcox.test)	Kruskal–Wallis (omnibus)
Dichotomous / categorical	χ^2 or Fisher exact on the contingency table	χ^2 or Fisher exact on the full contingency table (omnibus)
What the *p* value means	The two groups differ	*At least one* group differs — it does not say which
Matching	One pairing: group vs reference	Each non-reference group is matched to the reference **separately** (reference-group pairwise), producing one pairing per group
GEE / regression	One effect estimate	**One row per group versus the reference**, each with its own effect, CI and *p*
Adjusted columns in tables	One line per outcome	One line **per comparison** inside the cell
Figures	Pairwise statistic, with the test named, printed above each panel	Omnibus statistic plus each group against the reference, Holm-adjusted and tagged [Holm]; every group stays on the same chart

The practical consequence. With more than two groups the omnibus *p* value tells you only that the groups are not all alike. The information about *which* groups differ comes from the GEE/regression rows, each of

which is an explicit comparison against the reference group. Choose the reference group deliberately — it is usually the control or the standard technique — because every estimate in the output is expressed relative to it.

Multiplicity — one policy. On the figures, each group-versus-reference *P* value is Holm-adjusted across the k − 1 comparisons of its panel and marked **[Holm]** (only when there are two or more such comparisons; with exactly two groups there is a single test and no "vs reference" line). This applies equally to the bar panels and to the double-angle astigmatism plots. The omnibus *P* values, and every table, GEE and regression *p* value, are **unadjusted**. With several groups and several outcomes the number of tests grows quickly, and the interpretation of borderline *p* values should account for that. Pre-specifying a primary outcome is the cleanest solution.

PART 3 · WALKTHROUGH

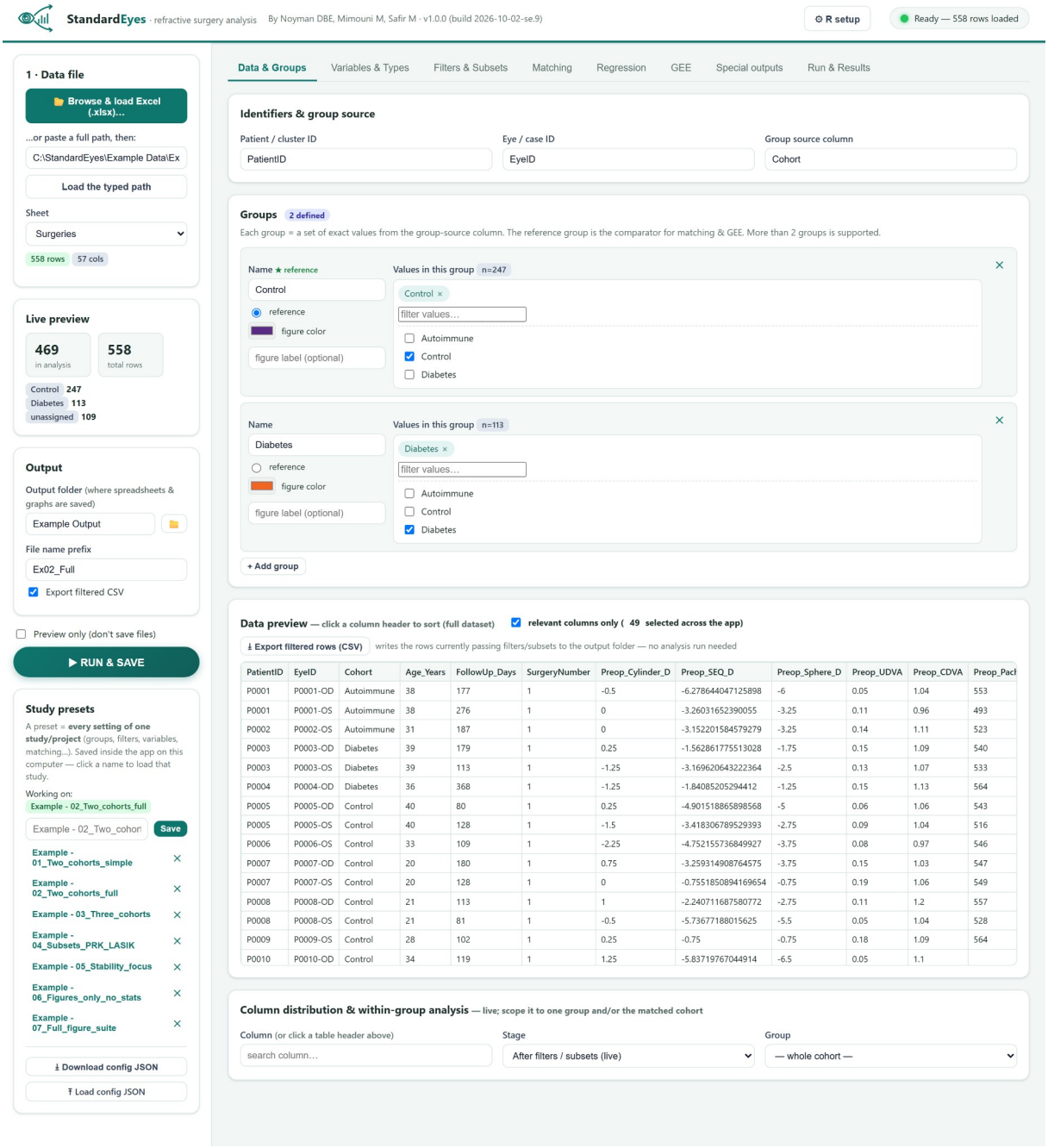
One continuous session on the example dataset: 558 rows, 57 columns, 320 patients. Steps 1–9 configure and run a **two-cohort** comparison (Control vs Diabetes) on all three procedures together, with the *Example - 02_Two_cohorts_full* preset. Steps 10–11 show what changes with **three cohorts**, and step 12 shows the R setup screen.

Step 1 · Data & Groups

Load the spreadsheet — browse for it, or paste a full path. The identifiers are then nominated: **PatientID** links the two eyes of one patient and becomes the clustering unit for GEE, **EyeID** identifies the row, and **Cohort** is the column whose values define the groups.

Each group is built from exact values of that column, given a colour and an optional figure label, and one group is marked as the **reference** — every effect estimate, matching pairing and adjusted model is expressed relative to it. Values left unassigned are excluded and counted separately, which is visible immediately in the sidebar.

The data preview below shows the file as loaded, and any column header can be clicked to inspect its distribution — scoped to one group and to the filtered or matched cohort.



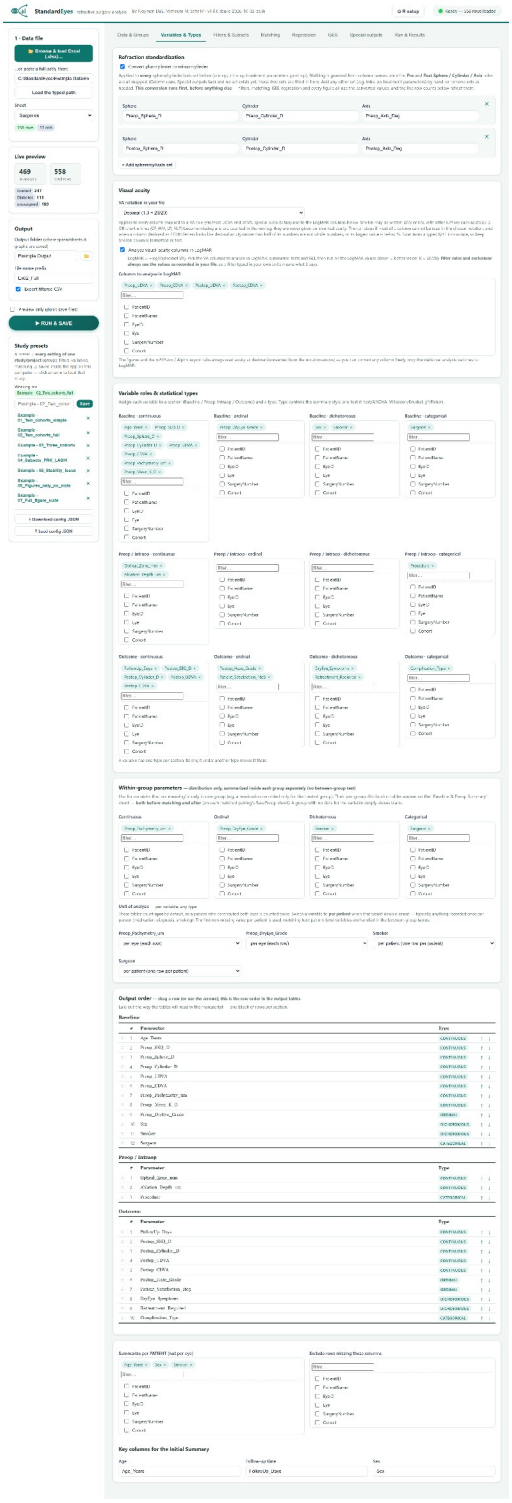
Step 2 · Variables & Types

Every variable is placed in **Baseline**, **Preop / Intraop** or **Outcome** and given one of four types. This single step determines the whole shape of the output: how the variable is summarized, which test is applied, and whether it is offered later as a covariate.

The example is typed to exercise all four: continuous (age, refraction, pachymetry, acuities), ordinal (dry-eye grade, haze grade, satisfaction 1–5), dichotomous (sex, smoker, dry-eye symptoms, retreatment) and categorical (surgeon, procedure, complication type). Age, sex and smoking are additionally flagged as **patient-level**, so they are counted once per patient rather than once per eye.

The same tab holds the two data-preparation steps that run before anything else: refraction standardization (plus → minus, applied to the preoperative and the postoperative sphere/cylinder/axis triple, 1.2) and the

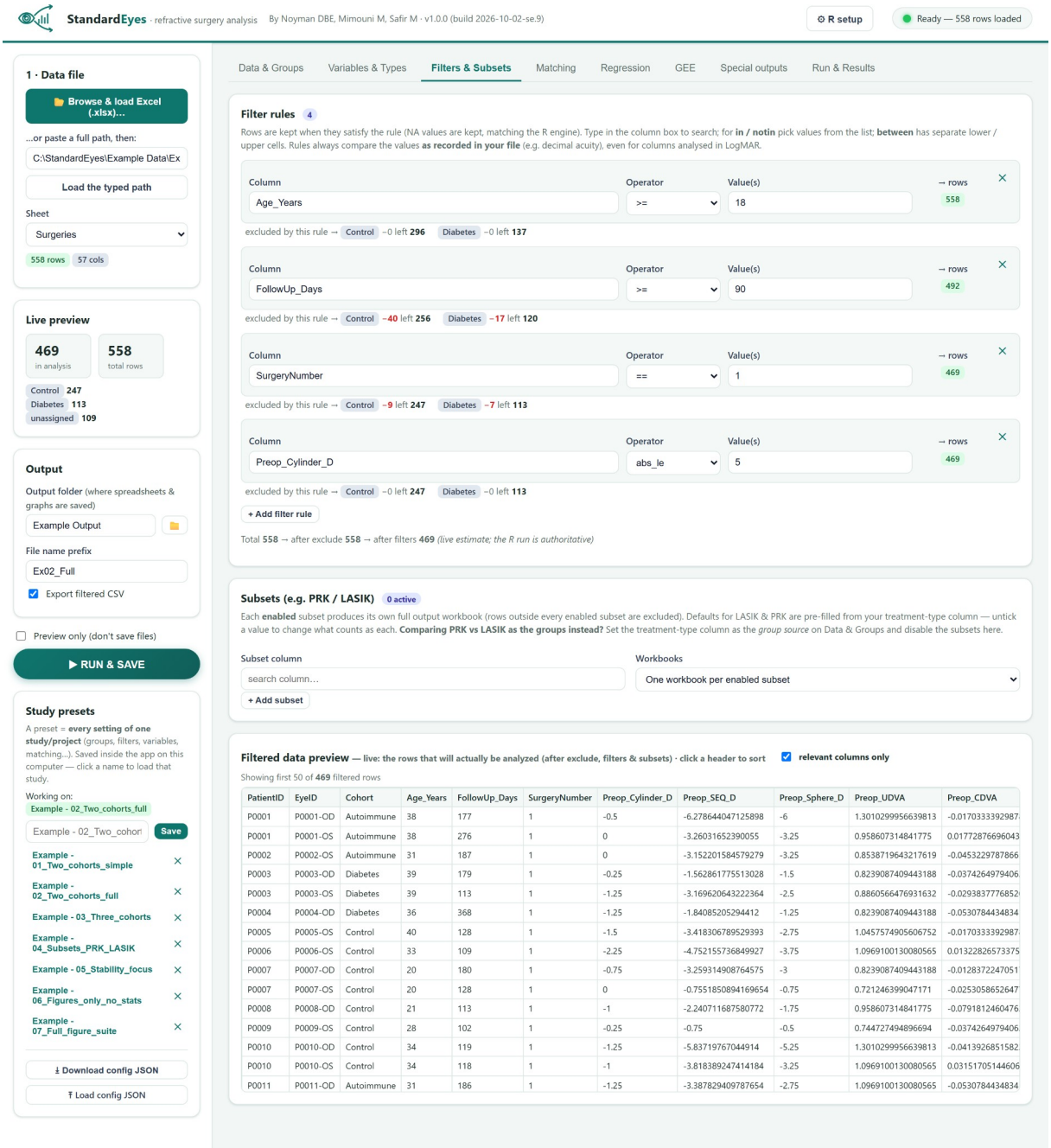
Visual acuity card — the notation the file records acuity in (decimal here) and the optional logMAR conversion (1.3).



Step 3 · Filters & Subsets

Inclusion rules are added one at a time — here age ≥ 18, follow-up ≥ 90 days, primary surgery only (`SurgeryNumber == 1`, which removes the retreatment rows), and preoperative cylinder ≤ 5.00 D by absolute value. The count remaining is displayed **per cohort** after each rule, so the cost of every criterion is visible before anything runs.

The subset column is set here too. It is left empty in this run (*Subsets: 0 active*), so one analysis covers every procedure. Choosing **Procedure**, as the *Example - 04_Subsets_PRK_LASIK* preset does, makes the software run a complete, independent analysis for PRK and for LASIK from one configuration.

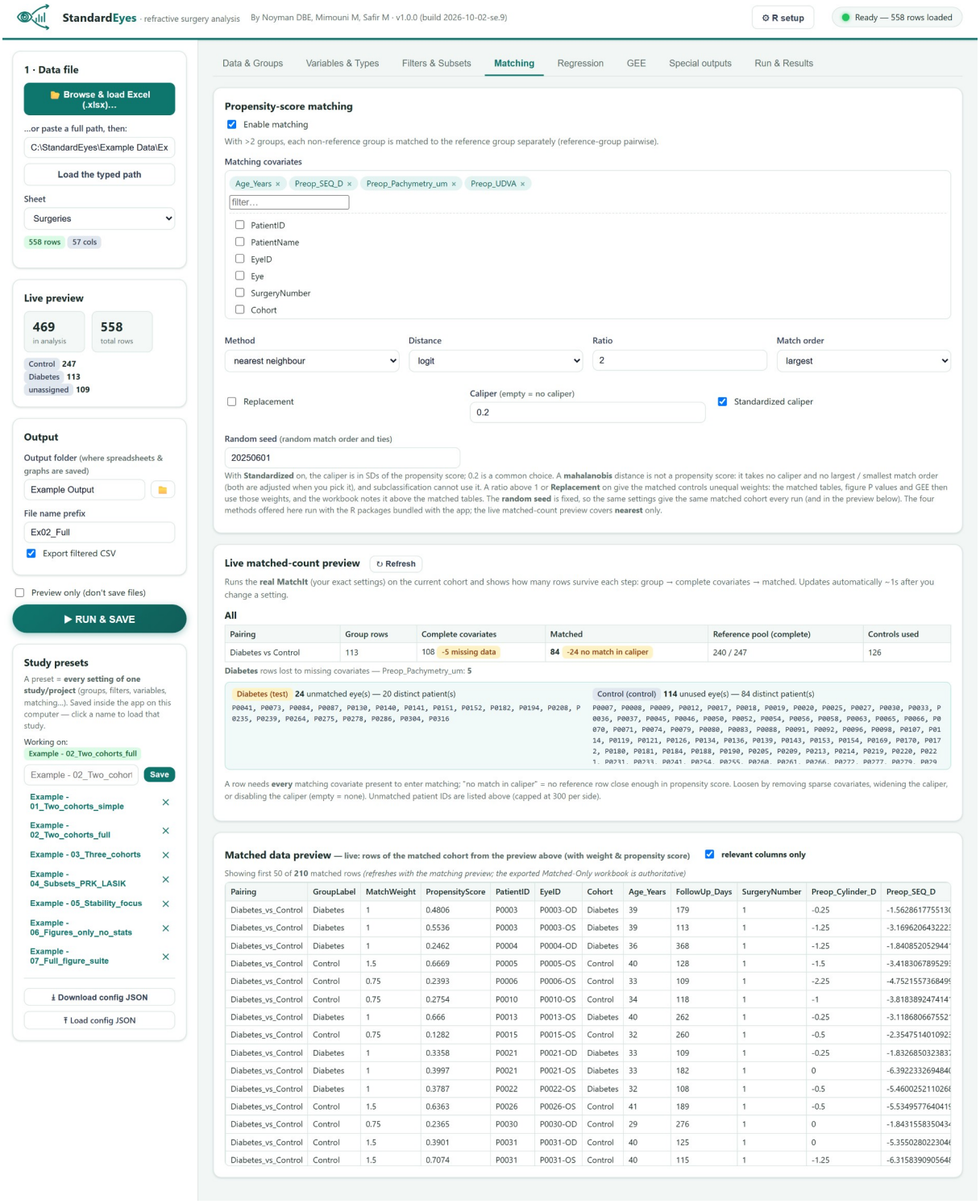


Step 4 · Matching

The matching covariates and every algorithm parameter are set here: method, distance, ratio, replacement, caliper, standardized caliper and match order.

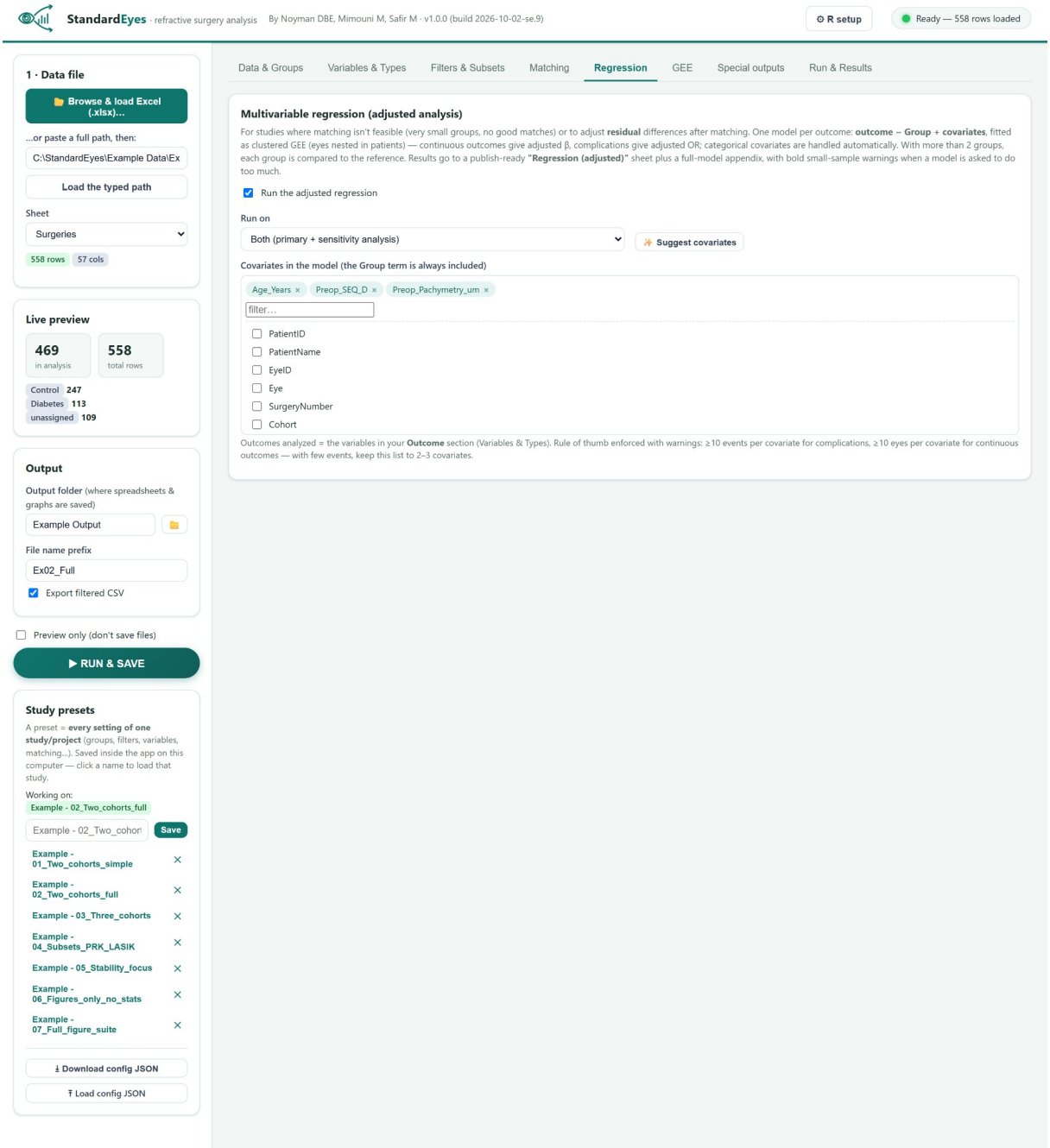
Beneath them the **live matched-count preview** runs the real algorithm, separately for each pairing and for each subset when subsets are set, and reports how many rows survive each step: group rows → rows with complete covariates → rows actually matched. In the capture below, 5 of the 113 Diabetes rows are lost to a missing pachymetry value and 24 more find no control within the caliper, so 84 Diabetes eyes are matched

to 126 control eyes, and the patient IDs of every unmatched eye are listed. The matched dataset itself (210 rows), with matching weights and propensity scores, is shown underneath.



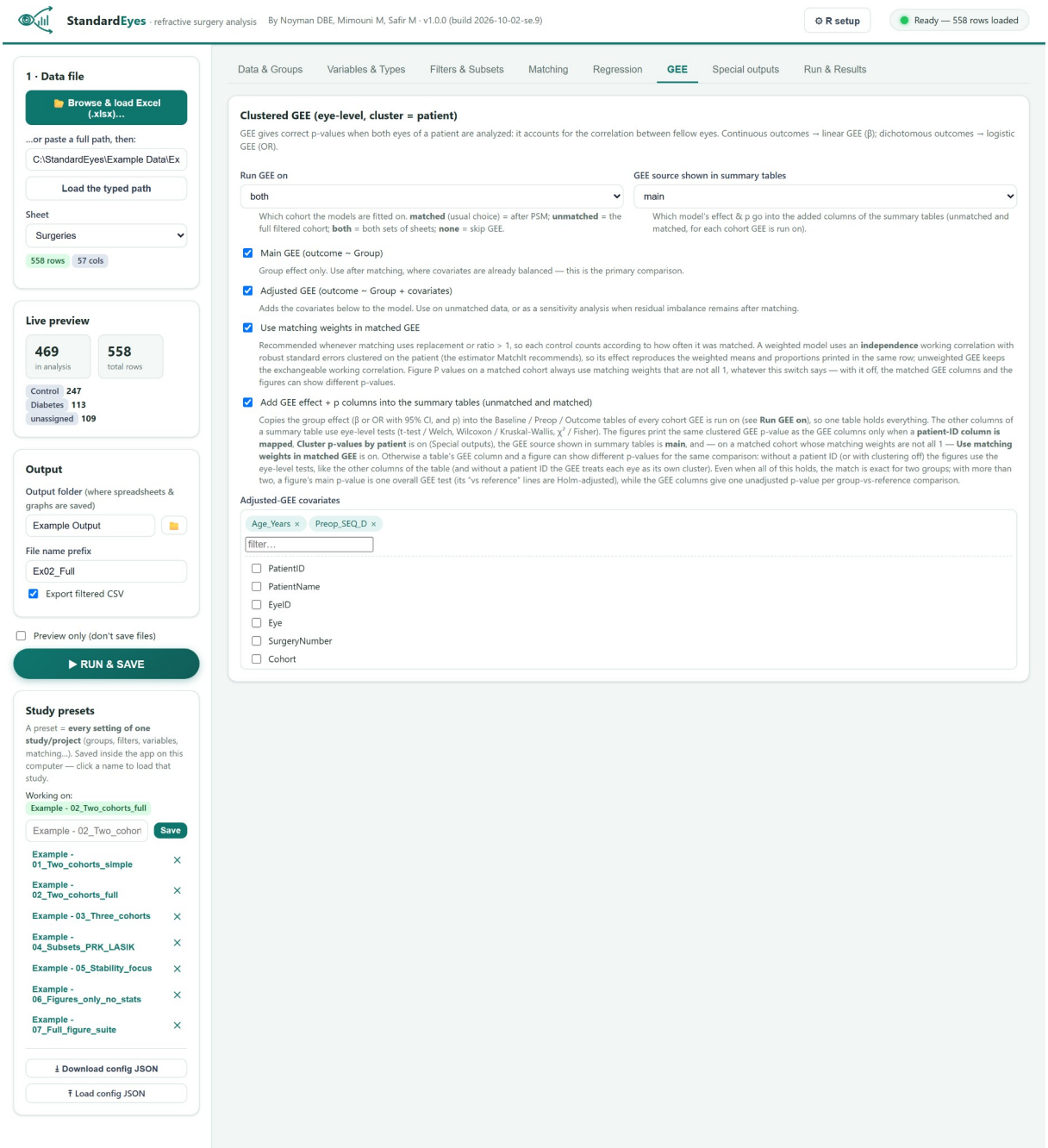
Step 5 · Regression

Covariates for the adjusted models are chosen here. **Suggest covariates** ranks the baseline variables by their largest standardized mean difference against the reference group and suggests those at 0.10 or above (the mapped age and sex variables are always suggested), so the choice is driven by imbalance rather than by its effect on the result.



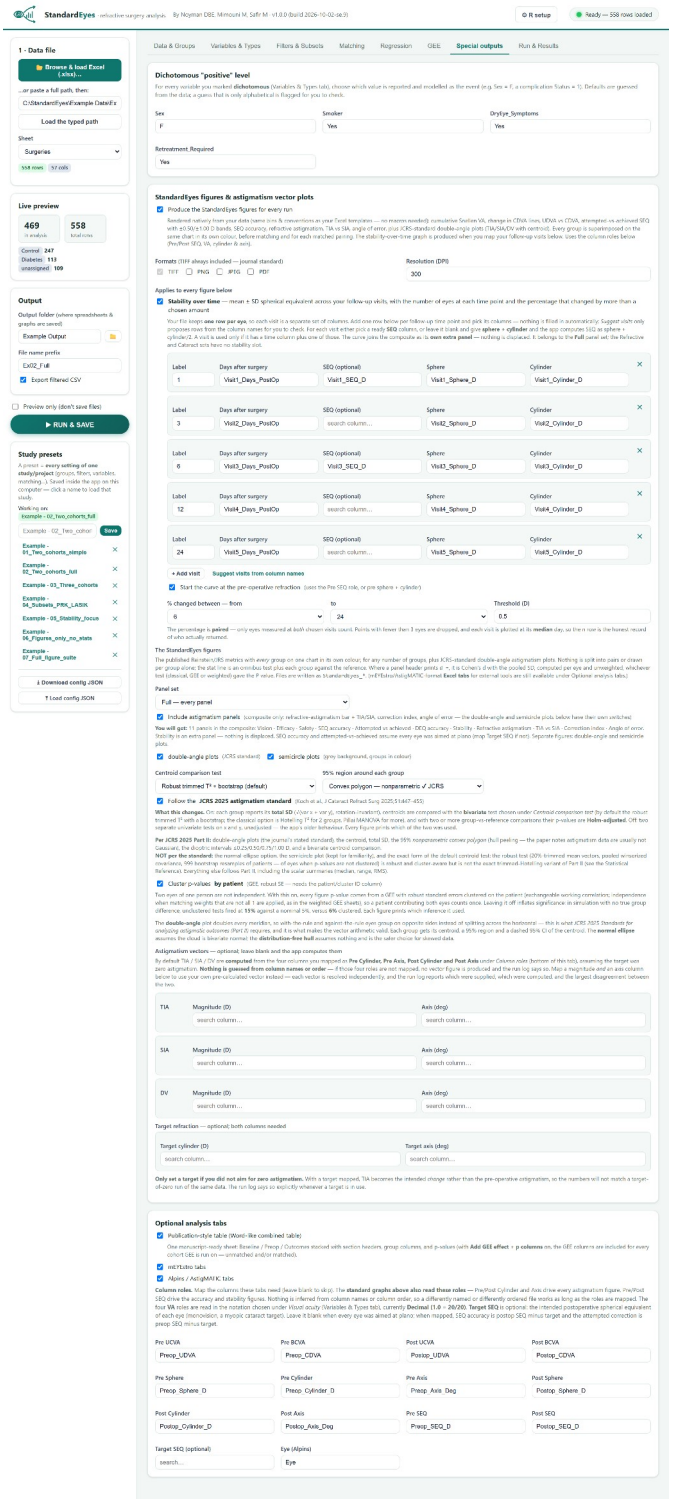
Step 6 · GEE

The clustered comparison. Choose which cohort(s) to model, whether to fit the unadjusted model, the adjusted model or both, whether matching weights are carried into the model, and which of the two supplies the extra columns merged into the summary tables.



Step 7 · Special outputs

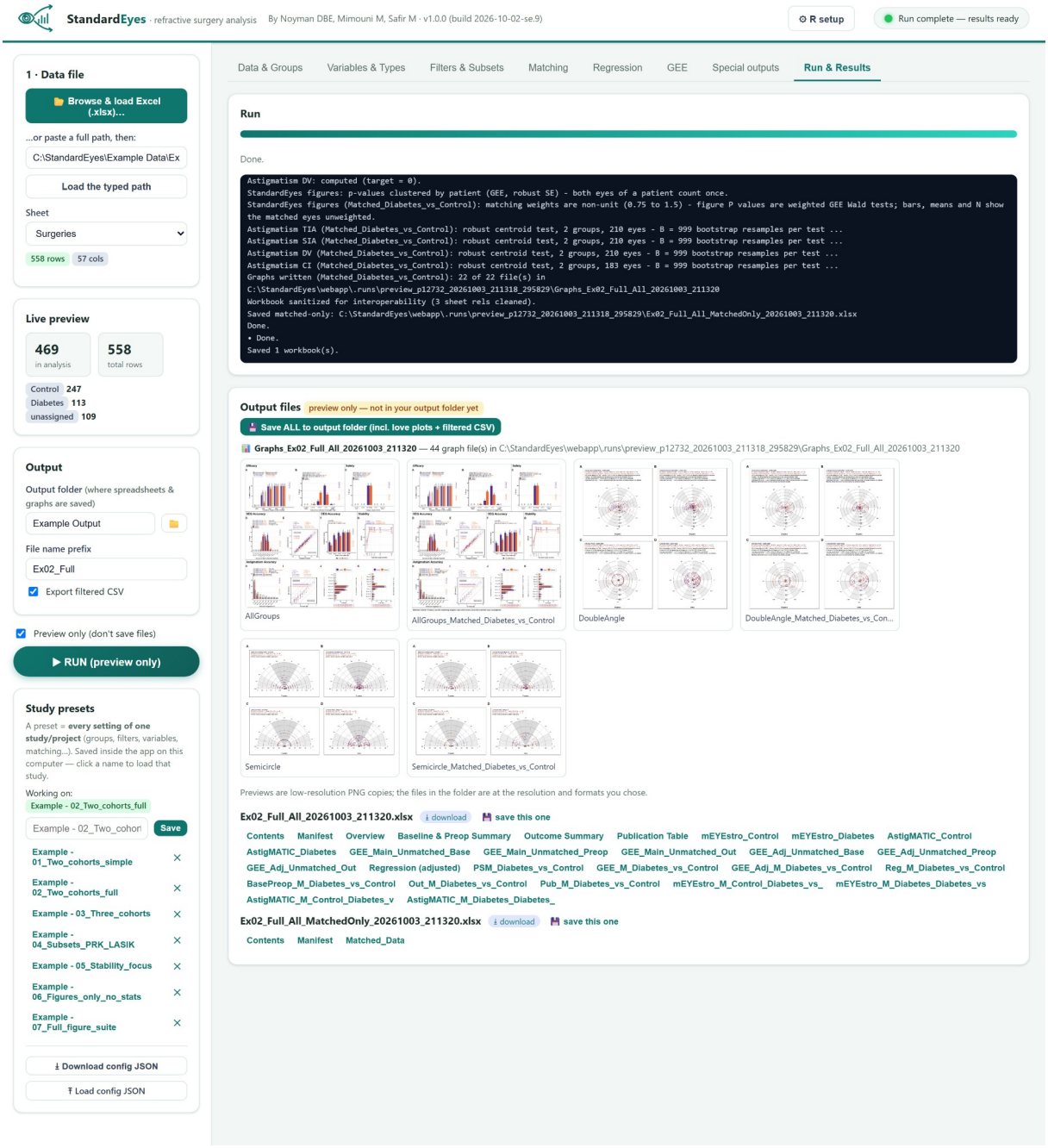
The dichotomous "positive" levels, every figure option with its formats and resolution, and the optional analysis sheets. The column roles the figures need are mapped at the bottom — including the four acuity roles, read in the notation declared under *Visual acuity* on the Variables & Types tab (decimal for the example dataset; see 1.3), and the optional target SEQ.



Step 8 · Run & Results

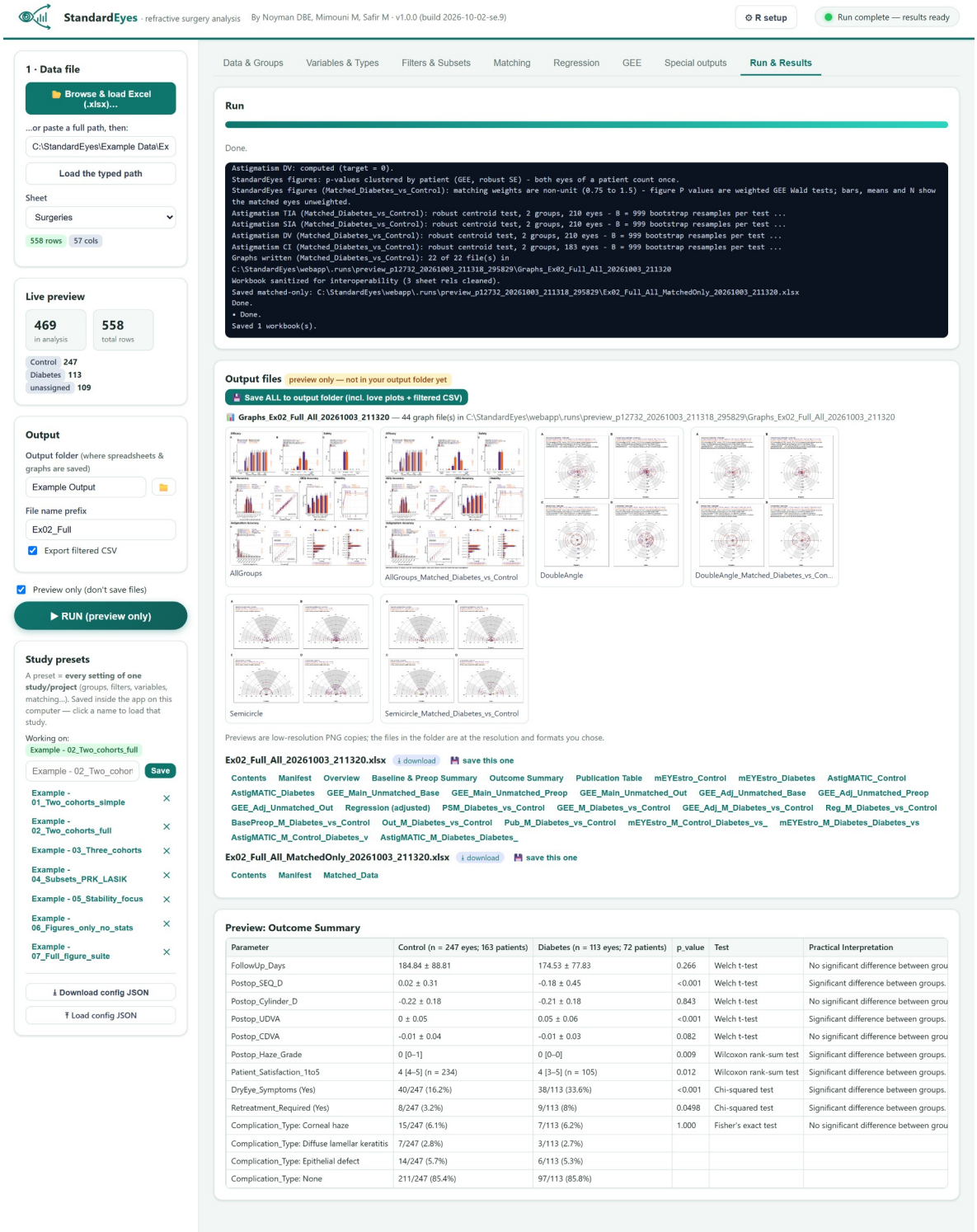
Run with **Preview only** to inspect everything first, or run and save directly. Progress and the full R log are shown while the engine works — including how many rows each filter removed, how each acuity column was read, every file written, and notes such as the stability graph being skipped when no follow-up visit is mapped. A running analysis can be stopped with **Cancel**; only one run is active at a time. The figure folders are listed with a gallery of preview images of every composite figure.

The capture below is a finished preview-only run. Note the amber badge, *preview only — not in your output folder yet*: nothing has been written to the output folder. The figure folder is listed with its file count (44 graph files, 22 for the whole cohort and 22 for the matched pairing) and a gallery of its six composite previews, followed by the workbook with all of its sheets and the matched-only workbook. **Save ALL to output folder** then writes the workbooks, love plots, figures and filtered CSV in one action.



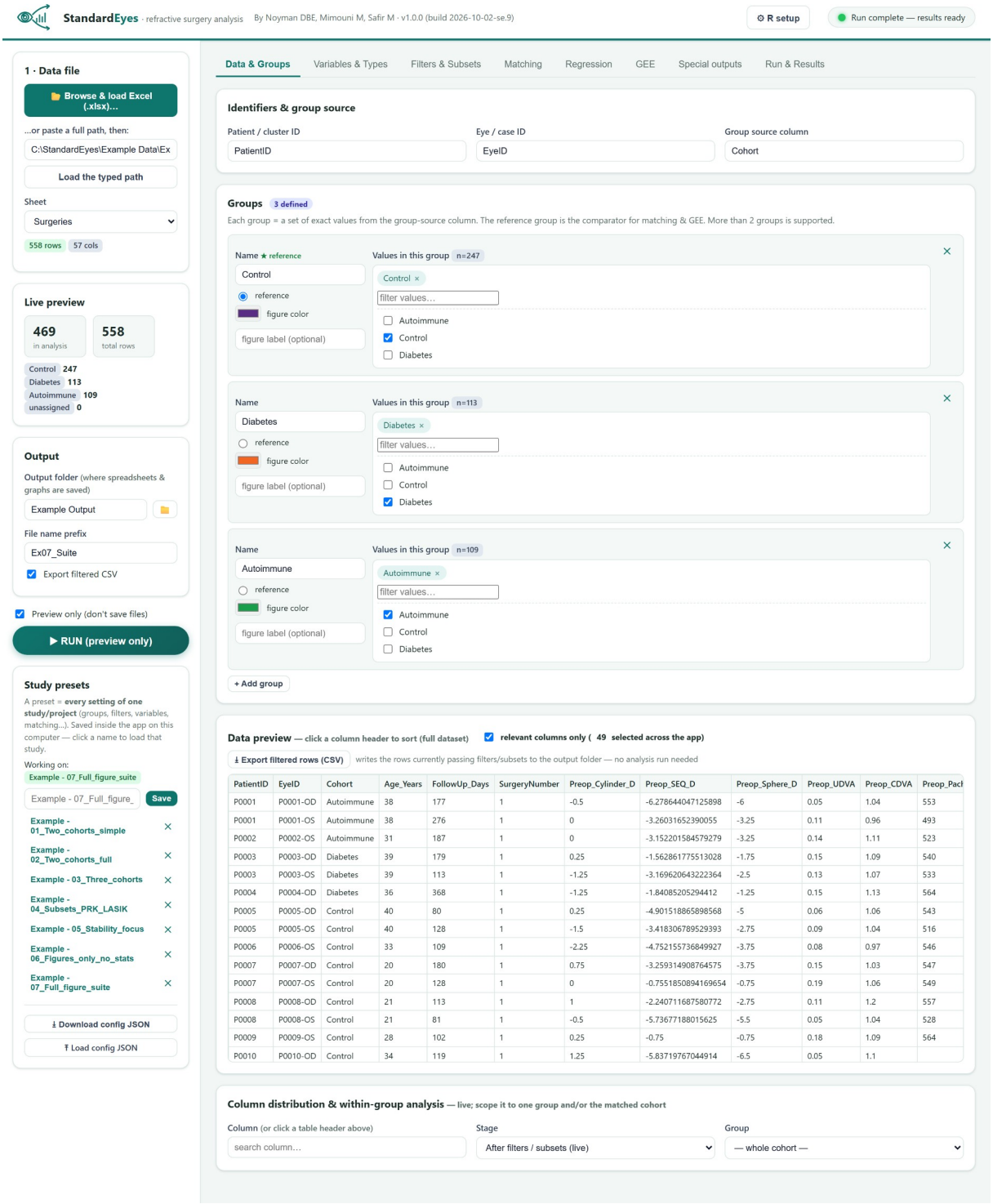
Step 9 · Browsing the results inside the app

Clicking any sheet name listed under a workbook opens that sheet in a *Preview* table beneath the list, before anything is saved, so the analysis can be checked without leaving the tool or opening Excel.



Step 10 · Three cohorts

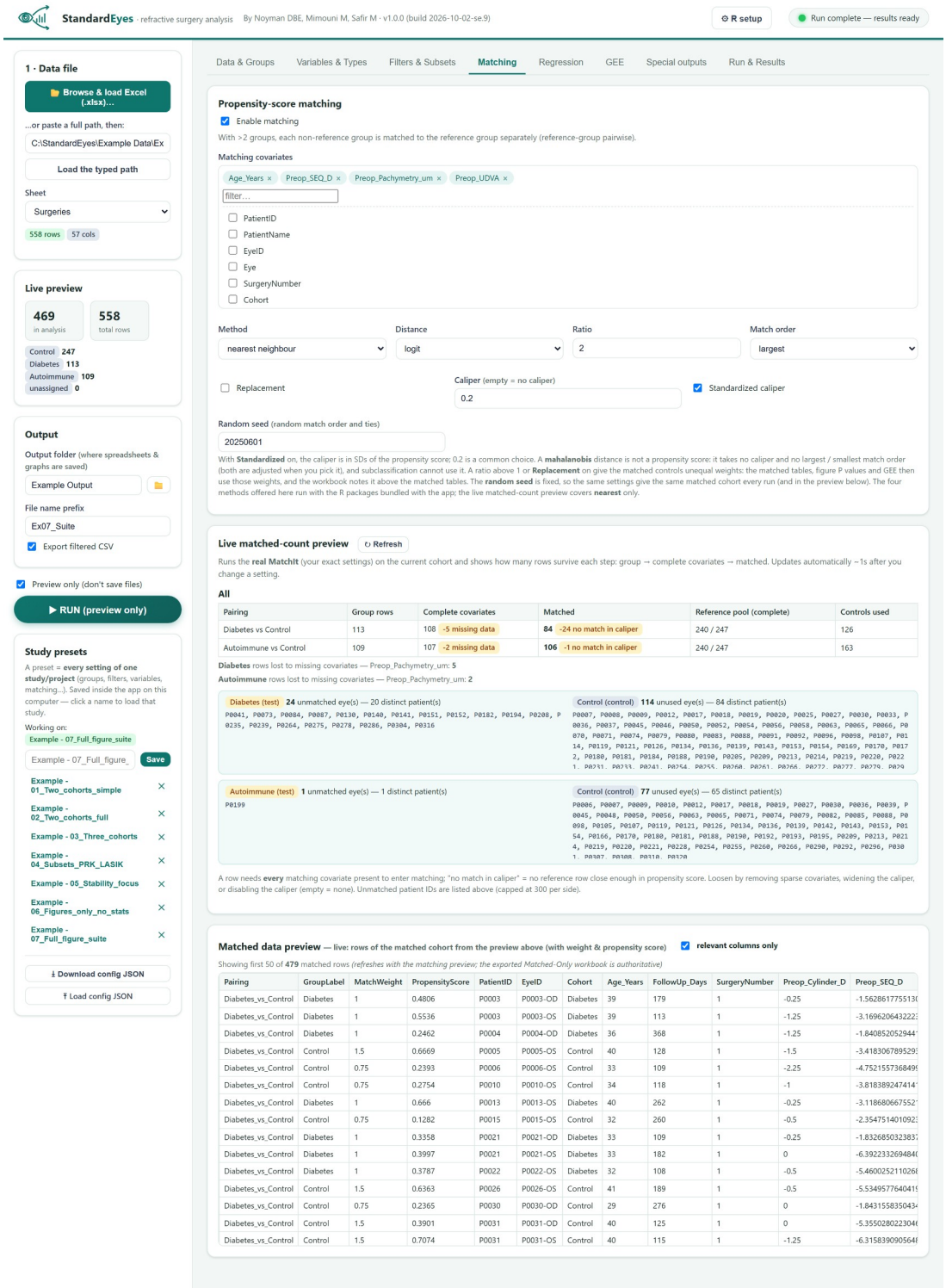
Loading a three-cohort preset adds the third group: Control, Diabetes and Autoimmune, with Control still the reference. The sidebar now counts 109 Autoimmune eyes and no unassigned rows. The capture uses the *Example - 07_Full_figure_suite* preset, which runs the same analysis as *Example - 03_Three_cohorts*. Apart from the third group, the file prefix and the within-group tables (none in these presets), its settings are those of steps 2–7.



PatientID	EyeID	Cohort	Age_Years	FollowUp_Days	SurgeryNumber	Preop_Cylinder_D	Preop_SEQ_D	Preop_Sphere_D	Preop_UDVA	Preop_CDVA	Preop_Pacl
P0001	P0001-OD	Autoimmune	38	177	1	-0.5	-6.278644047125898	-6	0.05	1.04	553
P0001	P0001-OS	Autoimmune	38	276	1	0	-3.26031652390055	-3.25	0.11	0.96	493
P0002	P0002-OS	Autoimmune	31	187	1	0	-3.152201584579279	-3.25	0.14	1.11	523
P0003	P0003-OD	Diabetes	39	179	1	0.25	-1.562861775513028	-1.75	0.15	1.09	540
P0003	P0003-OS	Diabetes	39	113	1	-1.25	-3.169620643222364	-2.5	0.13	1.07	533
P0004	P0004-OD	Diabetes	36	368	1	-1.25	-1.84085205294412	-1.25	0.15	1.13	564
P0005	P0005-OD	Control	40	80	1	0.25	-4.901518865898568	-5	0.06	1.06	543
P0005	P0005-OS	Control	40	128	1	-1.5	-3.418306789529393	-2.75	0.09	1.04	516
P0006	P0006-OS	Control	33	109	1	-2.25	-4.752155736849927	-3.75	0.08	0.97	546
P0007	P0007-OD	Control	20	180	1	0.75	-3.259314908764575	-3.75	0.15	1.03	547
P0007	P0007-OS	Control	20	128	1	0	-0.7551850894169654	-0.75	0.19	1.06	549
P0008	P0008-OD	Control	21	113	1	1	-2.240711687580772	-2.75	0.11	1.2	557
P0008	P0008-OS	Control	21	81	1	-0.5	-5.73677188015625	-5.5	0.05	1.04	528
P0009	P0009-OS	Control	28	102	1	0.25	-0.75	-0.75	0.18	1.09	564
P0010	P0010-OD	Control	34	119	1	1.25	-5.83719767044914	-6.5	0.05	1.1	

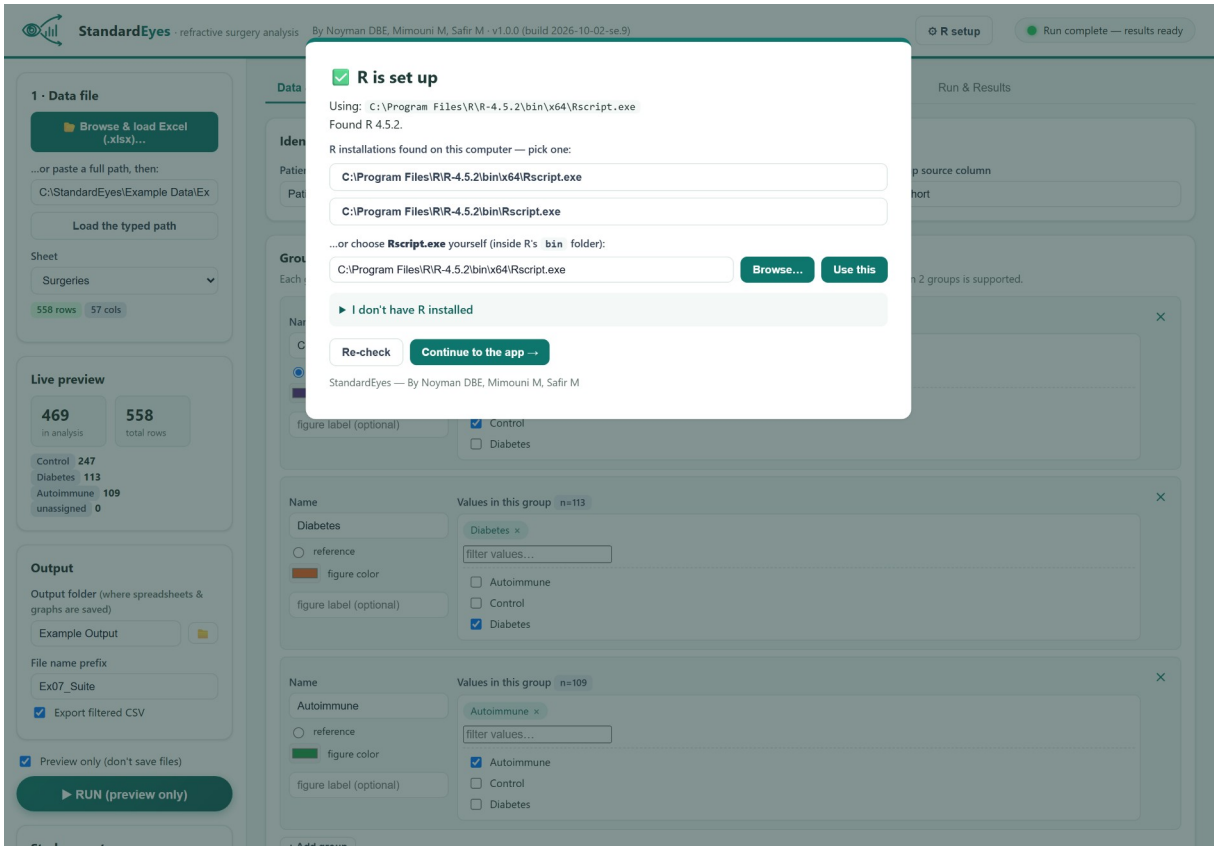
Step 11 · Matching with three cohorts

With more than two groups, each non-reference cohort is matched to the reference **separately**, producing one pairing per cohort. The live preview reports both pairings, one row each: Diabetes vs Control (84 of 113 eyes matched, 126 controls used) and Autoimmune vs Control (106 of 109 matched, 163 controls used). The run will produce a complete set of matched outputs for each.



Step 12 · The R setup screen

The only external requirement is R. If it is not found automatically, this screen opens; it can also be reopened at any time from the header. It lists the R installations found on the machine, accepts a path chosen with a file picker or typed in, and validates the choice immediately by reporting the R version.

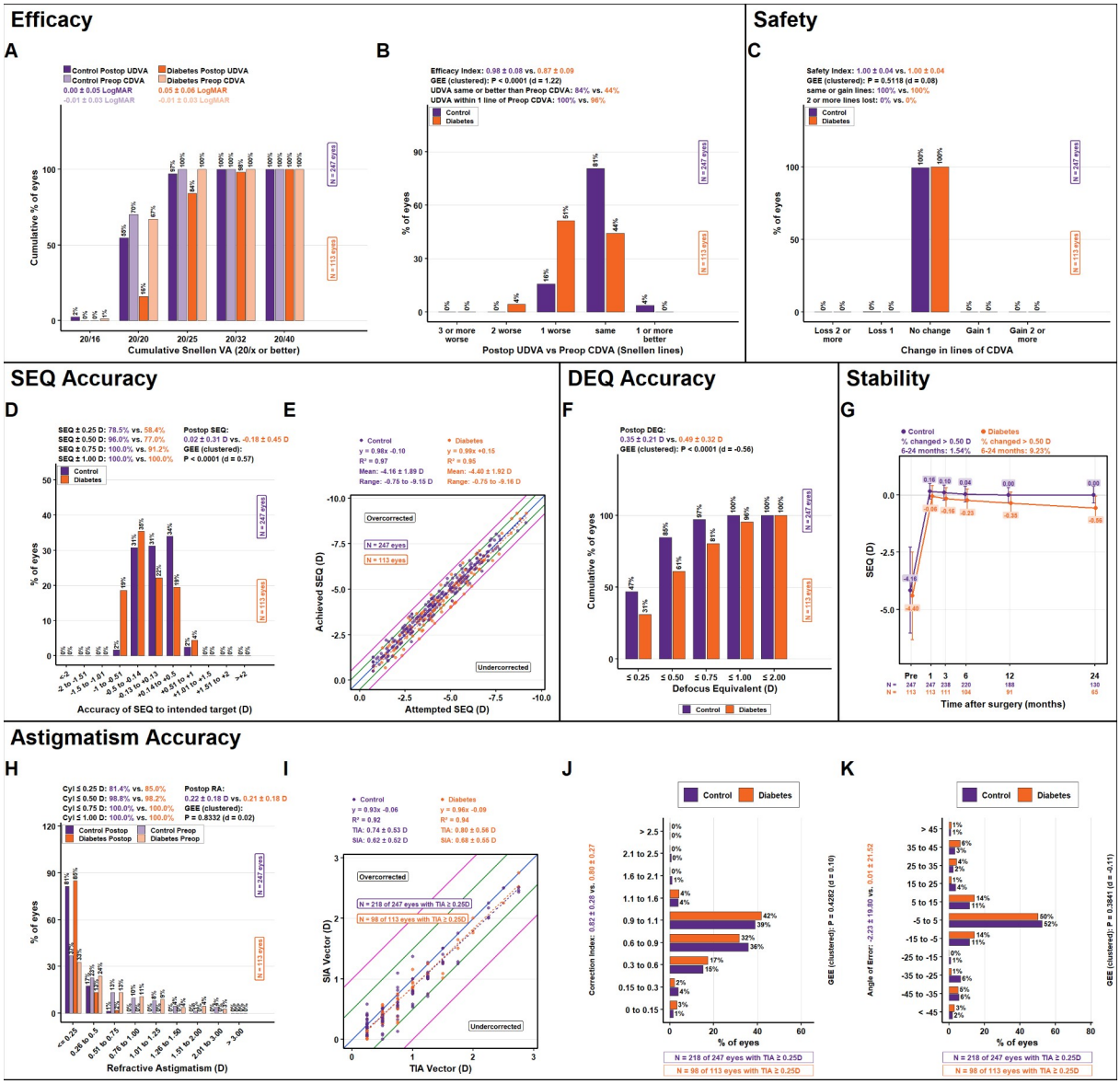


PART 4 · THE OUTPUT FIGURES

All figures below are unmodified outputs of the two example runs: the two-cohort run of Part 3 (the composite, the single panels, the balance plot and the stability panel) and the three-cohort run of steps 10–11 (the semicircle plots, the three-cohort composite and the double-angle plots). Every panel is written both as its own file and inside the composite. The interface always writes TIFF, at 300 dpi unless you change it, and PNG, JPEG and PDF can be added.

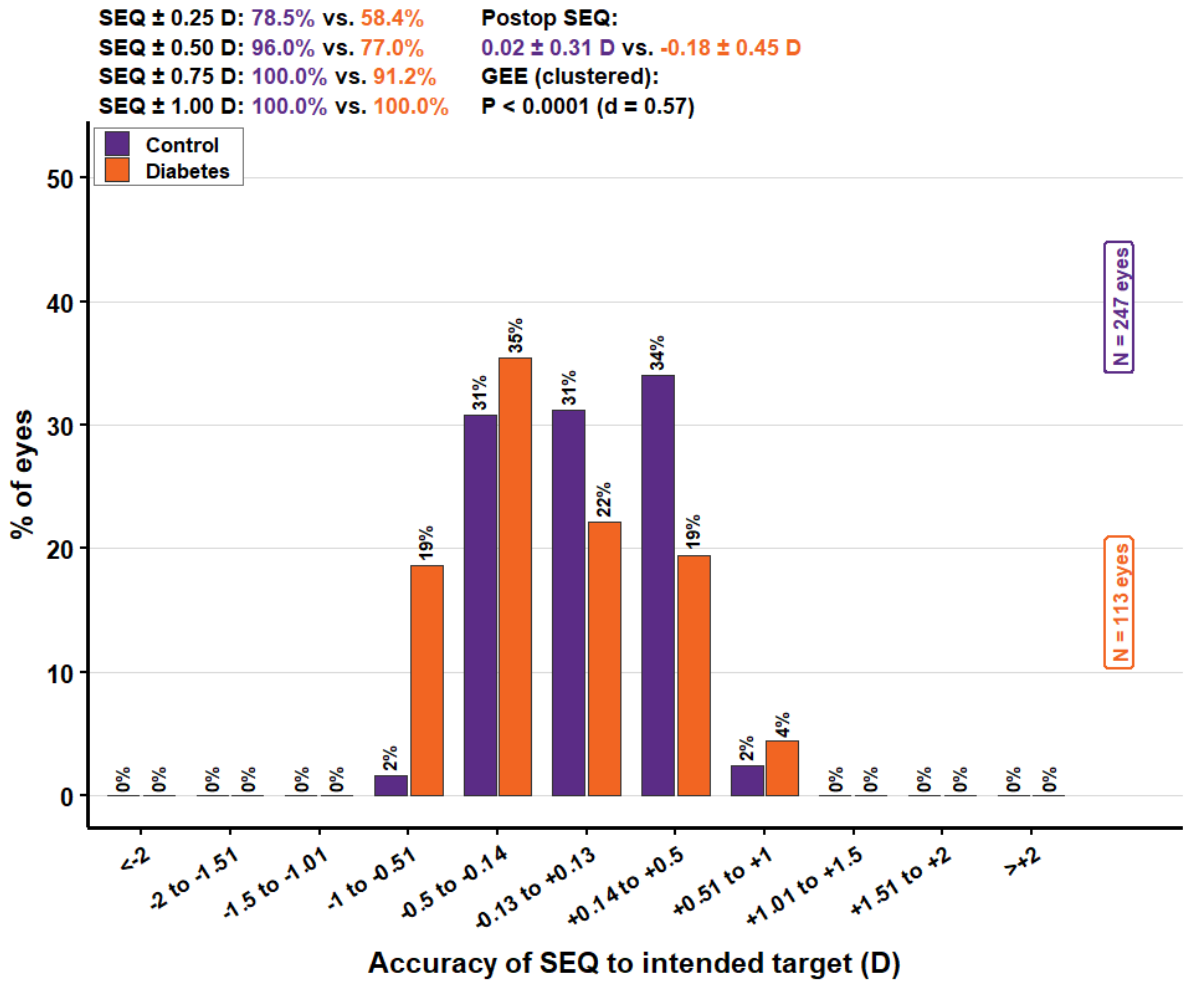
The StandardEyes composite

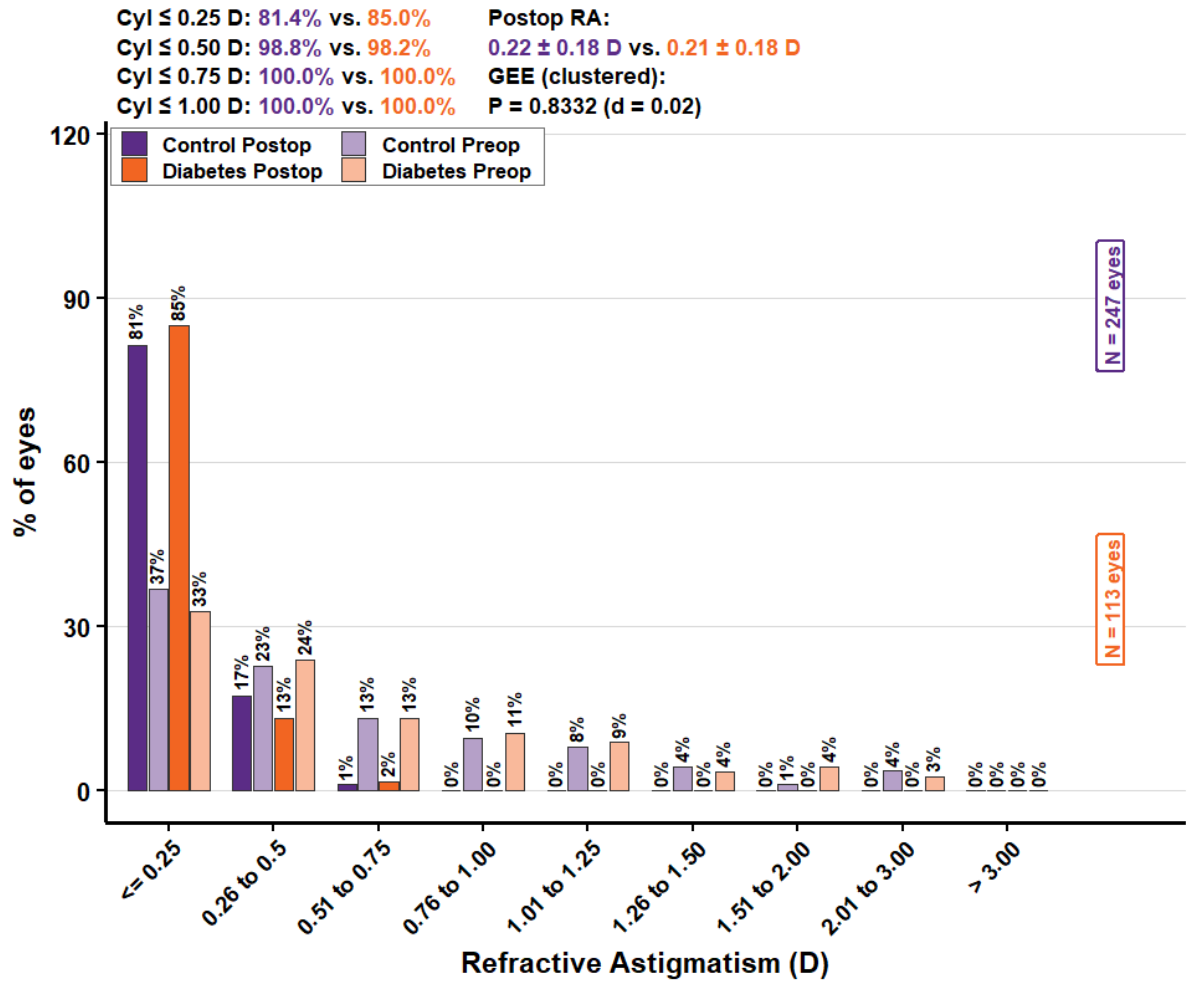
Both cohorts on shared axes in every panel, with the number of eyes per cohort in a box beside each bar panel (inside the plot area of the attempted-versus-achieved and TIA-versus-SIA scatter panels, beneath the axis in the stability panel, and beneath the correction-index and angle-of-error panels). The TIA-versus-SIA, correction-index and angle-of-error counts are the eyes with TIA ≥ 0.25 D. The between-cohort statistic and *P* value are printed above panels B, C, D, F and H and beside panels J and K. Section headers group the panels into Efficacy, Safety, SEQ Accuracy, DEQ Accuracy, Stability and Astigmatism Accuracy.

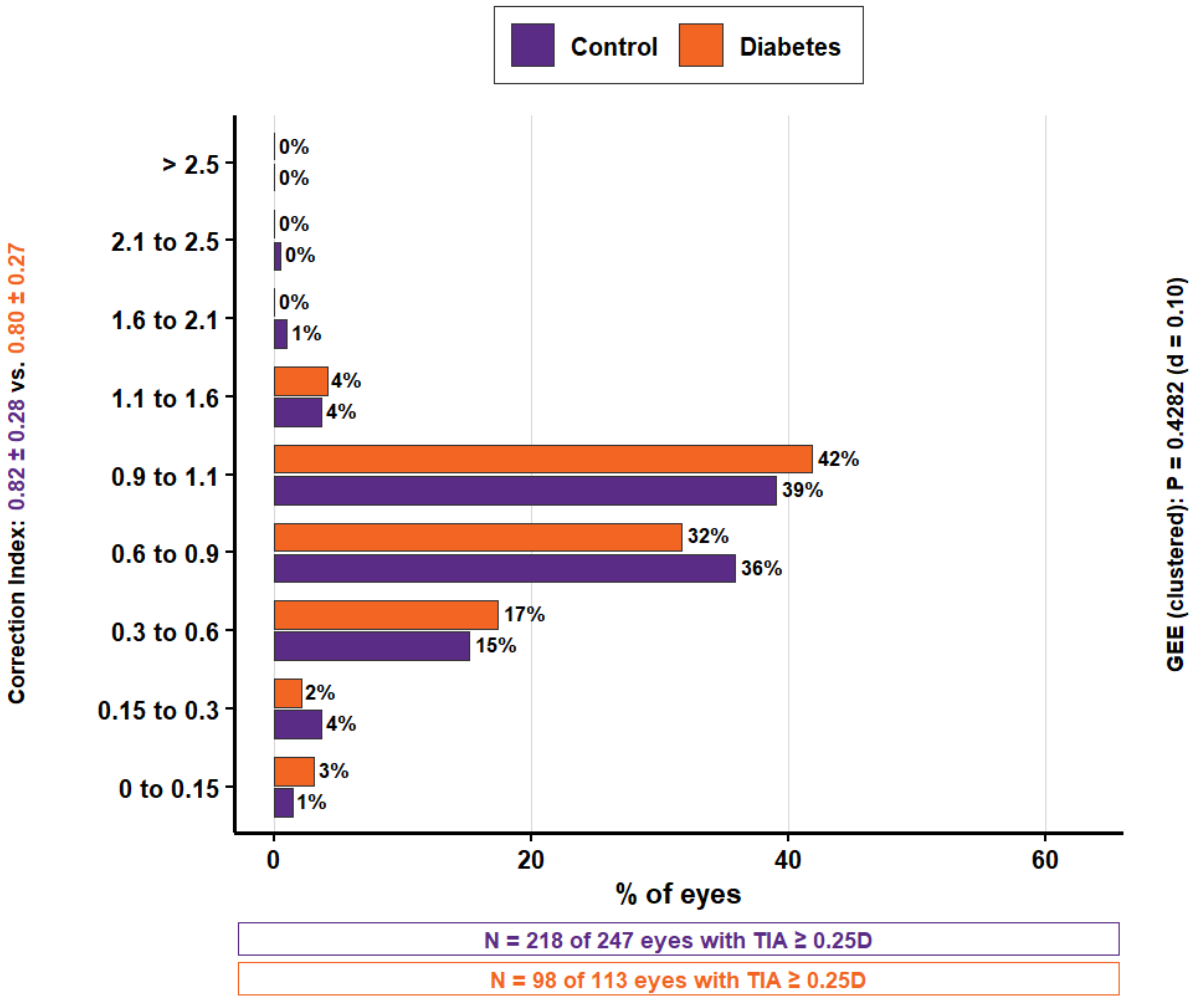


Individual panels

Each panel is also written on its own, at full size, ready to be used as a single figure.

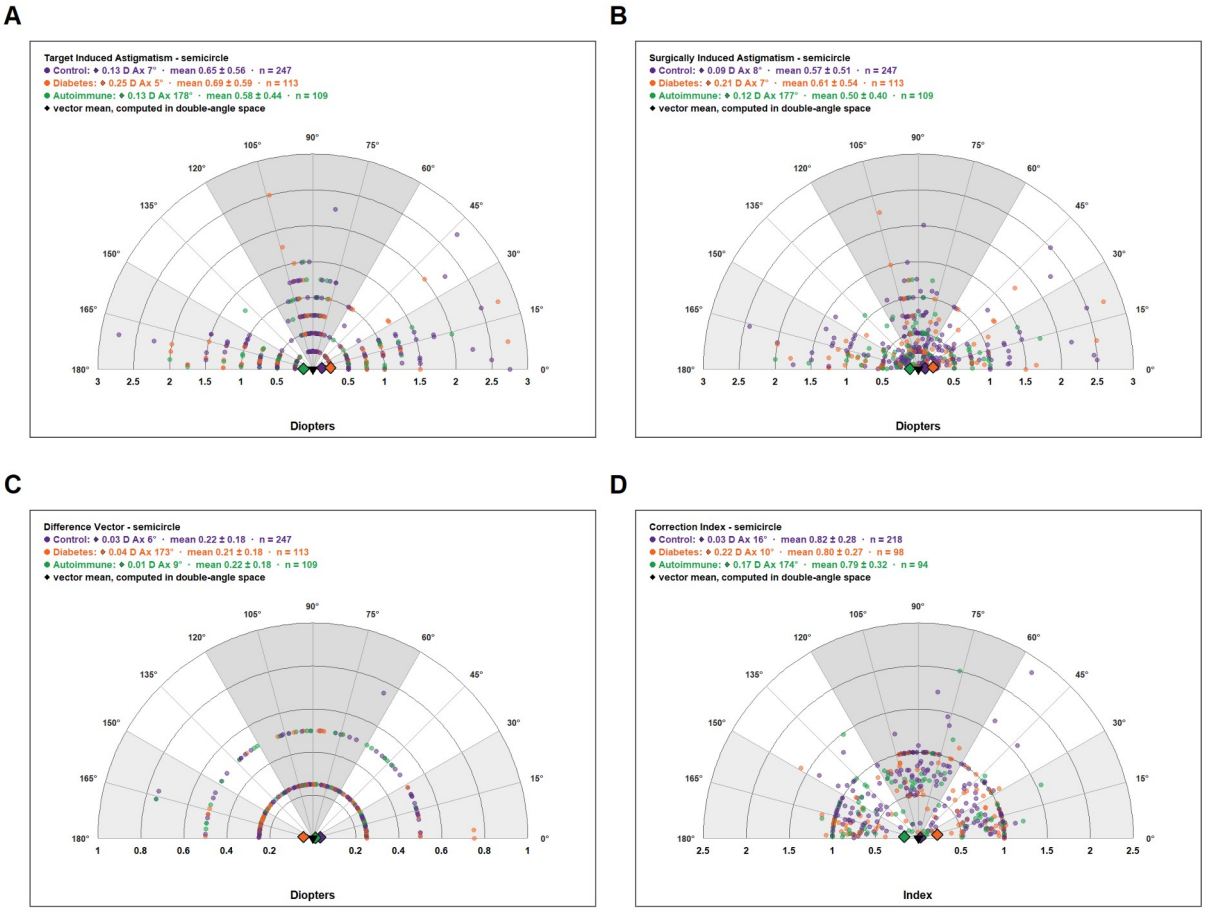

SEQ ± 0.25 D: 78.5% vs. 58.4%
SEQ ± 0.50 D: 96.0% vs. 77.0%
SEQ ± 0.75 D: 100.0% vs. 91.2%
SEQ ± 1.00 D: 100.0% vs. 100.0%
Postop SEQ:
0.02 ± 0.31 D vs. -0.18 ± 0.45 D
GEE (clustered):
P < 0.0001 (d = 0.57)
Control
Diabetes
N = 247 eyes
N = 113 eyes
% of eyes
0
10
20
30
40
50
0% 0%
0% 0%
0% 0%
2% 19%
31% 35%
31% 22%
34% 19%
2% 4%
0% 0%
0% 0%
0% 0%
<-2
-2 to -1.51
-1.5 to -1.01
-1 to -0.51
-0.5 to -0.14
-0.13 to +0.13
+0.14 to +0.5
+0.51 to +1
+1.01 to +1.5
+1.51 to +2
>+2
Accuracy of SEQ to intended target (D)


Cyl ≤ 0.25 D: 81.4% vs. 85.0%
Cyl ≤ 0.50 D: 98.8% vs. 98.2%
Cyl ≤ 0.75 D: 100.0% vs. 100.0%
Cyl ≤ 1.00 D: 100.0% vs. 100.0%
Postop RA:
0.22 ± 0.18 D vs. 0.21 ± 0.18 D
GEE (clustered):
P = 0.8332 (d = 0.02)
Control Postop
Control Preop
Diabetes Postop
Diabetes Preop
N = 247 eyes
N = 113 eyes
% of eyes
120
90
60
30
0
81%
37%
85%
33%
17%
23%
13%
24%
1%
13%
2%
13%
0%
10%
0%
11%
0%
8%
0%
9%
0%
4%
0%
4%
0%
1%
0%
4%
0%
4%
0%
3%
0%
0%
0%
0%
<= 0.25
0.26 to 0.5
0.51 to 0.75
0.76 to 1.00
1.01 to 1.25
1.26 to 1.50
1.51 to 2.00
2.01 to 3.00
> 3.00
Refractive Astigmatism (D)



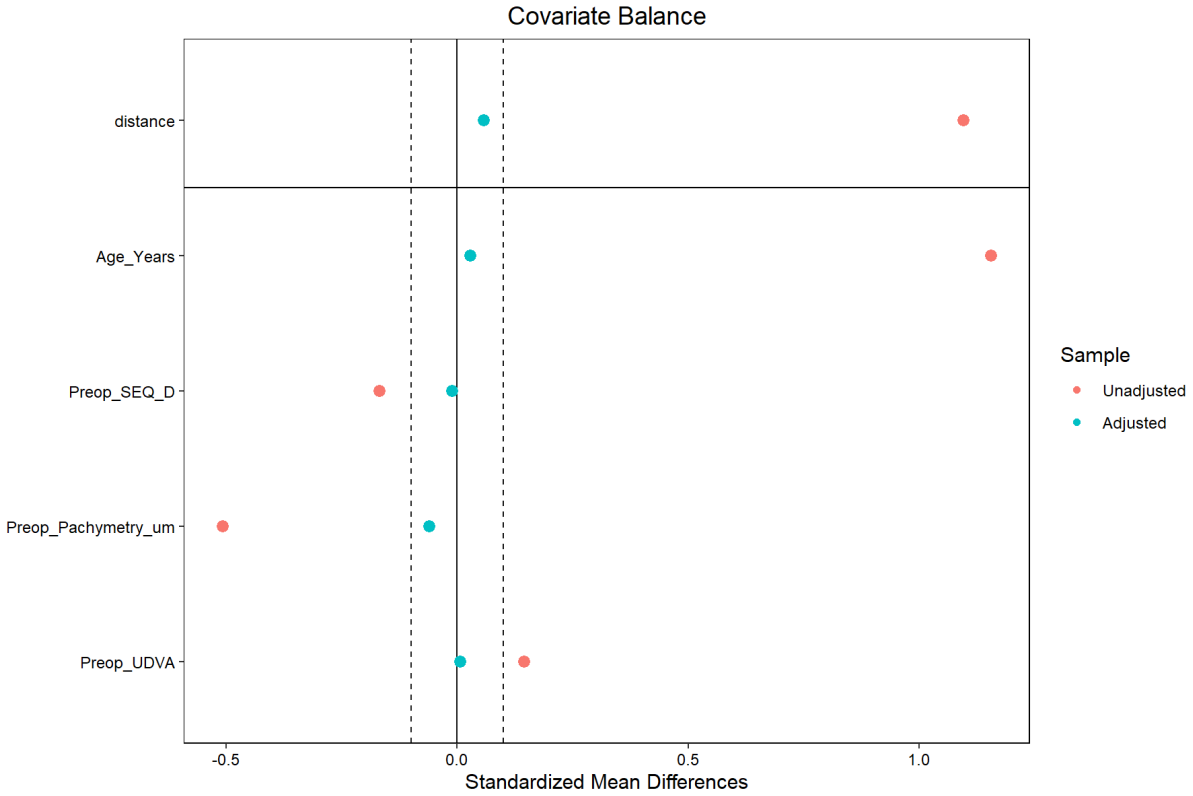
Semicircular vector plots

The Alpins set — target induced astigmatism, surgically induced astigmatism, difference vector and correction index — as semicircular polar diagrams with all cohorts superimposed on a grey-shaded background, vector means computed in double-angle space and back-transformed for display.



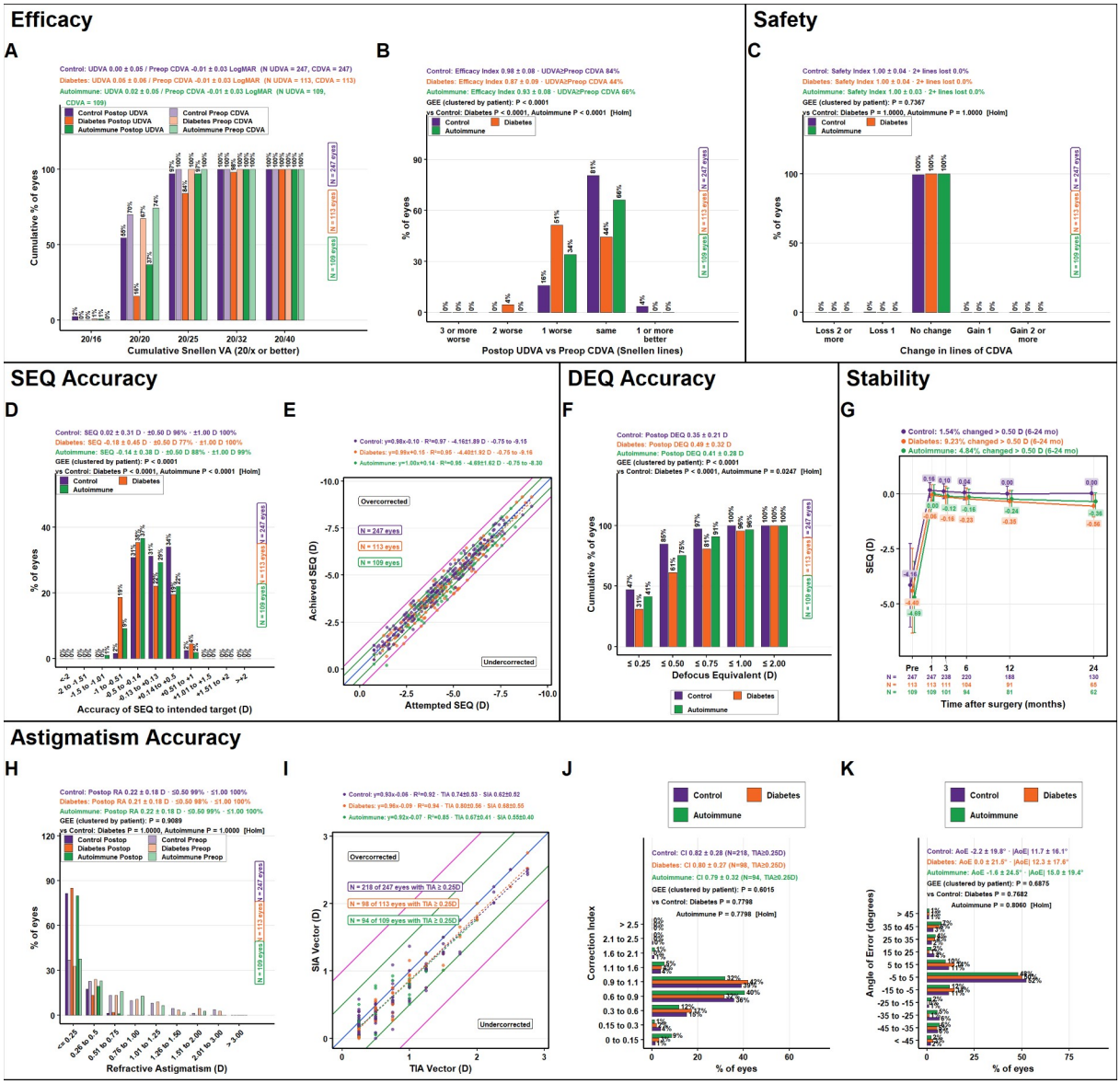
Covariate balance

Produced for each matched pairing. Age (SMD 1.16) and corneal thickness (−0.51) were severely imbalanced before matching, and after matching every covariate is inside the 0.1 threshold, preoperative spherical equivalent included (−0.17 before, −0.01 after). That is not guaranteed. In the PRK subset of the same study (*Example - 04_Subsets_PRK_LASIK*) preoperative spherical equivalent is still −0.13 after matching, which is exactly why this plot is worth reading before trusting a matched comparison.



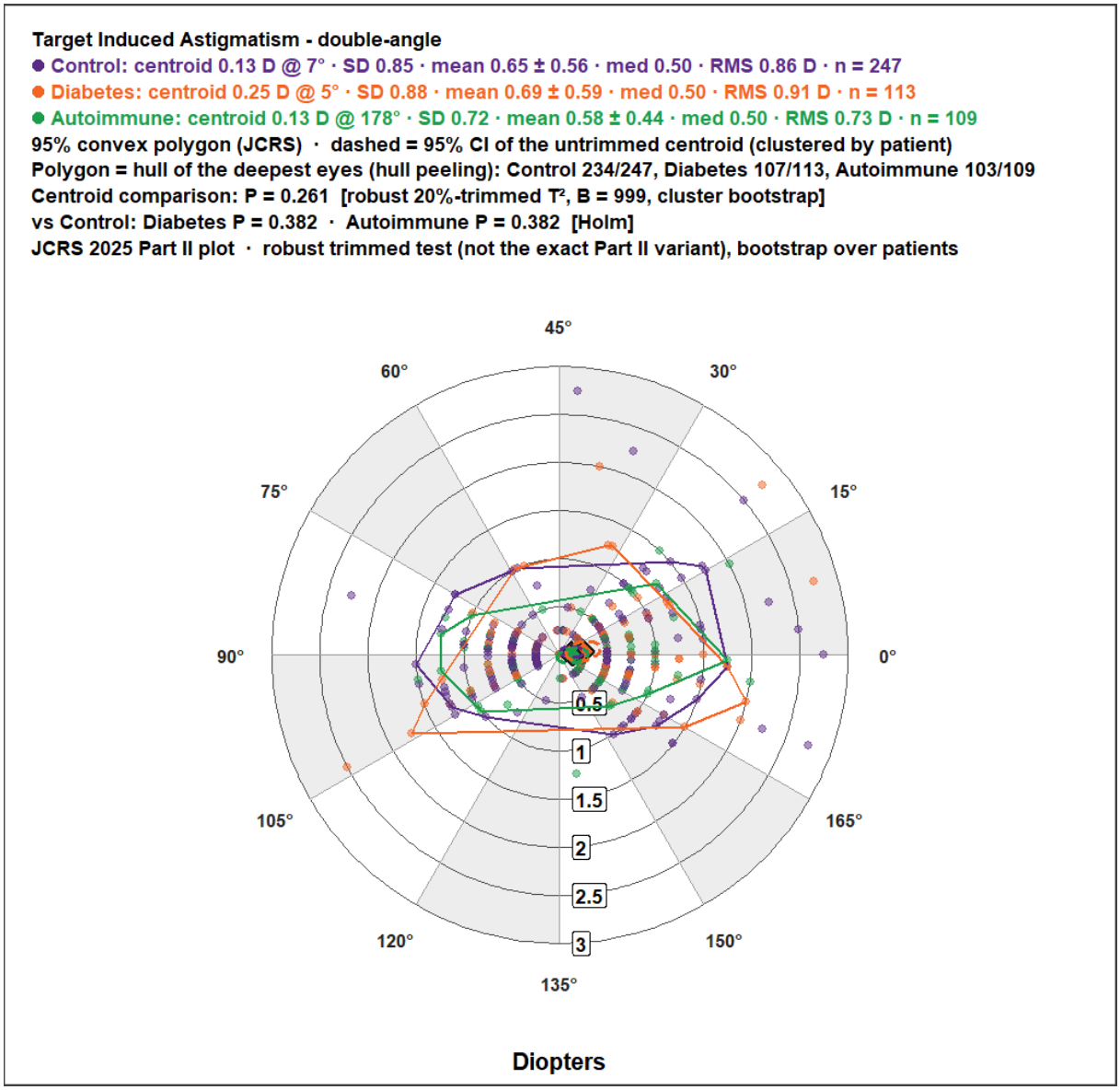
The composite with three cohorts

All cohorts share every panel, so nothing is split into pairs. Stability sits between DEQ accuracy and the astigmatism block, and the astigmatism panels follow.



The JCRS double-angle astigmatism plot

Meridians are doubled, so with-the-rule and against-the-rule eyes fall on opposite sides instead of splitting across the horizontal. Each cohort gets its centroid, a 95% nonparametric convex polygon (enclosing exactly $\lfloor 0.95\,n \rfloor$ of its eyes, counted in the lines above the plot, e.g. Control 234/247) and a dashed 95% confidence region for the centroid (Hotelling T^2 radius). The same lines name the statistical method used, here the robust 20%-trimmed T^2 with B = 999 cluster-bootstrap resamples ($P = 0.261$).

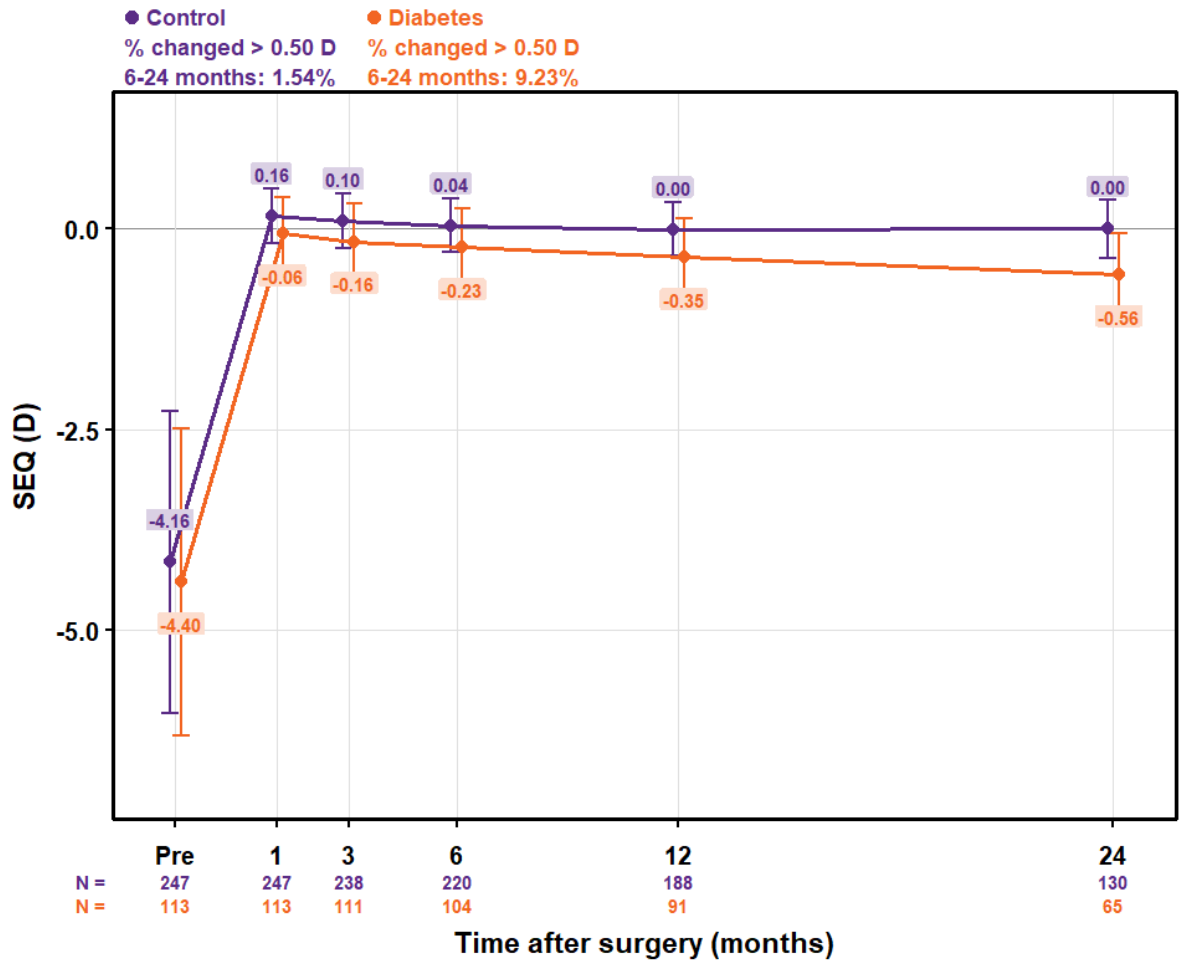


The four vectors are also composed into one 2x2 double-angle figure:



Stability over time

The stability curve as its own full-size panel — mean ± SD spherical equivalent at every mapped visit, the number of eyes beneath the axis, and the paired percentage changing by more than the chosen threshold.



PART 5 · THE OUTPUT FILES AND SHEETS

5.1 The files a run produces

File / folder	Contents
<prefix>_<subset>_<timestamp>.xlsx	The **main workbook** for that subset — every analysis sheet below.
<prefix>_<subset>_MatchedOnly_<timestamp>.xlsx	The matched rows themselves: one workbook per subset, holding every pairing (a *Pairing* column says which), with a Manifest that lists the pairings and counts the eyes per pairing and as distinct eyes (a reference eye matched in two pairings counts once). Its *Matched_Data* columns carry the headers as typed in your file (a converted acuity column ends in _logMAR or _decimal, as in the filtered CSV).
Graphs_<prefix>_<subset>_<timestamp>\	Every figure, in each format chosen: each panel separately plus the composites, once for the whole cohort and once for each matched pairing (22 files each in the example). A _preview subfolder holds low-resolution PNG copies of the composites for the in-app gallery.
LovePlot_<prefix>_<subset>_<pairing>_<timestamp>.png	Covariate balance before and after matching, one file per matched pairing, named with the same prefix, subset and time stamp as its workbook (a run without subsets is named All, the pooled run of a subset analysis Pooled), so a later run into the same folder never replaces it — e.g. LovePlot_Ex02_Full_All_Diabetes_vs_Control_20261002_205654.png.

File / folder	Contents
Filtered_Dataset_<timestamp>.csv	The rows left after the exclusions and filter rules, before they are split into groups and subsets, so rows of an unassigned group value or of no subset are included (in the example, 469 rows, 109 of them Autoimmune). The headers are as typed in your file (a converted acuity column ends in _logMAR or _decimal).

One main workbook and one figure folder are produced **per run**: one per subset when a subset column is set (a complete PRK set and a complete LASIK set with *Example - 04_Subsets_PRK_LASIK*), and a single `All` set otherwise, as in the run of Part 3. Each figure folder holds one set drawn on the whole cohort, every group on the same chart, and one more set for each matched pairing (one pairing with two cohorts, two with three).

5.2 Sheet by sheet

Contents and Manifest

Every workbook opens on a **Contents** sheet with one row per sheet: a link to it, its full name (Excel cuts sheet names at 31 characters, so `mEYEstro_M_Control_Diabetes_vs_` is listed in full as `mEYEstro_M_Control_Diabetes_vs_Control`) and a one-line description of what it holds.

The **Manifest** sheet comes second and records the conditions of the run: StandardEyes version and build (a run started from the command line, like the example workbooks, records them as "not recorded (run from the command line)"), R and package versions, the input file and its checksum, the VA notation and logMAR conversion, the matching method actually run (e.g. *nearest-neighbour on the propensity score (MatchIt)*, or *skipped - no matching covariate found in the file*) with its settings and any covariates not found in the file, whether the matching weights were unit or non-unit and, when the matched models use non-unit weights, their working correlation (*independence (weighted)*), the GEE and regression settings, the astigmatism test with its trimming fraction, bootstrap size and seed, and the random seeds used.

Overview

Cohort description and quality control: numbers of patients and eyes, age, follow-up (mean, SD, median and IQR), the sex distribution counted **per patient**, the group sizes, and a QC block showing how each source value was assigned to a group. A note at the top names the acuity columns analysed in logMAR and in decimal. Read this first — if a group is the wrong size, the reason is visible here.

Notes

Note

Visual acuity analysed in logMAR (lower = better): Preop_UDVA, Preop_CDVA, Postop_UDVA, Postop_CDVA - converted from the recorded notation.

Initial Summary

Description	Value
Number of Patients	235
Number of Eyes	360
Mean Age (per patient)	33.29
SD Age (per patient)	7.38
Age Range (per patient)	18 - 54
Mean Follow-up Time (per eye)	181.6
SD Follow-up Time (per eye)	85.54
Median Follow-up Time [IQR] (per eye)	170 [115–259.25]

Sex (per patient)

Sex	Count	Percentage
F	138	58.7
M	97	41.3

Groups

GroupLabel	Count	Percentage
Control	247	68.6
Diabetes	113	31.4

QC - group assignment by source value

Group_Source_Value	GroupLabel	N
Control	Control	247
Diabetes	Diabetes	113

Baseline & Preop Summary

Every baseline and preoperative variable, each group in its own column, with the *p* value, **the name of the test used** and a plain-language interpretation, followed here by the clustered GEE effect and its *p* value (*Add GEE columns* is on). Note how the type assigned in step 2 drives the presentation: continuous variables as mean ± SD with a Welch test, the ordinal dry-eye grade as median [IQR] with a rank test, dichotomous variables as n (%), and the categorical surgeon variable as one row per level. The within-group sections chosen in step 2 close the sheet, one block per group with no between-group test, and say whether each row was analysed per eye or per patient.

Baseline

Parameter	Control (n = 247 eyes; 163 patients)	Diabetes (n = 113 eyes; 72 patients)	p_value	Test	Practical Interpretation	GEE effect (95% CI)	GEE p value
Age_Years [per patient]	30.94 ± 6.34	38.58 ± 6.83	<0.001	Welch t-test	Significant difference between groups.	β = 7.627 [5.793, 9.461]	<0.001
Preop_SEQ_D	-4.16 ± 1.89	-4.4 ± 1.92	0.256	Welch t-test	No significant difference between groups.	β = -0.234 [-0.663, 0.196]	0.286
Preop_Sphere_D	-3.83 ± 1.89	-4.05 ± 1.95	0.317	Welch t-test	No significant difference between groups.	β = -0.205 [-0.642, 0.232]	0.358
Preop_Cylinder_D	-0.65 ± 0.56	-0.69 ± 0.59	0.546	Welch t-test	No significant difference between groups.	β = -0.040 [-0.161, 0.082]	0.521
Preop_UDVA	1.08 ± 0.23	1.1 ± 0.23	0.275	Welch t-test	No significant difference between groups.	β = 0.027 [-0.024, 0.079]	0.295
Preop_CDVA	-0.01 ± 0.03	-0.01 ± 0.03	0.098	Welch t-test	No significant difference between groups.	β = 0.006 [-0.001, 0.012]	0.093
Preop_Pachymetry_um	546.34 ± 26.34 (n = 240)	533.62 ± 25.04 (n = 108)	<0.001	Welch t-test	Significant difference between groups.	β = -12.434 [-18.508, -6.360]	<0.001
Preop_Mean_K_D	43.59 ± 1.4 (n = 243)	43.58 ± 1.47 (n = 110)	0.937	Welch t-test	No significant difference between groups.	β = -0.005 [-0.321, 0.311]	0.973
Preop_DryEye_Grade	0 [0–1]	1 [0–2]	0.048	Wilcoxon rank-sum test	Significant difference between groups.		
Sex (F) [per patient]	96/163 (58.9%)	42/72 (58.3%)	0.936	Chi-squared test	No significant difference between groups.	OR = 0.979 [0.557, 1.718]	0.940
Smoker (Yes) [per patient]	38/163 (23.3%)	13/72 (18.1%)	0.367	Chi-squared test	No significant difference between groups.	OR = 0.732 [0.363, 1.477]	0.384
Surgeon: Surgeon A	104/247 (42.1%)	37/113 (32.7%)	0.210	Chi-squared test	No significant difference between groups.		
Surgeon: Surgeon B	86/247 (34.8%)	43/113 (38.1%)					
Surgeon: Surgeon C	57/247 (23.1%)	33/113 (29.2%)					

Preoperative / intraoperative

Parameter	Control (n = 247 eyes; 163 patients)	Diabetes (n = 113 eyes; 72 patients)	p_value	Test	Practical Interpretation	GEE effect (95% CI)	GEE p value
Optical_Zone_mm	6.51 ± 0.29	6.49 ± 0.32	0.464	Welch t-test	No significant difference between groups.	β = -0.026 [-0.091, 0.039]	0.433
Ablation_Depth_um	64.05 ± 26.74	67.11 ± 27.86	0.330	Welch t-test	No significant difference between groups.	β = 2.946 [-3.219, 9.111]	0.349
Procedure: LASIK	102/247 (41.3%)	48/113 (42.5%)	0.338	Chi-squared test	No significant difference between groups.		
Procedure: PRK	108/247 (43.7%)	42/113 (37.2%)					
Procedure: SMILE	37/247 (15%)	23/113 (20.4%)					

Within-group only - Control (n = 247 eyes; 163 patients; no between-group test)

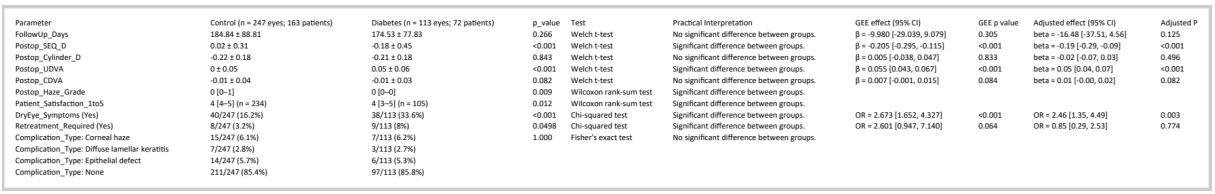
Parameter	n	Summary	Analyzed by
Preop_Pachymetry_um	240	546.34 ± 26.34 (median 547 [529.75-565.25], range 470-	eye
Preop_DryEye_Grade	247	0 [0-1]	eye
Smoker (Yes)	163	38 (23.3%)	patient
Surgeon: Surgeon A	163	70 (42.9%)	patient
Surgeon: Surgeon B	163	57 (35%)	patient
Surgeon: Surgeon C	163	36 (22.1%)	patient

Within-group only - Diabetes (n = 113 eyes; 72 patients; no between-group test)

Parameter	n	Summary	Analyzed by
Preop_Pachymetry_um	108	533.62 ± 25.04 (median 532 [518-550.25], range 470-592	eye
Preop_DryEye_Grade	113	1 [0-2]	eye
Smoker (Yes)	72	13 (18.1%)	patient
Surgeon: Surgeon A	72	23 (31.9%)	patient
Surgeon: Surgeon B	72	28 (38.9%)	patient
Surgeon: Surgeon C	72	21 (29.2%)	patient

Outcome Summary

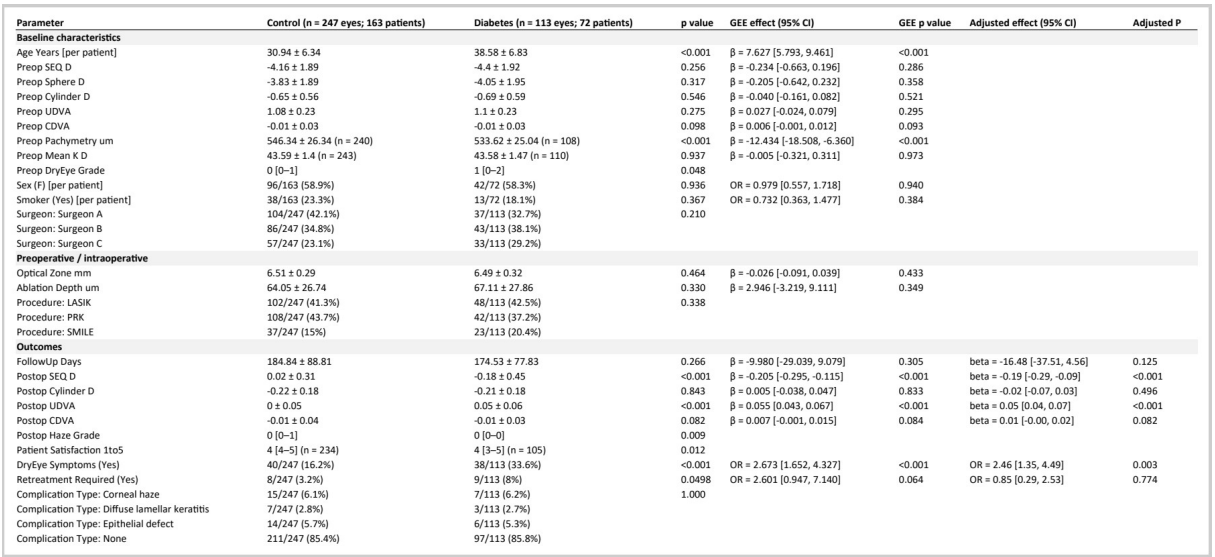
The same layout for the outcomes, with the adjusted effect and adjusted *p* merged in as the final two columns. Crude and adjusted estimates sit on one line, so the effect of adjustment is immediately visible. Postoperative SEQ survives it ($p < 0.001$ crude and adjusted), while the higher retreatment rate in the Diabetes group does not ($p = 0.0498$ crude, 0.774 adjusted). Note also that the complication-type row was switched automatically to **Fisher's exact test** because the expected counts were small.



Parameter	Control (n = 247 eyes; 163 patients)	Diabetes (n = 113 eyes; 72 patients)	p_value	Test	Practical Interpretation	GEE effect (95% CI)	GEE p value	Adjusted effect (95% CI)	Adjusted P
FollowUp_Days	184.84 ± 88.81	174.53 ± 77.83	0.266	Welch t-test	No significant difference between groups.	β = -9.980 [-29.039, 9.079]	0.305	beta = -16.48 [-37.51, 4.56]	0.125
Postop_SEQ_D	0.02 ± 0.31	-0.18 ± 0.45	<0.001	Welch t-test	Significant difference between groups.	β = -0.205 [-0.295, -0.115]	<0.001	beta = -0.19 [-0.29, -0.09]	<0.001
Postop_Cylinder_D	-0.22 ± 0.18	-0.21 ± 0.18	0.843	Welch t-test	No significant difference between groups.	β = 0.005 [-0.038, 0.047]	0.833	beta = -0.02 [-0.07, 0.03]	0.496
Postop_UDVA	0 ± 0.05	0.05 ± 0.06	<0.001	Welch t-test	Significant difference between groups.	β = 0.055 [0.043, 0.067]	<0.001	beta = 0.05 [0.04, 0.07]	<0.001
Postop_CDVA	-0.01 ± 0.04	-0.01 ± 0.03	0.082	Welch t-test	No significant difference between groups.	β = 0.007 [-0.001, 0.015]	0.084	beta = 0.01 [-0.00, 0.02]	0.082
Postop_Haze_Grade	0 [0–1]	0 [0–0]	0.009	Wilcoxon rank-sum test	Significant difference between groups.				
Patient_Satisfaction_1to5	4 [4–5] (n = 234)	4 [3–5] (n = 105)	0.012	Wilcoxon rank-sum test	Significant difference between groups.				
DryEye_Symptoms (Yes)	40/247 (16.2%)	38/113 (33.6%)	<0.001	Chi-squared test	Significant difference between groups.	OR = 2.673 [1.652, 4.327]	<0.001	OR = 2.46 [1.35, 4.49]	0.003
Retreatment_Required (Yes)	8/247 (3.2%)	9/113 (8%)	0.0498	Chi-squared test	Significant difference between groups.	OR = 2.601 [0.947, 7.140]	0.064	OR = 0.85 [0.29, 2.53]	0.774
Complication_Type: Corneal haze	15/247 (6.1%)	7/113 (6.2%)	1.000	Fisher's exact test	No significant difference between groups.				
Complication_Type: Diffuse lamellar keratitis	7/247 (2.8%)	3/113 (2.7%)							
Complication_Type: Epithelial defect	14/247 (5.7%)	6/113 (5.3%)							
Complication_Type: None	211/247 (85.4%)	97/113 (85.8%)							

Publication Table

A single manuscript-ready sheet: baseline, preoperative and outcome blocks stacked under section headers with group columns and *p* values, intended to be lifted into a manuscript with minimal reformatting. Because the Test and Interpretation columns are left out here, a test that could not be performed is printed as **NE** (not estimable) and explained in a footnote that names the variable and the reason (in the PRK-only run of *Example - 04_Subsets_PRK_LASIK*, for example, "variable is constant in this cohort: Procedure"), so it cannot be mistaken for a row that carries no *p* value by design (the second and later levels of a categorical variable). The table shown below, from the run of Part 3, has no NE cell.



Parameter	Control (n = 247 eyes; 163 patients)	Diabetes (n = 113 eyes; 72 patients)	p value	GEE effect (95% CI)	GEE p value	Adjusted effect (95% CI)	Adjusted P
Baseline characteristics							
Age Years [per patient]	30.94 ± 6.34	38.58 ± 6.83	<0.001	β = 7.627 [5.793, 9.461]	<0.001		
Preop SEQ D	-4.16 ± 1.89	-4.4 ± 1.92	0.256	β = -0.234 [-0.663, 0.196]	0.286		
Preop Sphere D	-3.83 ± 1.89	-4.05 ± 1.95	0.317	β = -0.205 [-0.642, 0.232]	0.358		
Preop Cylinder D	-0.65 ± 0.56	-0.69 ± 0.59	0.546	β = -0.040 [-0.161, 0.082]	0.521		
Preop UDVA	1.08 ± 0.23	1.1 ± 0.23	0.275	β = 0.027 [-0.024, 0.079]	0.295		
Preop CDVA	-0.01 ± 0.03	-0.01 ± 0.03	0.098	β = 0.006 [-0.001, 0.012]	0.093		
Preop Pachymetry um	546.34 ± 26.34 (n = 240)	533.62 ± 25.04 (n = 108)	<0.001	β = -12.434 [-18.508, -6.360]	<0.001		
Preop Mean K D	43.59 ± 1.4 (n = 243)	43.58 ± 1.47 (n = 110)	0.937	β = -0.005 [-0.321, 0.311]	0.973		
Preop DryEye Grade	0 [0–1]	1 [0–2]	0.048				
Sex (F) [per patient]	96/163 (58.9%)	42/72 (58.3%)	0.936	OR = 0.979 [0.557, 1.718]	0.940		
Smoker (Yes) [per patient]	38/163 (23.3%)	13/72 (18.1%)	0.367	OR = 0.732 [0.363, 1.477]	0.384		
Surgeon: Surgeon A	104/247 (42.1%)	37/113 (32.7%)	0.210				
Surgeon: Surgeon B	86/247 (34.8%)	43/113 (38.1%)					
Surgeon: Surgeon C	57/247 (23.1%)	33/113 (29.2%)					
Preoperative / intraoperative							
Optical Zone mm	6.51 ± 0.29	6.49 ± 0.32	0.464	β = -0.026 [-0.091, 0.039]	0.433		
Ablation Depth um	64.05 ± 26.74	67.11 ± 27.86	0.330	β = 2.946 [-3.219, 9.111]	0.349		
Procedure: LASIK	102/247 (41.3%)	48/113 (42.5%)	0.338				
Procedure: PRK	108/247 (43.7%)	42/113 (37.2%)					
Procedure: SMILE	37/247 (15%)	23/113 (20.4%)					
Outcomes							
FollowUp Days	184.84 ± 88.81	174.53 ± 77.83	0.266	β = -9.980 [-29.039, 9.079]	0.305	beta = -16.48 [-37.51, 4.56]	0.125
Postop SEQ D	0.02 ± 0.31	-0.18 ± 0.45	<0.001	β = -0.205 [-0.295, -0.115]	<0.001	beta = -0.19 [-0.29, -0.09]	<0.001
Postop Cylinder D	-0.22 ± 0.18	-0.21 ± 0.18	0.843	β = 0.005 [-0.038, 0.047]	0.833	beta = -0.02 [-0.07, 0.03]	0.496
Postop UDVA	0 ± 0.05	0.05 ± 0.06	<0.001	β = 0.055 [0.043, 0.067]	<0.001	beta = 0.05 [0.04, 0.07]	<0.001
Postop CDVA	-0.01 ± 0.04	-0.01 ± 0.03	0.082	β = 0.007 [-0.001, 0.015]	0.084	beta = 0.01 [-0.00, 0.02]	0.082
Postop Haze Grade	0 [0–1]	0 [0–0]	0.009				
Patient Satisfaction 1to5	4 [4–5] (n = 234)	4 [3–5] (n = 105)	0.012				
DryEye Symptoms (Yes)	40/247 (16.2%)	38/113 (33.6%)	<0.001	OR = 2.673 [1.652, 4.327]	<0.001	OR = 2.46 [1.35, 4.49]	0.003
Retreatment Required (Yes)	8/247 (3.2%)	9/113 (8%)	0.0498	OR = 2.601 [0.947, 7.140]	0.064	OR = 0.85 [0.29, 2.53]	0.774
Complication Type: Corneal haze	15/247 (6.1%)	7/113 (6.2%)	1.000				
Complication Type: Diffuse lamellar keratitis	7/247 (2.8%)	3/113 (2.7%)					
Complication Type: Epithelial defect	14/247 (5.7%)	6/113 (5.3%)					
Complication Type: None	211/247 (85.4%)	97/113 (85.8%)					

PSM sheet (one per pairing)

Everything about the matching in one place: the settings used, the covariates, the balance before and after (standardized mean differences for continuous and binary covariates alike) with the software's verdict against the 0.1 threshold, and the propensity score and matching weight of every matched row.

PSM settings

Setting	Value
Matching method	nearest
Distance	logit
Ratio	2
Replacement	FALSE
Caliper	0.2
Standardized caliper	TRUE
Matching order	largest
Random seed (matching)	20250601
Reference group	Control
Patient ID (clustering for GEE only; not used in matching)	PatientID

Matching covariates

PSM_Covariates
Age_Years
Preop_SEQ_D
Preop_Pachymetry_um
Preop_UDVA

Covariate balance (cobalt; standardized differences for every covariate)

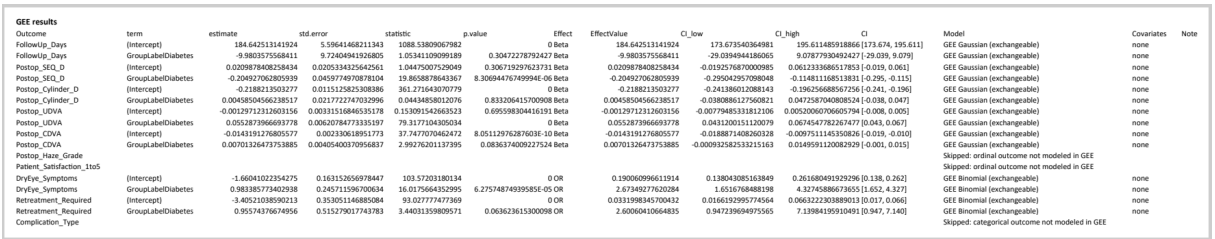
Variable	Type	Diff.Un	Diff.Adj	M.Threshold
distance	Distance	1.09600009126413	0.058299487318906	Balanced, <0.1
Age_Years	Contin.	1.15541607672935	0.0287498961353735	Balanced, <0.1
Preop_SEQ_D	Contin.	-0.167547509804362	-0.0106329885546305	Balanced, <0.1
Preop_Pachymetry_um	Contin.	-0.507811040132507	-0.0598968935902042	Balanced, <0.1
Preop_UDVA	Contin.	0.14446364942738	0.00674053736257519	Balanced, <0.1

Propensity-score distribution (matched rows)

PatientID	GroupLabel	PropensityScore	MatchingWeight
P0003	Diabetes	0.480645702059998	1
P0003	Diabetes	0.5535894377763	1

GEE sheets

The clustered models: for each outcome the term, estimate, robust standard error, test statistic, *p* value, the back-transformed effect, its confidence interval and the model family. Separate sheets are written for the unadjusted and adjusted models, and for the unmatched and matched cohorts: on the unmatched cohort one sheet per table section (`GEE_Main_Unmatched_Base`, `_Preop` and `_Out`, and the same three with `GEE_Adj_`), and on the matched cohort one sheet per pairing holding all three sections (`GEE_M_Diabetes_vs_Control`, `GEE_Adj_M_Diabetes_vs_Control`). A note row above the table says when the models are weighted by matching weights that are not all 1 ("working correlation: independence (weighted)"), counts the outcomes that could not be modelled — the reason is in the Model column; ordinal and categorical outcomes, which GEE does not model by design, are not counted — and points to the few-patient cautions in the Note column.



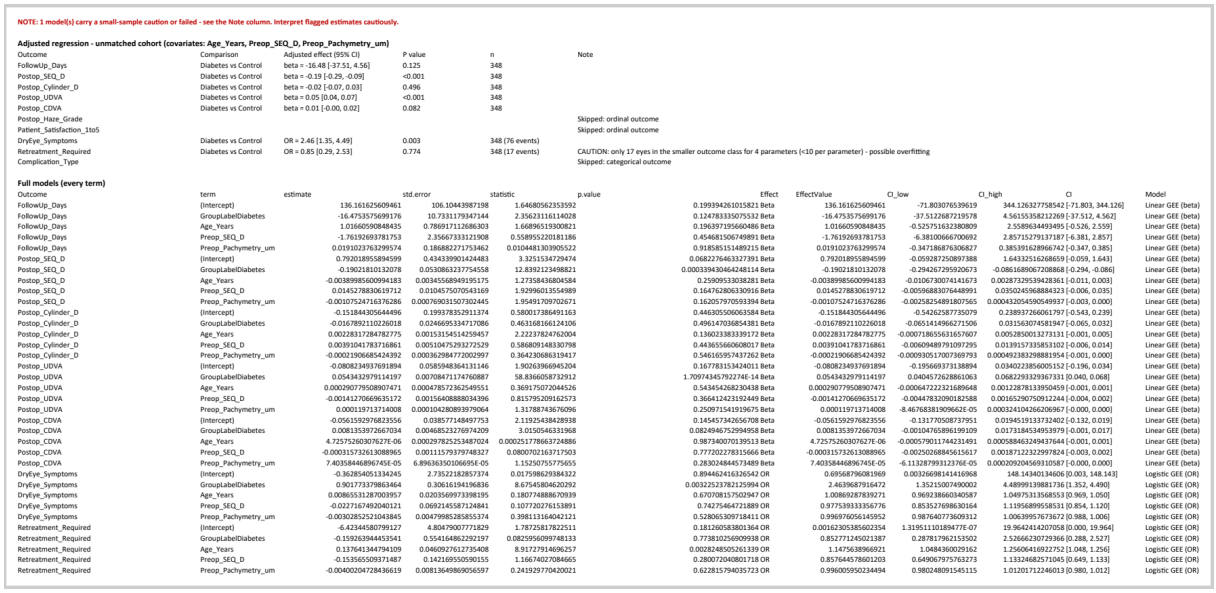
GEE results

Outcome	term	estimate	std.error	statistic	p.value	Effect	EffectValue	CI_low	CI_high	CI	Model	Covariates	Note
FollowUp_Days	(Intercept)	184.642513141924	5.5964146821343	1088.53809067982		0 Beta	184.642513141924	173.673540364981	195.611485918866	[173.674, 195.611]	GEE Gaussian (exchangeable)	none	
FollowUp_Days	GroupLabelDiabetes	-9.9803575568411	9.7240494192805	1.05341109099189	0.30472278792427	Beta	-9.9803575568411	-29.0394944186065	9.07877930492427	[-29.039, 9.079]	GEE Gaussian (exchangeable)	none	
Postop_SEQ_D	(Intercept)	0.0209878408258434	0.0205334325642561	1.04475007529049	0.306719297623731	Beta	0.0209878408258434	-0.0192576870000985	0.061233368651785 3	[-0.019, 0.061]	GEE Gaussian (exchangeable)	none	
Postop_SEQ_D	GroupLabelDiabetes	-0.204927062805939	0.0459774970878104	19.8658878643367	8.30694476749994E-06	Beta	-0.204927062805939	-0.295042957098048	-0.114811168513831	[-0.295, -0.115]	GEE Gaussian (exchangeable)	none	
Postop_Cylinder_D	(Intercept)	-0.218821350327 7	0.0115125825308886	361.271643070779		0 Beta	-0.218821350327 7	-0.241186012088143	-0.196256688567256	[-0.241, -0.196]	GEE Gaussian (exchangeable)	none	
Postop_Cylinder_D	GroupLabelDiabetes	0.00458504566238517	0.0217722747032996	0.044348580120 76	0.833206415700908	Beta	0.00458504566238517	-0.038088612756082 1	0.047258704080852 4	[-0.038, 0.047]	GEE Gaussian (exchangeable)	none	
Postop_UDVA	(Intercept)	-0.00129712312603156	0.0033151684653517 8	0.153091542663523	0.695598304416191	Beta	-0.00129712312603156	-0.007794853318121 06	0.005200607060579 4	[-0.008, 0.005]	GEE Gaussian (exchangeable)	none	
Postop_UDVA	GroupLabelDiabetes	0.0552873966693778	0.0062078477333519 7	79.3177104305034		0 Beta	0.0552873966693778	0.043120051122079	0.067454778226747 7	[0.043, 0.067]	GEE Gaussian (exchangeable)	none	
Postop_CDVA	(Intercept)	-0.0143191276805577	0.0023306189517 73	37.7477070462472	8.05112976287603E-10	Beta	-0.0143191276805577	-0.018887140826032 8	-0.009751114535082 6	[-0.019, -0.010]	GEE Gaussian (exchangeable)	none	
Postop_CDVA	GroupLabelDiabetes	0.00701326473753885	0.0040540037095683 7	2.99276201137395	0.0836374009227524	Beta	0.00701326473753885	-0.000932582533215163	0.014959112008292 9	[-0.001, 0.015]	GEE Gaussian (exchangeable)	none	
Postop_Haze_Grade											Skipped: ordinal outcome not modeled in GEE		
Patient_Satisfaction_1to5											Skipped: ordinal outcome not modeled in GEE		
DryEye_Symptoms	(Intercept)	-1.66041022354275	0.163152656978447	103.57203180134		0 OR	0.190060996611914	0.138043085163849	0.261680491929296	[0.138, 0.262]	GEE Binomial (exchangeable)	none	
DryEye_Symptoms	GroupLabelDiabetes	0.983385773402938	0.245711596700634	16.0175664352995	6.27574874939585E-05	OR	2.67349277620284	1.6516768488198	4.32745886673655	[1.652, 4.327]	GEE Binomial (exchangeable)	none	
Retreatment_Required	(Intercept)	-3.40521038590213	0.353051146885084	93.027777477369		0 OR	0.0331998345700432	0.016619295774564	0.066322230388901 3	[0.017, 0.066]	GEE Binomial (exchangeable)	none	
Retreatment_Required	GroupLabelDiabetes	0.95574376674956	0.515279017743783	3.44031359809571	0.0636236153000 98	OR	2.60060410664835	0.947239694975565	7.13984195910491	[0.947, 7.140]	GEE Binomial (exchangeable)	none	
Complication_Type											Skipped: categorical outcome not modeled in GEE		

Regression (adjusted) and full models

The publication-ready adjusted table — outcome, comparison, effect with 95% CI, *p*, n — followed by the complete model output. Cautions appear in the Note column: small events-per-variable, groups with few

patients, dropped constant covariates, outcomes below the minimum data of every clustered model (6 usable eyes and 3 patients, see 1.9) or without data in a group, or models that failed. The title names the covariates used, and any covariate not found in the file is listed separately.



NOTE: 1 model(s) carry a small-sample caution or failed - see the Note column. Interpret flagged estimates cautiously.

Adjusted regression - unmatched cohort (covariates: Age_Years, Preop_SEQ_D, Preop_Pachymetry_um)

Outcome	Comparison	Adjusted effect (95% CI)	P value	n	Note
FollowUp_Days	Diabetes vs Control	beta = -16.48 [-37.51, 4.56]	0.125	348	
Postop_SEQ_D	Diabetes vs Control	beta = -0.19 [-0.29, -0.09]	<0.001	348	
Postop_Cylinder_D	Diabetes vs Control	beta = -0.02 [-0.07, 0.03]	0.496	348	
Postop_UDVA	Diabetes vs Control	beta = 0.05 [0.04, 0.07]	<0.001	348	
Postop_CDVA	Diabetes vs Control	beta = 0.01 [-0.00, 0.02]	0.082	348	
Postop_Haze_Grade					Skipped: ordinal outcome
Patient_Satisfaction_1to5					Skipped: ordinal outcome
DryEye_Symptoms	Diabetes vs Control	OR = 2.46 [1.35, 4.49]	0.003	348 (76 events)	
Retreatment_Required	Diabetes vs Control	OR = 0.85 [0.29, 2.53]	0.774	348 (17 events)	CAUTION: only 17 eyes in the smaller outcome class for 4 parameters (<10 per parameter) - possible overfitting
Complication_Type					Skipped: categorical outcome

Full models (every term)

Outcome	term	estimate	std.error	statistic	p.value	Effect	EffectValue	CI_low	CI_high	CI	Model
FollowUp_Days	(Intercept)	136.161625609461	106.10443987198	1.646805623535 92		0.199394261015821 Beta	136.161625609461	-71.803076539619	344.126327758542	[-71.803, 344.126]	Linear GEE (beta)
FollowUp_Days	GroupLabelDiabetes	-16.4753575699176	10.7331179347144	2.356231166144028		0.124783335075532 Beta	-16.4753575699176	-37.5122687219578	4.56155358212269	[-37.512, 4.562]	Linear GEE (beta)
FollowUp_Days	Age_Years	1.01660590848435	0.786917112686303	1.668651930082 1		0.196397195660486 Beta	1.01660590848435	-0.525751632380809	2.55896244493495	[-0.526, 2.559]	Linear GEE (beta)
FollowUp_Days	Preop_SEQ_D	-1.76192693781753	2.356673331219 08	0.558955220181186		0.454681506749891 Beta	-1.76192693781753	-6.38100666700692	2.85715279137187	[-6.381, 2.857]	Linear GEE (beta)
FollowUp_Days	Preop_Pachymetry_um	0.0191023763299574	0.186882271753462	0.010448130390552 2		0.918585151489215 Beta	0.0191023763299574	-0.347186876306827	0.385391628966742	[-0.347, 0.385]	Linear GEE (beta)
Postop_SEQ_D	(Intercept)	0.792018955894599	0.434339901424483	3.3251534729474		0.068227646332739 1 Beta	0.792018955894599	-0.059287250897388	1.64332516268659	[-0.059, 1.643]	Linear GEE (beta)
Postop_SEQ_D	GroupLabelDiabetes	-0.190218101320 78	0.053086323775 4558	12.83921234988 21		0.000339430464248114 Beta	-0.190218101320 78	-0.294267295920673	-0.086168906720 8868	[-0.294, -0.086]	Linear GEE (beta)
Postop_SEQ_D	Age_Years	-0.00389985600994183	0.003455584945 95175	1.273584368045 84		0.25909513038281 Beta	-0.00389985600994183	-0.010673007414 1673	0.00287329539428361	[-0.011, 0.003]	Linear GEE (beta)
Postop_SEQ_D	Preop_SEQ_D	0.014527883061971 2	0.010457507054 3169	1.929960135549 89		0.164762806330916 Beta	0.014527883061971 2	-0.005968830764489 91	0.03502459688843 23	[-0.006, 0.035]	Linear GEE (beta)
Postop_SEQ_D	Preop_Pachymetry_um	-0.0010752471637628 6	0.000769031507302445	1.954917097026 71		0.162057907059339 4 Beta	-0.0010752471637628 6	-0.002582548918075 65	0.00043205459054 9937	[-0.003, 0.000]	Linear GEE (beta)
Postop_Cylinder_D	(Intercept)	-0.151844305644496	0.199378352911374	0.580017386491163		0.446305506063584 Beta	-0.151844305644496	-0.54262587735079	0.23893726061797	[-0.543, 0.239]	Linear GEE (beta)
Postop_Cylinder_D	GroupLabelDiabetes	-0.0167892110226018	0.024669533471086	0.463168166124106		0.496147036854381 Beta	-0.0167892110226018	-0.065141496627150 6	0.03156307458194 7	[-0.065, 0.032]	Linear GEE (beta)
Postop_Cylinder_D	Age_Years	0.0022831728478275	0.001531545142595 47	2.222378247620 04		0.136023383339172 Beta	0.0022831728478275	-0.000718655631657607	0.00528500132731 31	[-0.001, 0.005]	Linear GEE (beta)
Postop_Cylinder_D	Preop_SEQ_D	0.003910417837168 61	0.005104752932725 29	0.586809148330798		0.443655660608017 Beta	0.003910417837168 61	-0.006094897910972 95	0.01391573385331 02	[-0.006, 0.014]	Linear GEE (beta)
Postop_Cylinder_D	Preop_Pachymetry_um	-0.0002190668542439 2	0.000362984772002997	0.364230686319417		0.546165957437262 Beta	-0.0002190668542439 2	-0.000930517007369793	0.00049238329888195 4	[-0.001, 0.000]	Linear GEE (beta)
Postop_UDVA	(Intercept)	-0.0808234937691894	0.058594836411146	1.902639669452 04		0.167783153424011 Beta	-0.0808234937691894	-0.19566937313889 4	0.03402238560051 52	[-0.196, 0.034]	Linear GEE (beta)
Postop_UDVA	GroupLabelDiabetes	0.054343297911419 7	0.007084711747608 87	58.836605873291 2		1.709743457922746-14 Beta	0.054343297911419 7	0.040457262886106 3	0.06822933293673 31	[0.040, 0.068]	Linear GEE (beta)
Postop_UDVA	Age_Years	0.000290779508907471	0.000478577362549551	0.369175072044526		0.543454268230438 Beta	0.000290779508907471	-0.000647222321689648	0.00122878133950459	[-0.001, 0.001]	Linear GEE (beta)
Postop_UDVA	Preop_SEQ_D	-0.001412706696351 72	0.001564088880343 96	0.815795209162573		0.366412423192449 Beta	-0.001412706696351 72	-0.004478320901825 88	0.00165290750912 244	[-0.004, 0.002]	Linear GEE (beta)
Postop_UDVA	Preop_Pachymetry_um	0.000119713714008	0.000104280893979064	1.317887436760 96		0.250971541919675 Beta	0.000119713714008	-8.467683819096620E-05	0.00032410426206967	[-0.000, 0.000]	Linear GEE (beta)
Postop_CDVA	(Intercept)	-0.056159297682355 6	0.038577148497753	2.119254384289 38		0.145457342656708 Beta	-0.056159297682355 6	-0.13177050873795 1	0.01945191337324 02	[-0.132, 0.019]	Linear GEE (beta)
Postop_CDVA	GroupLabelDiabetes	0.008135397266703 4	0.004685232769742 09	3.015056433196 8		0.082494675299495 8 Beta	0.008135397266703 4	-0.001047658969 9109	0.01731845349539 79	[-0.001, 0.017]	Linear GEE (beta)
Postop_CDVA	Age_Years	4.725752603076 27E-06	0.000297825253487024	0.000251778663724886		0.987340070139513 Beta	4.725752603076 27E-06	-0.000579011744231491	0.00058846324934 7644	[-0.001, 0.001]	Linear GEE (beta)
Postop_CDVA	Preop_SEQ_D	-0.000315732613088965	0.001115793794832 7	0.080070216371503		0.777202278315666 Beta	-0.000315732613088965	-0.002502688456151 7	0.00187122322997 824	[-0.003, 0.002]	Linear GEE (beta)
Postop_CDVA	Preop_Pachymetry_um	7.403584468967 45E-05	6.896363501066 95E-05	1.152507557756 55		0.283024844573489 Beta	7.403584468967 45E-05	-6.113287993123 76E-05	0.00020920456931 0587	[-0.000, 0.000]	Linear GEE (beta)
DryEye_Symptoms	(Intercept)	-0.362854051334245	2.735221828573 74	0.017598629384322		0.894462416326542 OR	0.695687960819 69	0.003266981414169 68	148.14340134606 [0.003, 148.143]		Logistic GEE (OR)
DryEye_Symptoms	GroupLabelDiabetes	0.901773379863464	0.306161941968 36	8.675458046202 92		0.003225237821259 94 OR	2.463968791647 2	1.352150074900 02	4.48999139881736	[1.352, 4.490]	Logistic GEE (OR)
DryEye_Symptoms	Age_Years	0.008655312870039 57	0.020356997339819 5	0.180774888670939		0.670708157502947 OR	1.008692783927 1	0.969238660340587	1.04975313568553	[0.969, 1.050]	Logistic GEE (OR)
DryEye_Symptoms	Preop_SEQ_D	-0.022716749200401 21	0.069214558712484 1	0.107720276153891		0.747546472188 9 OR	0.977539333356776	0.853527698630164	1.11956899558531	[0.854, 1.120]	Logistic GEE (OR)
DryEye_Symptoms	Preop_Pachymetry_um	-0.003028525210438 45	0.004799852858553 74	0.398113164042121		0.528065309718411 OR	0.996976056145952	0.987640773609312	1.00639957673672	[0.988, 1.006]	Logistic GEE (OR)
Retreatment_Required	(Intercept)	-6.423445807991 27	4.804790077718 29	1.787258178225 11		0.181260583801364 OR	0.001623053856023 54	1.319511101894 77E-07	19.9642414207058	[0.000, 19.964]	Logistic GEE (OR)
Retreatment_Required	GroupLabelDiabetes	-0.159263944453541	0.554164862292197	0.082595609974813 3		0.773810256909938 OR	0.852771245021387	0.287817962153502	2.52666230729366	[0.288, 2.527]	Logistic GEE (OR)
Retreatment_Required	Age_Years	0.137641344794109	0.046092761273540 8	8.917279146962 57		0.002824850526133 9 OR	1.147563896692 1	1.048436002916 2	1.25606416922752	[1.048, 1.256]	Logistic GEE (OR)
Retreatment_Required	Preop_SEQ_D	-0.153565509371487	0.142169550590155	1.166740270866 65		0.280072040801718 OR	0.857644578601203	0.649067975763273	1.13324682571045	[0.649, 1.133]	Logistic GEE (OR)
Retreatment_Required	Preop_Pachymetry_um	-0.004002047284366 19	0.008136498690565 97	0.241929770420021		0.622815794035723 OR	0.996005950234494	0.980248091545115	1.01201712246013	[0.980, 1.012]	Logistic GEE (OR)

mEYEstro and AstigMATIC tabs

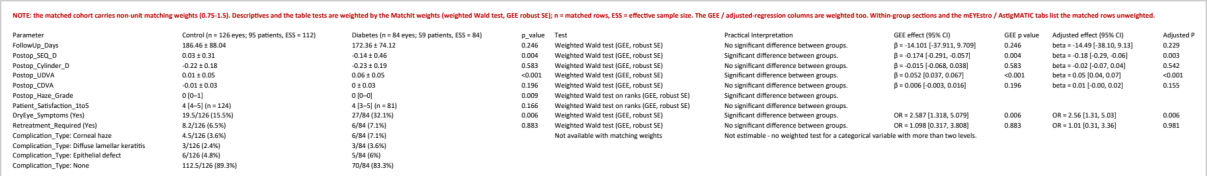
Worksheets laid out for direct entry into those tools, one sheet per group, for users who prefer to reproduce the figures in the original software. The values are already computed and standardized. The mEYEstro tab carries minus-cylinder refractions and decimal acuity with Snellen equivalents: columns E–S (preoperative refraction and CDVA, intended refraction, postoperative refraction, CDVA and UDVA, in the template's order) are copied into the mEYEstro template, and the grey columns A–D (group, decimal acuities) and the blue column T (postoperative SEQ) are for reference only. The AstigMATIC tab's columns A–E (eye as OD/OS, then preoperative and postoperative cylinder and axis) load as they are.

GroupLabel	Pre_BCVA	Post_UCVA	Post_BCVA	Pre_Sph	Pre_Cyl	Pre_Axis	Preop_MR_Vertex_Distance	Preop_CDVA	Intended_Postop_MR_Sphere	Intended_Postop_MR_Cylinder	Intended_Postop_MR_Axis	Intended_Postop_MR_Vertex_Distance	Post_Sph	Post_Cyl	Post_Axis	Postop_MR_Vertex_Distance	Postop_CDVA	Postop_UDVA	Post_SEQ
Control	1.04	1.03	1.01	-2.75	-1.5	161	12	20+	0	0	0	12	0	-0.25	167	12	20	20+	0
Control	0.97	0.78	0.93	-3.75	-2.25	35	12	20-1	0	0	0	12	0	0	40	12	20-2	25	0
Control	1.03	1	1.03	-3	-0.75	78	12	20+	0	0	0	12	0	-0.25	110	12	20+	20	0
Control	1.06	1.02	1.08	-0.75	0	80	12	20+	0	0	0	12	0.25	0	89	12	16-3	20	0.25
Control	1.2	1.25	1.3	-1.75	-1	103	12	15-2	0	0	0	12	0.25	-0.5	78	12	15	15-1	0
Control	1.09	1.16	1.08	-0.5	-0.25	3	12	16-3	0	0	0	12	0.5	-0.25	177	12	16-3	15-3	0.25
Control	1.1	1.09	1.12	-5.25	-1.25	103	12	16-3	0	0	0	12	0.5	-0.5	75	12	16-3	16-3	0.25
Control	0.93	0.93	0.95	-3.25	-1	24	12	20-2	0	0	0	12	0.5	-0.5	36	12	20-1	20-2	0.25
Control	0.96	0.94	0.95	-1	-0.5	101	12	20-1	0	0	0	12	0.5	-0.25	98	12	20-1	20-1	0.25
Control	0.95	0.9	1.01	-5.25	-1.25	79	12	20-1	0	0	0	12	0	-0.75	64	12	20	20-2	-0.25
Control	1.06	1.05	1.13	-2	-0.5	162	12	20+	0	0	0	12	0.5	-0.25	138	12	15-3	20+	0.25
Control	0.98	1.05	0.97	-4.5	-1.5	20	12	20	0	0	0	12	-0.25	0	24	12	20-1	20+	-0.25
Control	0.97	0.83	0.93	-5.5	-1	102	12	20-1	0	0	0	12	-0.25	0	88	12	20-2	25+	-0.25
Control	1.16	1	1.16	-1.5	-0.25	3	12	15-3	0	0	0	12	0.5	-0.25	152	12	15-3	20	0.5
Control	0.83	0.7	0.8	-7	0	8	12	25+	0	0	0	12	0	-0.25	75	12	25	25-3	-0.25
Control	1.13	1.05	1.14	-3.25	-1.5	92	12	15-3	0	0	0	12	0	-0.25	97	12	15-3	20+	0
Control	1.15	1.04	1.15	-5.75	-0.75	17	12	15-3	0	0	0	12	-0.5	-0.25	20	12	15-3	20+	-0.75
Control	1.06	1.01	1.09	-2.75	-0.25	7	12	20+	0	0	0	12	0	0	178	12	16-3	20	0
Control	1	0.92	0.95	-1.5	0	97	12	20	0	0	0	12	-0.5	-0.25	122	12	20-1	20-2	-0.5
Control	1.05	0.9	1.06	-6	-1.5	161	12	20+	0	0	0	12	0.5	-0.25	12	12	20+	20-2	0.25
Control	1.11	1.14	1.08	-4.25	-2.5	3	12	16-3	0	0	0	12	0	0	9	12	16-3	15-3	0
Control	1.1	1.09	1.12	-5.25	-0.5	13	12	16-3	0	0	0	12	0	-0.25	173	12	16-3	16-3	0
Control	1.07	0.97	1.1	-2.75	-0.25	178	12	20+	0	0	0	12	1	-0.25	162	12	16-3	20-1	0.75
Control	1.02	0.99	0.96	-3.75	0	159	12	20	0	0	0	12	0.5	0	159	12	20-1	20	0.5
Control	0.95	1.14	0.95	-1.75	0	161	12	20-1	0	0	0	12	0	-0.25	7	12	20-1	15-3	0
Control	0.97	1.06	0.98	-3.75	-1.75	15	12	20-1	0	0	0	12	0.75	-0.5	24	12	20	20+	0.5
Control	1.04	0.91	1.04	-5.25	0	3	12	20+	0	0	0	12	0.25	-0.5	91	12	20+	20-2	0
Control	1.13	1.18	1.16	-5.75	-1.25	100	12	15-3	0	0	0	12	0	-0.25	103	12	15-3	15-2	0
Control	0.92	0.87	0.92	-3.25	0	3	12	20-2	0	0	0	12	0	-0.25	140	12	20-2	20-3	0
Control	0.89	0.81	0.85	-2.5	0	1	12	20-3	0	0	0	12	0.5	-0.5	23	12	25+	25	0.25
Control	1.12	1.15	1.17	-0.75	0	163	12	16-3	0	0	0	12	0	-0.25	149	12	15-3	15-3	0
Control	0.97	1	0.95	-5	-1	159	12	20-1	0	0	0	12	0.25	-0.5	169	12	20-1	20	0
Control	1.02	1.2	1.05	-2.5	-0.5	168	12	20	0	0	0	12	0	-0.25	4	12	20+	15-2	0
Control	1.15	1.16	1.12	-2.5	-0.5	164	12	15-3	0	0	0	12	1	-0.25	170	12	16-3	15-3	0.75
Control	1.11	1.25	1.14	-2	-0.25	10	12	16-3	0	0	0	12	0	-0.5	28	12	15-3	15-1	-0.25
Control	1.01	0.91	1	-3.25	-0.25	47	12	20	0	0	0	12	0.5	-0.5	48	12	20	20-2	0.25
Control	1.01	1.14	0.98	-6.75	-0.5	7	12	20	0	0	0	12	0	-0.25	21	12	20	15-3	-0.25
Control	1.03	0.93	1.03	-7.25	-0.5	180	12	20+	0	0	0	12	0.5	-0.5	16	12	20+	20-2	0.25
Control	1.13	0.9	1.12	-2	-0.75	85	12	15-3	0	0	0	12	0.5	0	137	12	16-3	20-2	0.5
Control	0.98	0.86	0.95	-7.25	-0.75	14	12	20	0	0	0	12	0	-0.5	167	12	20-1	20-3	-0.25
Control	0.88	0.83	0.82	-4.5	-1.25	75	12	20-3	0	0	0	12	0.25	-0.5	50	12	25+	25+	0
Control	1.11	1.08	1.14	-4.75	-1.25	17	12	16-3	0	0	0	12	0	0	18	12	15-3	16-3	0
Control	1.1	1.22	1.12	-4.25	-2.75	170	12	16-3	0	0	0	12	0.25	-0.5	13	12	16-3	15-2	0
Control	0.99	0.92	0.98	-0.5	-0.5	23	12	20	0	0	0	12	0.5	-0.25	49	12	20	20-2	0.5
Control	1.08	1.15	1.04	-4.75	-0.25	172	12	16-3	0	0	0	12	0.5	-0.25	15	12	20+	15-3	0.25
Control	1.06	1.09	1.13	-3.25	-0.25	97	12	20+	0	0	0	12	0.5	-0.25	122	12	15-3	16-3	0.25
Control	1.02	0.94	1.02	-2.75	-2.25	7	12	20	0	0	0	12	-0.5	-0.25	153	12	20	20-1	-0.5
Control	1.06	1.11	1.06	-3	-0.75	165	12	20+	0	0	0	12	0	-0.5	12	12	20+	16-3	-0.25
Control	1.02	0.96	0.96	-4.25	-0.5	76	12	20	0	0	0	12	0	-0.25	62	12	20-1	20-1	0
Control	0.9	0.91	0.91	-3.5	0	170	12	20-2	0	0	0	12	0	-0.25	165	12	20-2	20-2	0
Control	1.06	0.95	1.06	-3.25	-1.25	84	12	20+	0	0	0	12	0.5	-0.25	62	12	20+	20-1	0.5
Control	1.11	1.06	1.09	-6.75	-0.75	81	12	16-3	0	0	0	12	-0.5	-0.25	34	12	16-3	20+	-0.5
Control	1.01	1.07	1.02	-2.5	-0.5	149	12	20	0	0	0	12	0	-0.25	161	12	20	20+	0
Control	0.92	0.75	0.93	-8.25	-0.75	172	12	20-2	0	0	0	12	-0.5	-0.25	171	12	20-2	25-1	-0.5
Control	0.99	0.96	1.01	-5	-0.25	161	12	20	0	0	0	12	0	0	80	12	20	20-1	0
Control	0.91	0.93	0.9	-1.25	-0.25	166	12	20-2	0	0	0	12	0	0	141	12	20-2	20-2	0
Control	1.01	0.95	1.01	-7.75	0	168	12	20	0	0	0	12	0.25	0	12	12	20	20-1	0.25
Control	0.93	0.96	0.94	-5.75	-1	83	12	20-2	0	0	0	12	0	-0.25	78	12	20-1	20-1	0
Control	0.91	0.89	0.88	-4	0	100	12	20-2	0	0	0	12	0	0	124	12	20-3	20-3	0
Control	1.04	1.09	1.03	-3.25	-0.5	82	12	20+	0	0	0	12	0.5	-0.5	86	12	20+	16-3	0.25
Control	1.03	1.08	1.06	-1	-0.5	80	12	20+	0	0	0	12	0	0	11	12	20+	16-3	0
Control	1.04	1.1	1.06	-1.25	-1	180	12	20+	0	0	0	12	-0.5	0	178	12	20+	16-3	-0.5
Control	0.98	1.08	1.02	-0.5	-0.75	7	12	20	0	0	0	12	0	-0.25	24	12	20	16-3	0
Control	1.19	1.16	1.23	-2	0	170	12	15-2	0	0	0	12	0.25	-0.5	23	12	15-2	15-3	0
Control	1.07	1.09	1.1	-4.25	-0.25	175	12	20+	0	0	0	12	0	-0.25	23	12	16-3	16-3	0
Control	1.05	0.96	1.02	-4.5	-0.25	165	12	20+	0	0	0	12	0.5	-0.25	164	12	20	20-1	0.25
Control	1.13	1.05	1.14	-6	-0.5	86	12	15-3	0	0	0	12	0	-0.25	106	12	15-3	20+	-0.25
Control	0.99	1.09	1.05	-5.75	-0.75	171	12	20	0	0	0	12	0.25	-0.5	129	12	20+	16-3	0
Control	1	1.07	0.97	-6.25	-0.25	91	12	20	0	0	0	12	-0.25	0	164	12	20-1	20+	-0.25
Control	0.93	0.87	0.94	-5	-0.75	88	12	20-2	0	0	0	12	0.5	-0.5	148	12	20-1	20-3	0.25
Control	1.06	1.13	1.05	-1.25	-0.75	99	12	20+	0	0	0	12	-0.5	-0.25	75	12	20+	15-3	-0.5
Control	0.86	0.83	0.84	-3.75	-1.25	23	12	20-3	0	0	0	12	0.5	-0.75	166	12	25+	25+	0
Control	0.99	0.82	0.97	-4.75	0	5	12	20	0	0	0	12	0.5	0	177	12	20-1	25+	0.5
Control	0.94	0.98	1	-0.75	-0.5	16	12	20-1	0	0	0	12	-0.25	0	128	12	20	20	-0.25
Control	0.96	0.98	0.97	-6.25	-0.25	67	12	20-1	0	0	0	12	0.5	-0.25	8	12	20-1	20	0.5
Control	1.07	1.11	1.12	-2.5	-0.75	100	12	20+	0	0	0	12	-0.25	0	86	12	16-3	16-3	-0.25
Control	0.99	1.01	1	-4	-0.25	89	12	20	0	0	0	12	0	-0.25	106	12	20	20	0
Control	1.1	1.13	1.14	-5.25	-1.5	96	12	16-3	0	0	0	12	-0.5	0	21	12	15-3	15-3	-0.5

Eye	preoperative cylinder	preoperative axis	postoperative cylinder	postoperative axis
OS	-1.5	161	-0.25	167
OS	-2.25	35	0	40
OD	-0.75	78	-0.25	110
OS	0	80	0	89
OD	-1	103	-0.5	78
OS	-0.25	3	-0.25	177
OD	-1.25	103	-0.5	75
OS	-1	24	-0.5	36
OD	-0.5	101	-0.25	98
OS	-1.25	79	-0.75	64
OS	-0.5	162	-0.25	138
OD	-1.5	20	0	24
OS	-1	102	0	88
OD	-0.25	3	-0.25	152
OS	0	8	-0.25	75
OD	-1.5	92	-0.25	97
OS	-0.75	17	-0.25	20
OD	-0.25	7	0	178
OS	0	97	-0.25	122
OD	-1.5	161	-0.25	12
OS	-2.5	3	0	9
OS	-0.5	13	-0.25	173
OD	-0.25	178	-0.25	162
OS	0	159	0	159
OD	0	161	-0.25	7
OS	-1.75	15	-0.5	24
OD	0	3	-0.5	91
OS	-1.25	100	-0.25	103
OS	0	3	-0.25	140
OD	0	1	-0.5	23
OS	0	163	-0.25	149

Matched-cohort sheets

The summary, publication, GEE, regression, mEYEstro and AstigMATIC sheets are repeated for each matched pairing, with `_M_` and the pairing in the sheet name (`BasePreop_M_Diabetes_vs_Control`, `Out_M_…`, `Pub_M_…`, `GEE_M_…`, `GEE_Adj_M_…`, `Reg_M_…`, `mEYEstro_M_…`, `AstigMATIC_M_…`) and the matched group sizes in the headers. When the matching weights are not all 1, a note row above each matched table says that its cells and tests are weighted, and the headers add the effective sample size (see 1.8). Comparing the unmatched and matched outcome sheets side by side is the clearest way to see what matching changed.

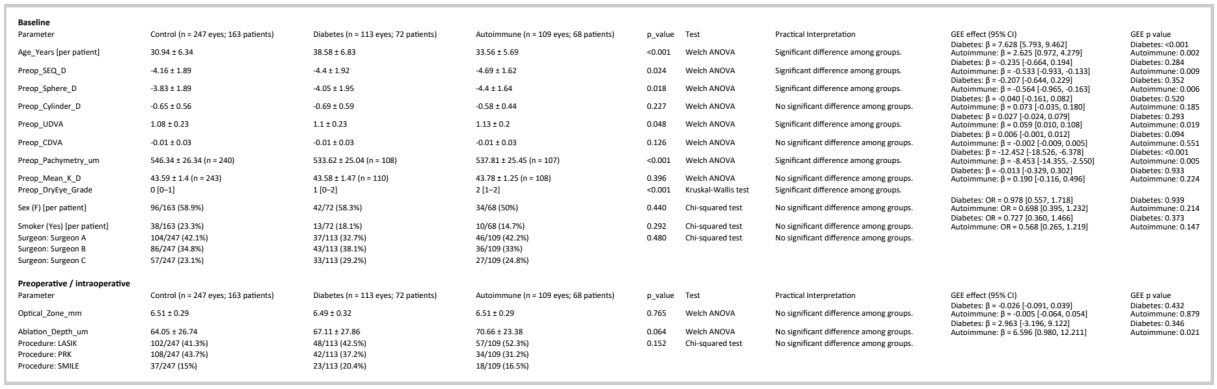


NOTE: the matched cohort carries non-unit matching weights (0.75-1.5). Descriptives and the table tests are weighted by the Matchit weights (weighted Wald test, GEE robust SE); n = matched rows, ESS = effective sample size. The GEE / adjusted-regression columns are weighted too. Within-group sections and the mEYEstro / AstigMATIC tabs list the matched rows unweighted.

Parameter	Control (n = 126 eyes; 95 patients, ESS = 112)	Diabetes (n = 84 eyes; 59 patients, ESS = 84)	p_value	Test	Practical Interpretation	GEE effect (95% CI)	GEE p value	Adjusted effect (95% CI)	Adjusted P
FollowUp_Days	186.46 ± 88.04	172.36 ± 74.12	0.246	Weighted Wald test (GEE, robust SE)	No significant difference between groups.	β = -14.101 [-37.911, 9.709]	0.246	beta = -14.49 [-38.10, 9.13]	0.229
Postop_SEQ_D	0.03 ± 0.31	-0.14 ± 0.46	0.004	Weighted Wald test (GEE, robust SE)	Significant difference between groups.	β = -0.174 [-0.291, -0.057]	0.004	beta = -0.18 [-0.29, -0.06]	0.003
Postop_Cylinder_D	-0.22 ± 0.18	-0.23 ± 0.19	0.583	Weighted Wald test (GEE, robust SE)	No significant difference between groups.	β = -0.015 [-0.068, 0.038]	0.583	beta = -0.02 [-0.07, 0.04]	0.542
Postop_UDVA	0.01 ± 0.05	0.06 ± 0.05	<0.001	Weighted Wald test (GEE, robust SE)	Significant difference between groups.	β = 0.052 [0.037, 0.067]	<0.001	beta = 0.05 [0.04, 0.07]	<0.001
Postop_CDVA	-0.01 ± 0.03	0 ± 0.03	0.196	Weighted Wald test (GEE, robust SE)	No significant difference between groups.	β = 0.006 [-0.003, 0.016]	0.196	beta = 0.01 [-0.00, 0.02]	0.155
Postop_Haze_Grade	0 [0–1]	0 [0–0]	0.009	Weighted Wald test on ranks (GEE, robust SE)	Significant difference between groups.				
Patient_Satisfaction_1to5	4 [4–5] (n = 124)	4 [3–5] (n = 81)	0.166	Weighted Wald test on ranks (GEE, robust SE)	No significant difference between groups.				
DryEye_Symptoms (Yes)	19.5/126 (15.5%)	27/84 (32.1%)	0.006	Weighted Wald test (GEE, robust SE)	Significant difference between groups.	OR = 2.587 [1.318, 5.079]	0.006	OR = 2.56 [1.31, 5.03]	0.006
Retreatment_Required (Yes)	8.2/126 (6.5%)	6/84 (7.1%)	0.883	Weighted Wald test (GEE, robust SE)	No significant difference between groups.	OR = 1.098 [0.317, 3.808]	0.883	OR = 1.01 [0.31, 3.36]	0.981
Complication_Type: Corneal haze	4.5/126 (3.6%)	6/84 (7.1%)		Not available with matching weights	Not estimable - no weighted test for a categorical variable with more than two levels.				
Complication_Type: Diffuse lamellar keratitis	3/126 (2.4%)	3/84 (3.6%)							
Complication_Type: Epithelial defect	6/126 (4.8%)	5/84 (6%)							
Complication_Type: None	112.5/126 (89.3%)	70/84 (83.3%)							

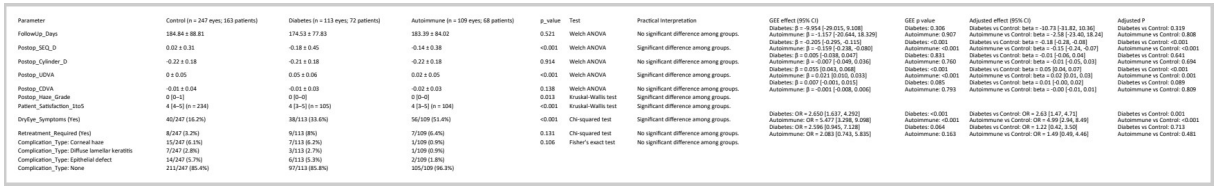
5.3 What changes with three cohorts

The baseline sheet now carries three group columns and a single **omnibus** *p* value per variable — Welch ANOVA for continuous variables, Kruskal–Wallis for ordinal ones — and the interpretation column reads "among groups" rather than "between groups".

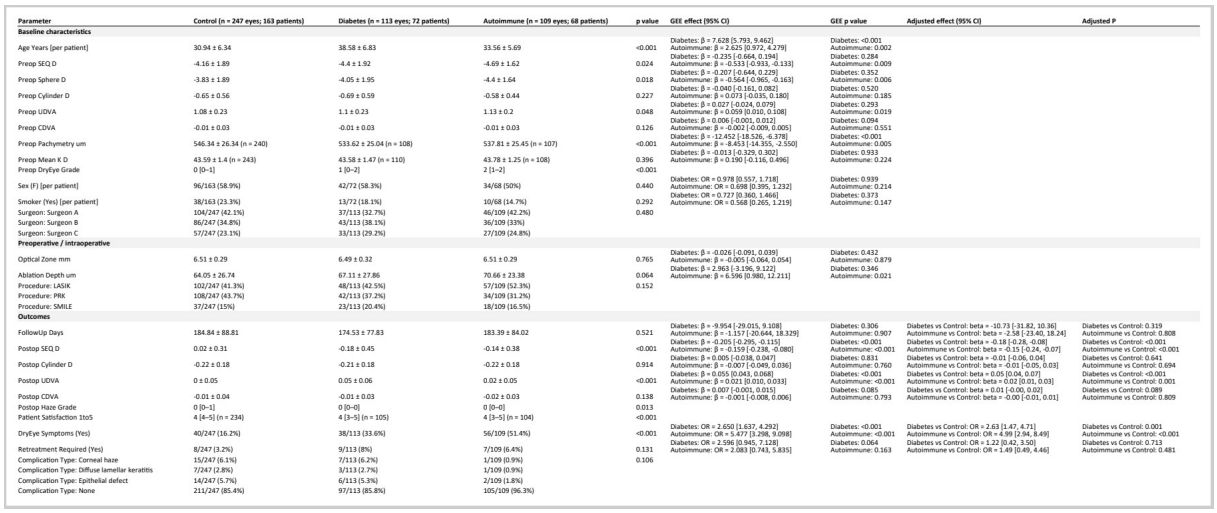


Baseline								
Parameter	Control (n = 247 eyes; 163 patients)	Diabetes (n = 113 eyes; 72 patients)	Autoimmune (n = 109 eyes; 68 patients)	p_value	Test	Practical Interpretation	GEE effect (95% CI)	GEE p value
Age_Years [per patient]	30.94 ± 6.34	38.58 ± 6.83	33.56 ± 5.69	<0.001	Welch ANOVA	Significant difference among groups.	Diabetes: β = 7.628 [5.793, 9.462] Autoimmune: β = 2.625 [0.972, 4.279]	Diabetes: <0.001 Autoimmune: 0.002
Preop_SEQ_D	-4.16 ± 1.89	-4.4 ± 1.92	-4.69 ± 1.62	0.024	Welch ANOVA	Significant difference among groups.	Diabetes: β = -0.235 [-0.664, 0.194] Autoimmune: β = -0.533 [-0.933, -0.133]	Diabetes: 0.284 Autoimmune: 0.009
Preop_Sphere_D	-3.83 ± 1.89	-4.05 ± 1.95	-4.4 ± 1.64	0.018	Welch ANOVA	Significant difference among groups.	Diabetes: β = -0.207 [-0.644, 0.229] Autoimmune: β = -0.564 [-0.965, -0.163]	Diabetes: 0.352 Autoimmune: 0.006
Preop_Cylinder_D	-0.65 ± 0.56	-0.69 ± 0.59	-0.58 ± 0.44	0.227	Welch ANOVA	No significant difference among groups.	Diabetes: β = -0.040 [-0.161, 0.082] Autoimmune: β = 0.073 [-0.035, 0.180]	Diabetes: 0.520 Autoimmune: 0.185
Preop_UDVA	1.08 ± 0.23	1.1 ± 0.23	1.13 ± 0.2	0.048	Welch ANOVA	Significant difference among groups.	Diabetes: β = 0.027 [-0.024, 0.079] Autoimmune: β = 0.059 [0.010, 0.108]	Diabetes: 0.293 Autoimmune: 0.019
Preop_CDVA	-0.01 ± 0.03	-0.01 ± 0.03	-0.01 ± 0.03	0.126	Welch ANOVA	No significant difference among groups.	Diabetes: β = 0.006 [-0.001, 0.012] Autoimmune: β = -0.002 [-0.009, 0.005]	Diabetes: 0.094 Autoimmune: 0.551
Preop_Pachymetry_um	546.34 ± 26.34 (n = 240)	533.62 ± 25.04 (n = 108)	537.81 ± 25.45 (n = 107)	<0.001	Welch ANOVA	Significant difference among groups.	Diabetes: β = -12.452 [-18.526, -6.378] Autoimmune: β = -8.453 [-14.355, -2.550]	Diabetes: <0.001 Autoimmune: 0.005
Preop_Mean_K_D	43.59 ± 1.4 (n = 243)	43.58 ± 1.47 (n = 110)	43.78 ± 1.25 (n = 108)	0.396	Welch ANOVA	No significant difference among groups.	Diabetes: β = -0.013 [-0.329, 0.302] Autoimmune: β = 0.190 [-0.116, 0.496]	Diabetes: 0.933 Autoimmune: 0.224
Preop_DryEye_Grade	0 [0–1]	1 [0–2]	2 [1–2]	<0.001	Kruskal-Wallis test	Significant difference among groups.		
Sex (F) [per patient]	96/163 (58.9%)	42/72 (58.3%)	34/68 (50%)	0.440	Chi-squared test	No significant difference among groups.	Diabetes: OR = 0.978 [0.557, 1.718] Autoimmune: OR = 0.698 [0.395, 1.232]	Diabetes: 0.939 Autoimmune: 0.214
Smoker (Yes) [per patient]	38/163 (23.3%)	13/72 (18.1%)	10/68 (14.7%)	0.292	Chi-squared test	No significant difference among groups.	Diabetes: OR = 0.727 [0.360, 1.466] Autoimmune: OR = 0.568 [0.265, 1.219]	Diabetes: 0.373 Autoimmune: 0.147
Surgeon: Surgeon A	104/247 (42.1%)	37/113 (32.7%)	46/109 (42.2%)	0.480	Chi-squared test	No significant difference among groups.		
Surgeon: Surgeon B	86/247 (34.8%)	43/113 (38.1%)	36/109 (33%)					
Surgeon: Surgeon C	57/247 (23.1%)	33/113 (29.2%)	27/109 (24.8%)					
Preoperative / intraoperative								
Parameter	Control (n = 247 eyes; 163 patients)	Diabetes (n = 113 eyes; 72 patients)	Autoimmune (n = 109 eyes; 68 patients)	p_value	Test	Practical Interpretation	GEE effect (95% CI)	GEE p value
Optical_Zone_mm	6.51 ± 0.29	6.49 ± 0.32	6.51 ± 0.29	0.765	Welch ANOVA	No significant difference among groups.	Diabetes: β = -0.026 [-0.091, 0.039] Autoimmune: β = -0.005 [-0.064, 0.054]	Diabetes: 0.432 Autoimmune: 0.879
Ablation_Depth_um	64.05 ± 26.74	67.11 ± 27.86	70.66 ± 23.38	0.064	Welch ANOVA	No significant difference among groups.	Diabetes: β = 2.963 [-3.196, 9.122] Autoimmune: β = 6.596 [0.980, 12.211]	Diabetes: 0.346 Autoimmune: 0.021
Procedure: LASIK	102/247 (41.3%)	48/113 (42.5%)	57/109 (52.3%)	0.152	Chi-squared test	No significant difference among groups.		
Procedure: PRK	108/247 (43.7%)	42/113 (37.2%)	34/109 (31.2%)					
Procedure: SMILE	37/247 (15%)	23/113 (20.4%)	18/109 (16.5%)					

In the outcome sheet the omnibus *p* is still a single value, but the adjusted-effect cell now contains **one line per comparison against the reference** — Diabetes vs Control and Autoimmune vs Control — each with its own estimate, confidence interval and *p*. This is where the information about *which* groups differ comes from.

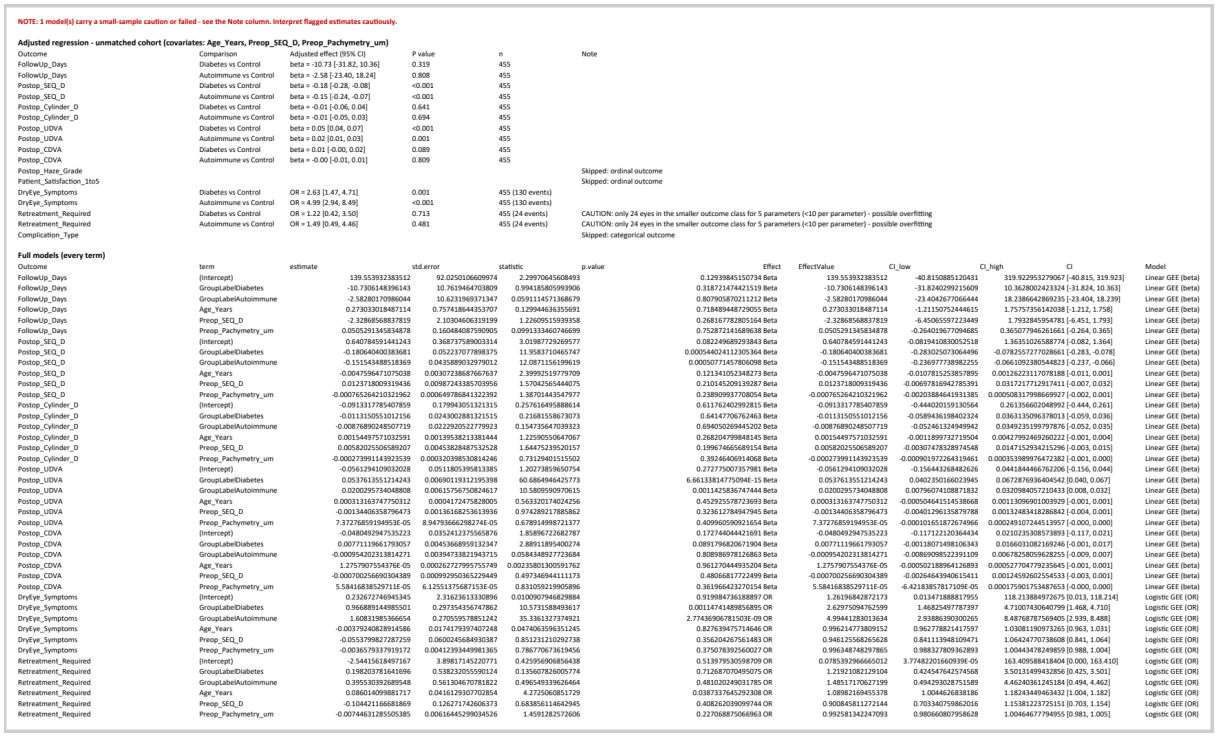


Parameter	Control (n = 247 eyes; 163 patients)	Diabetes (n = 113 eyes; 72 patients)	Autoimmune (n = 109 eyes; 68 patients)	p_value	Test	Practical Interpretation	GEE effect (95% CI)	GEE p value	Adjusted effect (95% CI)	Adjusted P
FollowUp_Days	184.84 ± 88.81	174.53 ± 77.83	183.39 ± 84.02	0.521	Welch ANOVA	No significant difference among groups.	Diabetes: β = -9.954 [-29.015, 9.108] Autoimmune: β = -1.157 [-20.644, 18.329]	Diabetes: 0.306 Autoimmune: 0.907	Diabetes vs Control: beta = -10.73 [-31.82, 10.36] Autoimmune vs Control: beta = -2.58 [-23.40, 18.24]	Diabetes vs Control: 0.319 Autoimmune vs Control: 0.808
Postop_SEQ_D	0.02 ± 0.31	-0.18 ± 0.45	-0.14 ± 0.38	<0.001	Welch ANOVA	Significant difference among groups.	Diabetes: β = -0.205 [-0.295, -0.115] Autoimmune: β = -0.159 [-0.238, -0.080]	Diabetes: <0.001 Autoimmune: <0.001	Diabetes vs Control: beta = -0.18 [-0.28, -0.08] Autoimmune vs Control: beta = -0.15 [-0.24, -0.07]	Diabetes vs Control: <0.001 Autoimmune vs Control: <0.001
Postop_Cylinder_D	-0.22 ± 0.18	-0.21 ± 0.18	-0.22 ± 0.18	0.914	Welch ANOVA	No significant difference among groups.	Diabetes: β = 0.005 [-0.038, 0.047] Autoimmune: β = -0.007 [-0.049, 0.036]	Diabetes: 0.831 Autoimmune: 0.760	Diabetes vs Control: beta = -0.01 [-0.06, 0.04] Autoimmune vs Control: beta = -0.01 [-0.05, 0.03]	Diabetes vs Control: 0.641 Autoimmune vs Control: 0.694
Postop_UDVA	0 ± 0.05	0.05 ± 0.06	0.02 ± 0.05	<0.001	Welch ANOVA	Significant difference among groups.	Diabetes: β = 0.055 [0.043, 0.068] Autoimmune: β = 0.021 [0.010, 0.033]	Diabetes: <0.001 Autoimmune: <0.001	Diabetes vs Control: beta = 0.05 [0.04, 0.07] Autoimmune vs Control: beta = 0.02 [0.01, 0.03]	Diabetes vs Control: <0.001 Autoimmune vs Control: 0.001
Postop_CDVA	-0.01 ± 0.04	-0.01 ± 0.03	-0.02 ± 0.03	0.138	Welch ANOVA	No significant difference among groups.	Diabetes: β = 0.007 [-0.001, 0.015] Autoimmune: β = -0.001 [-0.008, 0.006]	Diabetes: 0.085 Autoimmune: 0.793	Diabetes vs Control: beta = 0.01 [-0.00, 0.02] Autoimmune vs Control: beta = -0.00 [-0.01, 0.01]	Diabetes vs Control: 0.089 Autoimmune vs Control: 0.809
Postop_Haze_Grade	0 [0–1]	0 [0–0]	0 [0–0]	0.013	Kruskal-Wallis test	Significant difference among groups.				
Patient_Satisfaction_1to5	4 [4–5] (n = 234)	4 [3–5] (n = 105)	4 [3–5] (n = 104)	<0.001	Kruskal-Wallis test	Significant difference among groups.				
DryEye_Symptoms (Yes)	40/247 (16.2%)	38/113 (33.6%)	56/109 (51.4%)	<0.001	Chi-squared test	Significant difference among groups.	Diabetes: OR = 2.650 [1.637, 4.292] Autoimmune: OR = 5.477 [3.298, 9.098]	Diabetes: <0.001 Autoimmune: <0.001	Diabetes vs Control: OR = 2.63 [1.47, 4.71] Autoimmune vs Control: OR = 4.99 [2.94, 8.49]	Diabetes vs Control: 0.001 Autoimmune vs Control: <0.001
Retreatment_Required (Yes)	8/247 (3.2%)	9/113 (8%)	7/109 (6.4%)	0.131	Chi-squared test	No significant difference among groups.	Diabetes: OR = 2.596 [0.945, 7.128] Autoimmune: OR = 2.083 [0.743, 5.835]	Diabetes: 0.064 Autoimmune: 0.163	Diabetes vs Control: OR = 1.22 [0.42, 3.50] Autoimmune vs Control: OR = 1.49 [0.49, 4.46]	Diabetes vs Control: 0.713 Autoimmune vs Control: 0.481
Complication_Type: Corneal haze	15/247 (6.1%)	7/113 (6.2%)	1/109 (0.9%)	0.106	Fisher's exact test	No significant difference among groups.				
Complication_Type: Diffuse lamellar keratitis	7/247 (2.8%)	3/113 (2.7%)	1/109 (0.9%)							
Complication_Type: Epithelial defect	14/247 (5.7%)	6/113 (5.3%)	2/109 (1.8%)							
Complication_Type: None	211/247 (85.4%)	97/113 (85.8%)	105/109 (96.3%)							



Parameter	**Control (n = 247 eyes; 163 patients)**	**Diabetes (n = 113 eyes; 72 patients)**	**Autoimmune (n = 109 eyes; 68 patients)**	**p value**	**GEE effect (95% CI)**	**GEE p value**	**Adjusted effect (95% CI)**	**Adjusted P**
Baseline characteristics								
Age Years [per patient]	30.94 ± 6.34	38.58 ± 6.83	33.56 ± 5.69	<0.001	Diabetes: β = 7.628 [5.793, 9.462] Autoimmune: β = 2.625 [0.972, 4.279]	Diabetes: <0.001 Autoimmune: 0.002		
Preop SEQ D	-4.16 ± 1.89	-4.4 ± 1.92	-4.69 ± 1.62	0.024	Diabetes: β = -0.235 [-0.664, 0.194] Autoimmune: β = -0.533 [-0.933, -0.133]	Diabetes: 0.284 Autoimmune: 0.009		
Preop Sphere D	-3.83 ± 1.89	-4.05 ± 1.95	-4.4 ± 1.64	0.018	Diabetes: β = -0.207 [-0.644, 0.229] Autoimmune: β = -0.564 [-0.965, -0.163]	Diabetes: 0.352 Autoimmune: 0.006		
Preop Cylinder D	-0.65 ± 0.56	-0.69 ± 0.59	-0.58 ± 0.44	0.227	Diabetes: β = -0.040 [-0.161, 0.082] Autoimmune: β = 0.073 [-0.035, 0.180]	Diabetes: 0.520 Autoimmune: 0.185		
Preop UDVA	1.08 ± 0.23	1.1 ± 0.23	1.13 ± 0.2	0.048	Diabetes: β = 0.027 [-0.024, 0.079] Autoimmune: β = 0.059 [0.010, 0.108]	Diabetes: 0.293 Autoimmune: 0.019		
Preop CDVA	-0.01 ± 0.03	-0.01 ± 0.03	-0.01 ± 0.03	0.126	Diabetes: β = 0.006 [-0.001, 0.012] Autoimmune: β = -0.002 [-0.009, 0.005]	Diabetes: 0.094 Autoimmune: 0.551		
Preop Pachymetry um	546.34 ± 26.34 (n = 240)	533.62 ± 25.04 (n = 108)	537.81 ± 25.45 (n = 107)	<0.001	Diabetes: β = -12.452 [-18.526, -6.378] Autoimmune: β = -8.453 [-14.355, -2.550]	Diabetes: <0.001 Autoimmune: 0.005		
Preop Mean K D	43.59 ± 1.4 (n = 243)	43.58 ± 1.47 (n = 110)	43.78 ± 1.25 (n = 108)	0.396	Diabetes: β = -0.013 [-0.329, 0.302] Autoimmune: β = 0.190 [-0.116, 0.496]	Diabetes: 0.933 Autoimmune: 0.224		
Preop DryEye Grade	0 [0–1]	1 [0–2]	2 [1–2]	<0.001				
Sex (F) [per patient]	96/163 (58.9%)	42/72 (58.3%)	34/68 (50%)	0.440	Diabetes: OR = 0.978 [0.557, 1.718] Autoimmune: OR = 0.698 [0.395, 1.232]	Diabetes: 0.939 Autoimmune: 0.214		
Smoker (Yes) [per patient]	38/163 (23.3%)	13/72 (18.1%)	10/68 (14.7%)	0.292	Diabetes: OR = 0.727 [0.360, 1.466] Autoimmune: OR = 0.568 [0.265, 1.219]	Diabetes: 0.373 Autoimmune: 0.147		
Surgeon: Surgeon A	104/247 (42.1%)	37/113 (32.7%)	46/109 (42.2%)	0.480				
Surgeon: Surgeon B	86/247 (34.8%)	43/113 (38.1%)	36/109 (33%)					
Surgeon: Surgeon C	57/247 (23.1%)	33/113 (29.2%)	27/109 (24.8%)					
Preoperative / intraoperative								
Optical Zone mm	6.51 ± 0.29	6.49 ± 0.32	6.51 ± 0.29	0.765	Diabetes: β = -0.026 [-0.091, 0.039] Autoimmune: β = -0.005 [-0.064, 0.054]	Diabetes: 0.432 Autoimmune: 0.879		
Ablation Depth um	64.05 ± 26.74	67.11 ± 27.86	70.66 ± 23.38	0.064	Diabetes: β = 2.963 [-3.196, 9.122] Autoimmune: β = 6.596 [0.980, 12.211]	Diabetes: 0.346 Autoimmune: 0.021		
Procedure: LASIK	102/247 (41.3%)	48/113 (42.5%)	57/109 (52.3%)	0.152				
Procedure: PRK	108/247 (43.7%)	42/113 (37.2%)	34/109 (31.2%)					
Procedure: SMILE	37/247 (15%)	23/113 (20.4%)	18/109 (16.5%)					
Outcomes								
FollowUp Days	184.84 ± 88.81	174.53 ± 77.83	183.39 ± 84.02	0.521	Diabetes: β = -9.954 [-29.015, 9.108] Autoimmune: β = -1.157 [-20.644, 18.329]	Diabetes: 0.306 Autoimmune: 0.907	Diabetes vs Control: beta = -10.73 [-31.82, 10.36] Autoimmune vs Control: beta = -2.58 [-23.40, 18.24]	Diabetes vs Control: 0.319 Autoimmune vs Control: 0.808
Postop SEQ D	0.02 ± 0.31	-0.18 ± 0.45	-0.14 ± 0.38	<0.001	Diabetes: β = -0.205 [-0.295, -0.115] Autoimmune: β = -0.159 [-0.238, -0.080]	Diabetes: <0.001 Autoimmune: <0.001	Diabetes vs Control: beta = -0.18 [-0.28, -0.08] Autoimmune vs Control: beta = -0.15 [-0.24, -0.07]	Diabetes vs Control: <0.001 Autoimmune vs Control: <0.001
Postop Cylinder D	-0.22 ± 0.18	-0.21 ± 0.18	-0.22 ± 0.18	0.914	Diabetes: β = 0.005 [-0.038, 0.047] Autoimmune: β = -0.007 [-0.049, 0.036]	Diabetes: 0.831 Autoimmune: 0.760	Diabetes vs Control: beta = -0.01 [-0.06, 0.04] Autoimmune vs Control: beta = -0.01 [-0.05, 0.03]	Diabetes vs Control: 0.641 Autoimmune vs Control: 0.694
Postop UDVA	0 ± 0.05	0.05 ± 0.06	0.02 ± 0.05	<0.001	Diabetes: β = 0.055 [0.043, 0.068] Autoimmune: β = 0.021 [0.010, 0.033]	Diabetes: <0.001 Autoimmune: <0.001	Diabetes vs Control: beta = 0.05 [0.04, 0.07] Autoimmune vs Control: beta = 0.02 [0.01, 0.03]	Diabetes vs Control: <0.001 Autoimmune vs Control: 0.001
Postop CDVA	-0.01 ± 0.04	-0.01 ± 0.03	-0.02 ± 0.03	0.138	Diabetes: β = 0.007 [-0.001, 0.015] Autoimmune: β = -0.001 [-0.008, 0.006]	Diabetes: 0.085 Autoimmune: 0.793	Diabetes vs Control: beta = 0.01 [-0.00, 0.02] Autoimmune vs Control: beta = -0.00 [-0.01, 0.01]	Diabetes vs Control: 0.089 Autoimmune vs Control: 0.809
Postop Haze Grade	0 [0–1]	0 [0–0]	0 [0–0]	0.013				
Patient Satisfaction 1to5	4 [4–5] (n = 234)	4 [3–5] (n = 105)	4 [3–5] (n = 104)	<0.001				
DryEye Symptoms (Yes)	40/247 (16.2%)	38/113 (33.6%)	56/109 (51.4%)	<0.001	Diabetes: OR = 2.650 [1.637, 4.292] Autoimmune: OR = 5.477 [3.298, 9.098]	Diabetes: <0.001 Autoimmune: <0.001	Diabetes vs Control: OR = 2.63 [1.47, 4.71] Autoimmune vs Control: OR = 4.99 [2.94, 8.49]	Diabetes vs Control: 0.001 Autoimmune vs Control: <0.001
Retreatment Required (Yes)	8/247 (3.2%)	9/113 (8%)	7/109 (6.4%)	0.131	Diabetes: OR = 2.596 [0.945, 7.128] Autoimmune: OR = 2.083 [0.743, 5.835]	Diabetes: 0.064 Autoimmune: 0.163	Diabetes vs Control: OR = 1.22 [0.42, 3.50] Autoimmune vs Control: OR = 1.49 [0.49, 4.46]	Diabetes vs Control: 0.713 Autoimmune vs Control: 0.481
Complication Type: Corneal haze	15/247 (6.1%)	7/113 (6.2%)	1/109 (0.9%)	0.106				
Complication Type: Diffuse lamellar keratitis	7/247 (2.8%)	3/113 (2.7%)	1/109 (0.9%)					
Complication Type: Epithelial defect	14/247 (5.7%)	6/113 (5.3%)	2/109 (1.8%)					
Complication Type: None	211/247 (85.4%)	97/113 (85.8%)	105/109 (96.3%)					

The adjusted regression sheet lists one comparison for each group against the reference.



NOTE: 1 model(s) carry a small-sample caution or failed - see the Note column. Interpret flagged estimates cautiously.

Adjusted regression - unmatched cohort (covariates: Age_Years, Preop_SEQ_D, Preop_Pachymetry_um)

Outcome	Comparison	Adjusted effect (95% CI)	P value	n	Note
FollowUp_Days	Diabetes vs Control	beta = -10.73 [-31.82, 10.36]	0.319	455	
FollowUp_Days	Autoimmune vs Control	beta = -2.58 [-23.40, 18.24]	0.808	455	
Postop_SEQ_D	Diabetes vs Control	beta = -0.18 [-0.28, -0.08]	<0.001	455	
Postop_SEQ_D	Autoimmune vs Control	beta = -0.15 [-0.24, -0.07]	<0.001	455	
Postop_Cylinder_D	Diabetes vs Control	beta = -0.01 [-0.06, 0.04]	0.641	455	
Postop_Cylinder_D	Autoimmune vs Control	beta = -0.01 [-0.05, 0.03]	0.694	455	
Postop_UDVA	Diabetes vs Control	beta = 0.05 [0.04, 0.07]	<0.001	455	
Postop_UDVA	Autoimmune vs Control	beta = 0.02 [0.01, 0.03]	0.001	455	
Postop_CDVA	Diabetes vs Control	beta = 0.01 [-0.00, 0.02]	0.089	455	
Postop_CDVA	Autoimmune vs Control	beta = -0.00 [-0.01, 0.01]	0.809	455	
Postop_Haze_Grade					Skipped: ordinal outcome
Patient_Satisfaction_1to5					Skipped: ordinal outcome
DryEye_Symptoms	Diabetes vs Control	OR = 2.63 [1.47, 4.71]	0.001	455 (130 events)	
DryEye_Symptoms	Autoimmune vs Control	OR = 4.99 [2.94, 8.49]	<0.001	455 (130 events)	
Retreatment_Required	Diabetes vs Control	OR = 1.22 [0.42, 3.50]	0.713	455 (24 events)	CAUTION: only 24 eyes in the smaller outcome class for 5 parameters (<10 per parameter) - possible overfitting
Retreatment_Required	Autoimmune vs Control	OR = 1.49 [0.49, 4.46]	0.481	455 (24 events)	CAUTION: only 24 eyes in the smaller outcome class for 5 parameters (<10 per parameter) - possible overfitting
Complication_Type					Skipped: categorical outcome

Full models (every term)

Outcome	term	estimate	std.error	statistic	p.value	Effect	EffectValue	CI_low	CI_high	CI	Model
FollowUp_Days	(Intercept)	139.553932383512	92.0250106609974	2.29970645608493		0.129398451507348 Beta	139.553932383512	-40.8150885120431	319.922953279067	[-40.815, 319.923]	Linear GEE (beta)
FollowUp_Days	GroupLabelDiabetes	-10.7306148396143	10.7619464703809	0.994185805993906		0.318721474421519 Beta	-10.7306148396143	-31.8240299215609	10.3628002423324	[-31.824, 10.363]	Linear GEE (beta)
FollowUp_Days	GroupLabelAutoimmune	-2.58280170986044	10.6231969371347	0.059111457136879		0.807905870211212 Beta	-2.58280170986044	-23.4042677066444	18.2386642869235	[-23.404, 18.239]	Linear GEE (beta)
FollowUp_Days	Age_Years	0.273033018487114	0.757418644353707	0.129944636355691		0.718489448729055 Beta	0.273033018487114	-1.21150752444615	1.75757356142038	[-1.212, 1.758]	Linear GEE (beta)
FollowUp_Days	Preop_SEQ_D	-2.32868568837819	2.10304606319199	1.226095159939358		0.268167782805164 Beta	-2.32868568837819	-6.45065597223449	1.79328455954781	[-6.451, 1.793]	Linear GEE (beta)
FollowUp_Days	Preop_Pachymetry_um	0.0505291345834878	0.160484087590905	0.099133346074669		0.752872141689638 Beta	0.0505291345834878	-0.2640196770946	0.365077946261661	[-0.264, 0.365]	Linear GEE (beta)
Postop_SEQ_D	(Intercept)	0.640784591441243	0.368737589003314	3.01987729269577		0.082249689293843 Beta	0.640784591441243	-0.0819410830052518	1.36351026588774	[-0.082, 1.364]	Linear GEE (beta)
Postop_SEQ_D	GroupLabelDiabetes	-0.180640400383681	0.0522370778983375	11.9583710465747		0.000544024112305364 Beta	-0.180640400383681	-0.283025073064496	-0.0782557277028661	[-0.283, -0.078]	Linear GEE (beta)
Postop_SEQ_D	GroupLabelAutoimmune	-0.151543488518369	0.0435889032979012	12.0871156199619		0.000507714578026 Beta	-0.151543488518369	-0.23697773898255	-0.0661092380544823	[-0.237, -0.066]	Linear GEE (beta)
Postop_SEQ_D	Age_Years	-0.00475964710750 38	0.00307238687667637	2.39992519779709		0.121341052348273 Beta	-0.00475964710750 38	-0.0107815253857895	0.00126223111078188	[-0.011, 0.001]	Linear GEE (beta)
Postop_SEQ_D	Preop_SEQ_D	0.0123718009319436	0.00987243385703956	1.57042565444075		0.210145209139287 Beta	0.0123718009319436	-0.0069781694278539 1	0.0317217712917411	[-0.007, 0.032]	Linear GEE (beta)
Postop_SEQ_D	Preop_Pachymetry_um	-0.000765264210321962	0.000649786841322392	1.18701443547977		0.238099377078054 Beta	-0.000765264210321962	-0.0020388464193138 5	0.000508317998669927	[-0.002, 0.001]	Linear GEE (beta)
Postop_Cylinder_D	(Intercept)	-0.0913117785407859	0.179943051321315	0.257616495888614		0.611762402992815 Beta	-0.0913117785407859	-0.4440201591305 64	0.261356602048992	[-0.444, 0.261]	Linear GEE (beta)
Postop_Cylinder_D	GroupLabelDiabetes	-0.011315055101256	0.0243002881321515	0.216815586737 3		0.64147706762463 Beta	-0.011315055101256	-0.0589436198402324	0.0363135096378013	[-0.059, 0.036]	Linear GEE (beta)
Postop_Cylinder_D	GroupLabelAutoimmune	-0.008768902485077 19	0.022292052277923	0.154735647039323		0.694050269445202 Beta	-0.008768902485077 19	-0.0524613249449 42	0.0349235199797876	[-0.052, 0.035]	Linear GEE (beta)
Postop_Cylinder_D	Age_Years	0.00154497571032591	0.00139538213381444	1.12590550647067		0.268204799848145 Beta	0.00154497571032591	-0.0011899732719504	0.00427992469260222	[-0.001, 0.004]	Linear GEE (beta)
Postop_Cylinder_D	Preop_SEQ_D	0.00582025506589207	0.00453828487532528	1.64475239520157		0.199674665689154 Beta	0.00582025506589207	-0.0030747832897454 8	0.0147152934215296	[-0.003, 0.015]	Linear GEE (beta)
Postop_Cylinder_D	Preop_Pachymetry_um	-0.000273991143923539	0.000320398530814246	0.731294015515 02		0.392464069140 68 Beta	-0.000273991143923539	-0.0009019722643 1946 1	0.000533989976472382	[-0.001, 0.000]	Linear GEE (beta)
Postop_UDVA	(Intercept)	-0.056129410903202 8	0.051180539581 3385	1.20273859650754		0.272775007357981 Beta	-0.056129410903202 8	-0.156443268482626	0.0441844466762206	[-0.156, 0.044]	Linear GEE (beta)
Postop_UDVA	GroupLabelDiabetes	0.0537613551214243	0.0069011931219539 8	60.6864946425773		6.661338147750 94E-15 Beta	0.0537613551214243	0.040235016602 3945	0.067287693640542	[0.040, 0.067]	Linear GEE (beta)
Postop_UDVA	GroupLabelAutoimmune	0.0200295734048808	0.0065175675084617	10.5809590970615		0.001142583674744 Beta	0.0200295734048808	0.007960741088718 32	0.0320984057210433	[0.008, 0.032]	Linear GEE (beta)
Postop_UDVA	Age_Years	0.000313163747750312	0.000417247528800 5	0.563320174024256		0.452925578723693 Beta	0.000313163747750312	-0.000504641514538668	0.00113096901003929	[-0.001, 0.001]	Linear GEE (beta)
Postop_UDVA	Preop_SEQ_D	-0.001344063587964 73	0.001361682536136 96	0.974289217885862		0.323612784947945 Beta	-0.001344063587964 73	-0.0040129613587978 8	0.0013248341828684 2	[-0.004, 0.001]	Linear GEE (beta)
Postop_UDVA	Preop_Pachymetry_um	7.372768591949 53E-05	8.947936662982 74E-05	0.678914998721377		0.409960590921654 Beta	7.372768591949 53E-05	-0.000101651872674966	0.000249107244513957	[-0.000, 0.000]	Linear GEE (beta)
Postop_CDVA	(Intercept)	-0.048049294753523	0.035241237566587 6	1.85896722682787		0.172744044421691 Beta	-0.048049294753523	-0.117122120364434	0.0210235308573893	[-0.117, 0.021]	Linear GEE (beta)
Postop_CDVA	GroupLabelDiabetes	0.007711196617930 57	0.004536689591323 47	2.88911895400274		0.089179682067190 4 Beta	0.007711196617930 57	-0.001180714981063 43	0.0166031082169246	[-0.001, 0.017]	Linear GEE (beta)
Postop_CDVA	GroupLabelAutoimmune	-0.000954202313814271	0.003947338219437 15	0.058434892772368 4		0.808986978126863 Beta	-0.000954202313814271	-0.008690985223911 09	0.0067825805962855	[-0.009, 0.007]	Linear GEE (beta)
Postop_CDVA	Age_Years	1.275790755437 6E-05	0.000262727995755749	0.002358013005917 62		0.961270444935204 Beta	1.275790755437 6E-05	-0.000502188964126893	0.000527704779235645	[-0.001, 0.001]	Linear GEE (beta)
Postop_CDVA	Preop_SEQ_D	-0.000700256690304389	0.000992950365229449	0.497346944111173		0.480668177224998 Beta	-0.000700256690304389	-0.002646439406154 11	0.0012459260255453 3	[-0.003, 0.001]	Linear GEE (beta)
Postop_CDVA	Preop_Pachymetry_um	5.584168385297 11E-05	6.125513756871 53E-05	0.831059219905896		0.361966423270154 Beta	5.584168385297 11E-05	-6.421838578171 09E-05	0.000175901753487653	[-0.000, 0.000]	Linear GEE (beta)
DryEye_Symptoms	(Intercept)	0.232672746945345	2.31623613330896	0.010090794682988 4		0.919984736188897 OR	1.261968428721 73	0.013471888817955	118.213884972675	[0.013, 118.214]	Logistic GEE (OR)
DryEye_Symptoms	GroupLabelDiabetes	0.966889144985501	0.297354356747862	10.5731588493617		0.001147414898568 95 OR	2.629750947625 99	1.468254977873 97	4.710074306407 99	[1.468, 4.710]	Logistic GEE (OR)
DryEye_Symptoms	GroupLabelAutoimmune	1.608319853666654	0.270595578512 42	35.3361327374921		2.774369067815 03E-09 OR	4.994412830136 34	2.938863090300 265	8.487687875694 05	[2.939, 8.488]	Logistic GEE (OR)
DryEye_Symptoms	Age_Years	-0.003792408289145 86	0.017417939740724 8	0.047463596351245		0.827639475714646 OR	0.996214738091 52	0.962778821417597	1.030811909733 26	[0.963, 1.031]	Logistic GEE (OR)
DryEye_Symptoms	Preop_SEQ_D	-0.055379982728259	0.060024568493038 7	0.851231210292738		0.356204267561483 OR	0.946125568265628	0.841113948109471	1.064247707386 08	[0.841, 1.064]	Logistic GEE (OR)
DryEye_Symptoms	Preop_Pachymetry_um	-0.003657933791917 2	0.004123934499813 65	0.786770673619456		0.375078392560027 OR	0.996348748297865	0.988327809362893	1.004434782498 59	[0.988, 1.004]	Logistic GEE (OR)
Retreatment_Required	(Intercept)	-2.544156184971 67	3.898171452707 71	0.425956906856438		0.513979530598709 OR	0.078539296666501 2	3.774822016609399E-05	163.409588418404	[0.000, 163.410]	Logistic GEE (OR)
Retreatment_Required	GroupLabelDiabetes	0.198203781641696	0.538232055590124	0.135607826005774		0.712687070495075 OR	1.219210822910 4	0.424547642574568	3.503134994328 56	[0.425, 3.501]	Logistic GEE (OR)
Retreatment_Required	GroupLabelAutoimmune	0.395530392689548	0.561304670781822	0.496549339626464		0.481020249031785 OR	1.485171706271 99	0.494293028751589	4.462403612451 84	[0.494, 4.462]	Logistic GEE (OR)
Retreatment_Required	Age_Years	0.086014099881717	0.041612930770285 4	4.272506085172 9		0.038733764529230 8 OR	1.089821694553 78	1.004462683818 6	1.182434349463432	[1.004, 1.182]	Logistic GEE (OR)
Retreatment_Required	Preop_SEQ_D	-0.104421166681869	0.126271742606373	0.683856114642945		0.408262039099744 OR	0.900845811272144	0.703340759862016	1.153812237251 51	[0.703, 1.154]	Logistic GEE (OR)
Retreatment_Required	Preop_Pachymetry_um	-0.007446312855053 85	0.006164452590345 26	1.459128572606		0.227068875066963 OR	0.992581342247093	0.980660807958628	1.004667779495 5	[0.981, 1.005]	Logistic GEE (OR)

APPENDIX · READING THE OUTPUT WELL

1. **Check the Overview first.** Wrong group sizes are almost always a grouping or filter problem, and everything downstream inherits the error.

2. **Read the Test column.** It tells you, and your reviewer, exactly what produced each *p* value.

3. **If fellow eyes are present, prefer the clustered estimate.** The unclustered *p* values are anticonservative in that situation.

4. **Check post-matching balance before believing the matched comparison.** Matching can worsen balance on an individual covariate even while improving the propensity score overall.

5. **Do not report an odds ratio from a zero-event cell.** Give the counts and the exact test instead.

6. **Save the settings file (*Download config JSON*) and the filtered CSV with the manuscript.** Together they make the analysis exactly reproducible.

StandardEyes · By Noyman DBE, Mimouni M, Safir M